ON THE DEMAND-SUPPLY GEOMETRY OF REPRESENTATION COLLAPSE

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ABSTRACT

Preventing representation collapse remains a central challenge in Joint-Embedding Predictive Architectures (JEPAs). Endpoint-only constraints cannot fully prevent or repair collapse when intermediate representations lose sample distinctions. We introduce **SADL** (Supply And Demand Loss), an adaptive one-sided bridge that matches endpoint demand to the available capacity of the representation path. **SADL** compares demand from the output objective and realized endpoint geometry with the weaker of encoder excitation and cross-endpoint transmission capacity, applying a correction only when demand exceeds supply. It works with different output regularization functions without auxiliary networks, learned gates, or a separate schedule. In single-task and jointly trained four-task LeWorldModel (LeWM) and INTACT experiments, the collapse rate falls from 50% to 0% in a ten-seed collapse test. After only 1 epoch of training with **SADL**, encoder effective rank increases by 3.4–6.4× and latent prediction error decreases by 20–303× relative to the corresponding endpoint-only baselines. Under multiple regularization functions, INTACT also achieves over 95% mean search-free control success. Across four image datasets, each with multiple paired seeds, **SADL** delivers consistent seed-wise improvements in accuracy and stability at representative checkpoints, with its largest gains emerging early in training. These findings indicate that demand–supply matching along the representation path is a general principle for faster and more stable representation learning across control and image classification.

1 INTRODUCTION

Joint-Embedding Predictive Architectures (JEPAs) learn from images and video by predicting latent targets instead of reconstructing every pixel (Assran et al., 2023; Bardes et al., 2024). Predictability alone, however, does not ensure a useful representation: suppressing distinctions between samples can also make targets easier to predict. Existing anti-collapse objectives typically specify statistics or geometry at a projected endpoint Z (Zbontar et al., 2021; Bardes et al., 2022a; Balestriero & LeCun, 2025). Yet an apparently regular endpoint does not, on its own, certify that its encoder representation H retains sample variation, or that such variation survives the projection from H to Z. In a world model, losing these distinctions can turn different states or action consequences into nearly indistinguishable control inputs.

The projection interface makes this gap concrete. Endpoint gradients reach H through the Jacobian of the projector; they can shape upstream features, as prior analysis shows (Mialon et al., 2022), but their action depends on the directions that the projector transmits. In our audited endpoint-only INTACT–VISReg run, Official success falls from 55.67% at epoch one to 24.75% at epoch five, while the effective rank of H falls from 13.74 to 6.64 (Table 2). This example does not imply that every endpoint objective collapses. It shows why output-side regularization needs a further condition: the representation path must be able to supply and transmit the variation demanded at its endpoint.

This leads us to ask not which endpoint density is universally preferable, but *whether the endpoint's present demand matches the path's attainable supply*. The answer changes over training and across endpoint objectives. Upstream supply has two distinct limitations: the encoder may fail to excite sample-dependent directions, or the projector may fail to transmit them. We therefore in-

论表征坍塌的需求–供给几何

匿名作者 论文接受双盲评审

摘要

防止表征坍塌仍是联合嵌入预测架构（JEPA）的核心挑战。当中间表征丢失样本差异时，仅约束端点无法彻底预防或修复坍塌。我们提出 SADL（供给与需求损失），一种自适应的单侧桥接机制，将端点需求与表征路径的可用容量匹配。SADL 比较输出目标和已实现端点几何的需求与编码器激发、跨端点传输容量中较弱的一项，仅在需求超过供给时校正。它适用于不同输出正则化函数，无需辅助网络、学习门控或独立调度。在单任务与联合四任务 LeWorldModel（LeWM）、INTACT 实验的十种子坍塌测试中，坍塌率从 50% 降至 0%。使用 SADL 仅训练一轮，编码器有效秩较对应的仅端点基线增加 3.4–6.4×，潜变量预测误差下降 20–303×。在多种正则化函数下，INTACT 无搜索控制的平均成功率超过 95%。在四个图像数据集、各含多个配对种子的实验中，SADL 在代表性检查点上稳定改善逐种子的准确率与稳定性，最大增益出现在训练早期。这表明沿表征路径匹配需求与供给，是实现更快、更稳定表征学习的一般原则，适用于控制和图像分类。

1 引言

联合嵌入预测架构（JEPA）通过预测潜空间目标而非重建每个像素，从图像与视频中学习（Assran 等，2023；Bardes 等，2024）。但可预测性并不保证表征有用：压制样本间差异也会使目标更容易预测。现有防坍塌目标通常规定投影端点 Z 的统计量或几何结构（Zbontar 等，2021；Bardes 等，2022a；Balestriero 与 LeCun，2025）。然而，端点看似规整，本身并不能保证编码器表征 H 保留样本变化，也不能保证这种变化从 H 投影到 Z 后仍存在。在世界模型中，丢失这些差异可能使不同状态或动作后果变成几乎不可区分的控制输入。

投影接口具体体现了这一缺口。端点梯度通过投影器的雅可比矩阵到达 H；既有分析表明它们能塑造上游特征（Mialon 等，2022），但效果取决于投影器传输的方向。在经过审计的仅端点 INTACT–VISReg 运行中，官方成功率从第一轮的 55.67% 降到第五轮的 24.75%，而 H 的有效秩从 13.74 降到 6.64（表 2）。该例不意味着每个端点目标都会坍塌，而是说明输出端正则化需要额外条件：表征路径必须能够供给并传输端点所要求的变化。

因此，我们提出的问题不是哪一种端点密度普遍更优，而是端点当前的需求是否匹配路径可达到的供给。答案会随训练及端点目标变化。上游供给有两种独立限制：编码器可能无法激发依赖样本的方向，或投影器无法传输这些方向。因此我们引入

troduce **SADL** (**S**upply **A**nd **D**emand **L**oss), a general supply–demand bridge that can **Shape Any Demand-aligned Latent** representation. **SADL** reads demand from both the endpoint objective and the structure already attained at Z, measures supply through the weaker of encoder excitation and cross-endpoint transmission, and corrects H only while supply falls short. OT-based endpoint losses shape the distribution of Z; **SADL** instead repairs insufficient excitation and transmission along $H \to Z$, without optimizing a transport plan. The selected endpoint objective remains unchanged; **SADL** adds no auxiliary networks, learnable gates, or separate regularization schedule.

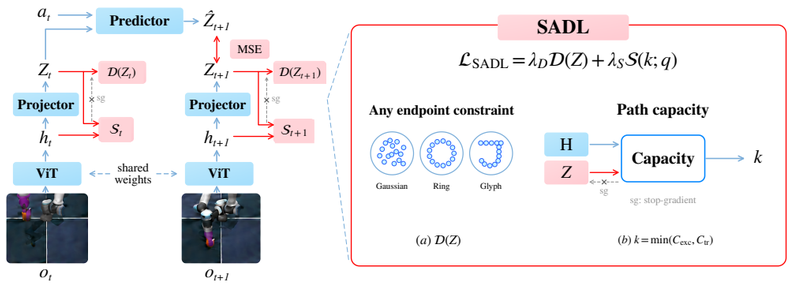


Figure 1: Overview of **SADL**. The endpoint constraint $\mathcal{D}(Z)$ and realized endpoint geometry $C_{\text{exc}}(\text{sg}(Z))$ jointly determine demand q; encoder excitation and cross-endpoint transmission determine path capacity k. Their mismatch is penalized by the one-sided supply loss $\mathcal{S}$ alongside the task objective. The figure displays the scalar dependence on q and k; Section 3.4 gives its gradient-routing arguments.

We evaluate **SADL** on LeWorldModel (LeWM) and INTACT with multiple endpoint constraints in single-task and jointly trained four-task control, and examine its generalization to image classification. Across more than ten tested endpoint target distributions and geometries, one epoch of joint four-task INTACT training yields at least 92% mean search-free success under the reported evaluation protocols (Appendix Figure 6). By epoch three, the INTACT curves reach approximately 96% mean Official Direct success (Figure 3). Across image self-supervised learning datasets, **SADL** also serves as a near-zero-overhead plug-in that accelerates convergence while yielding more stable and higher-quality representations.

Our contributions are:

- We explicitly separate endpoint statistical regularity, encoder excitation, and cross-endpoint transmission. Across endpoint choices, our analysis identifies a mismatch between endpoint demand and upstream path capacity as a mechanism of collapse; endpoint loss form alone does not diagnose it.
- We introduce **SADL**, a low-overhead, adaptive, one-sided correction loss function that attaches to any endpoint objective satisfying the stated conditions and targets an observed supply deficit rather than imposing a fixed upstream distribution.
- Across more than ten tested endpoint objectives and several representation-learning settings, we study early control, convergence, representation probes, and image classification. Across the evaluated tasks, the results support path-capacity matching as a transferable design principle and show faster convergence and more stable representations.

2 Related Work

Predictive representations and control world models. Latent-dynamics models learn compact states for planning or policy improvement (Hafner et al., 2019; 2023; Hansen et al., 2024); iCEM and self-predictive representations refine planning and predictive training (Pinneri et al., 2021; Schwarzer et al., 2021). I-JEPA and V-JEPA predict in learned image/video feature spaces (Assran et al., 2023; Bardes et al., 2024), with later video models extending toward planning and dense

SADL（供给与需求损失），一种通用供给–需求桥接机制，可塑造任意与需求对齐的潜表征。SADL 从端点目标及 Z 已达到的结构读取需求，以编码器激发、跨端点传输中较弱者衡量供给，仅在供给不足时校正 H。基于 OT 的端点损失塑造 Z 分布；SADL 则修复 $H \to Z$ 路径上不足的激发与传输，不优化传输计划。所选端点目标不变；SADL 不增加辅助网络、可学习门控或独立正则化调度。

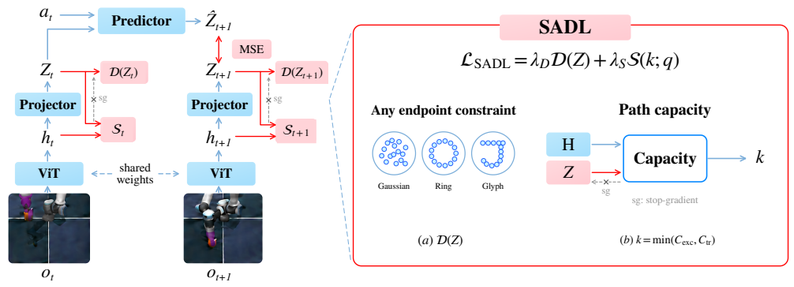


图 1：SADL 概览。端点约束 $\mathcal{D}(Z)$ 与已实现端点几何 $C_{\text{exc}}(\text{sg}(Z))$ 共同决定需求 q；编码器激发和跨端点传输决定路径容量 k。单侧供给损失 $\mathcal{S}$ 与任务目标一起惩罚二者失配。图中展示对 q、k 的标量依赖；第 3.4 节给出梯度路径参数。

我们在单任务和联合四任务控制中，以多个端点约束评估 LeWorldModel（LeWM）与 INTACT 上的 SADL，并检验其对图像分类的泛化。在十余种测试的端点目标分布和几何结构下，联合四任务 INTACT 训练一轮后，在所报告评估协议中均达到至少 92% 的平均无搜索成功率（附录图 6）。至第三轮，INTACT 曲线达到约 96% 的平均官方 Direct 成功率（图 3）。在图像自监督学习数据集上，SADL 也作为几乎无额外开销的插件加速收敛，产生更稳定、质量更高的表征。

我们的贡献如下：

- 我们显式区分端点统计规整性、编码器激发与跨端点传输。跨端点选择的分析将端点需求与上游路径容量失配辨识为坍塌机制；仅端点损失形式无法诊断它。
- 我们提出 SADL，一种低开销、自适应单侧校正损失，可附加于满足给定条件的任何端点目标，针对观测供给缺口，而不强制固定上游分布。
- 我们在十余种端点目标及多种表征学习设置中，研究早期控制、收敛、表征探针与图像分类。评估任务结果支持路径容量匹配作为可迁移设计原则，显示更快收敛与更稳定表征。

2 相关工作

预测表征与控制世界模型。潜动力学模型学习用于规划或策略改进的紧凑状态（Hafner 等，2019；2023；Hansen 等，2024）；iCEM 和自预测表征改进规划与预测训练（Pinneri 等，2021；Schwarzer 等，2021）。I-JEPA、V-JEPA 在学习的图像/视频特征空间中预测（Assran 等，2023；Bardes 等，2024），后续视频模型扩展到规划与密集

features (Assran et al., 2025; Mur-Labadia et al., 2026). Other JEPA-style world models explore compact dynamics, object-level latent masking, and value-guided planning (Zhou et al., 2024; Nam et al., 2026; Destrade et al., 2026). LeWM and INTACT provide the planning and search-free control interfaces studied here (Maes et al., 2026; Sun et al., 2026); PhyLatent distinguishes statistical non-collapse from dynamics-relevant information (Zeng et al., 2026). **SADL** focuses on the encoder–projector path capacity supporting a chosen endpoint objective.

Self-supervised objectives and collapse. Contrastive image learning uses pairwise or assignment relations (Chen et al., 2020; He et al., 2020; Caron et al., 2020), whereas BYOL, SimSiam, and DINO learn without explicit negative pairs through architectural or optimization asymmetries (Grill et al., 2020; Chen & He, 2021; Caron et al., 2021). Masked prediction and self-distillation provide other routes (He et al., 2022; Zhou et al., 2022; Assran et al., 2022; Baevski et al., 2022; Oquab et al., 2023). These methods differ in their prediction targets and collapse-avoidance mechanisms; **SADL** instead studies how upstream supply is transmitted to meet endpoint demand along the representation path.

Endpoint geometry and statistical regularization. Alignment and uniformity, whitening, Barlow Twins, VICReg, and VICRegL constrain embedding geometry through distributional or moment statistics (Wang & Isola, 2020; Ermolov et al., 2021; Zbontar et al., 2021; Bardes et al., 2022a;b; Balestriero & LeCun, 2025; Zhao et al., 2026). VISReg's shape term uses sliced W_2^2 matching, while QQWorld's quantile objective approximates one-dimensional W_2^2 (Wu et al., 2026; Yu et al., 2026). In the balanced setting, Earth Mover's Distance corresponds to W_1 for normalized discrete measures with a metric ground cost (Peyré & Cuturi, 2019; Rubner et al., 2000). The transport-based objectives above constrain endpoint geometry; **SADL** retains the selected loss but compares its demand with upstream excitation and cross-endpoint association, without optimizing a transport coupling.

Projection interfaces and intermediate regularization. DirectCLR and Guillotine Regularization show that projection and readout choices affect dimensional collapse and downstream utility (Jing et al., 2022; Bordes et al., 2023). Under specific projector conditions, endpoint variance–covariance control can induce pre-projection independence; VCReg and Seq-VCR also constrain intermediate representations (Mialon et al., 2022; Zhu et al., 2023; Arefin et al., 2024). Analyses of non-contrastive Siamese learning examine architectural and optimization routes around collapse (Zhang et al., 2022; Li et al., 2022), while the contrastive–non-contrastive duality concerns relations between loss families under stated assumptions (Garrido et al., 2023). Unlike fixed intermediate constraints, **SADL** conditions its correction on demand from the selected endpoint objective.

3 METHOD

For N aligned samples, an encoder f_θ and projector g_ϕ produce $H = f_\theta(X) \in \mathbb{R}^{N\times d_h}$ and $Z = g_\phi(H) \in \mathbb{R}^{N\times d_z}$. We retain the task loss $\mathcal{L}_{\text{task}}$ and an endpoint constraint $\mathcal{D}(Z) \geq 0$, where lower values indicate greater endpoint satisfaction and $0 < \mathcal{D}(0) < \infty$. For a transport-based $\mathcal{D}$, **SADL** calibrates q_R from its detached scalar value; the bridge solves no transport problem (Peyré & Cuturi, 2019; Wu et al., 2026; Yu et al., 2026). **SADL** reads a detached endpoint demand q, measures the available path supply k, and corrects H only when $k < q$. Figure 2 summarizes the demand and supply branches and their gradient routes.

3.1 ENDPOINT CONSTRAINTS AND REPRESENTATION-PATH CAPACITY

For the batch-level projector $Z = \mathcal{G}_\phi(H)$, backpropagation gives $\nabla_H \mathcal{D}(Z) = J_{\mathcal{G}_\phi}(H)^\top \nabla_Z \mathcal{D}(Z)$. Thus, for any perturbation ΔH,

$$|\langle \nabla_H \mathcal{D}(Z), \Delta H\rangle_F| \leq \|\nabla_Z \mathcal{D}(Z)\|_F \left\|J_{\mathcal{G}_\phi}(H)[\Delta H]\right\|_F . \tag{1}$$

The corrective signal along an upstream direction is limited by the projector's local response and vanishes in its Jacobian null space. This motivates a direct constraint on H, coordinated with the structure formed at Z. We characterize representation-path capacity through encoder excitation, which measures sample-dependent variation, and cross-endpoint transmission, which measures its

特征（Assran 等，2025；Mur-Labadia 等，2026）。其他 JEPA 式世界模型探索紧凑动力学、物体级潜空间掩码及价值引导规划（Zhou 等，2024；Nam 等，2026；Destrade 等，2026）。LeWM 和 INTACT 提供本文研究的规划与无搜索控制接口（Maes 等，2026；Sun 等，2026）；PhyLatent 区分统计非坍塌与动力学相关信息（Zeng 等，2026）。SADL 关注支持所选端点目标的编码器–投影器路径容量。

自监督目标与坍塌。对比式图像学习使用成对或分配关系（Chen 等，2020；He 等，2020；Caron 等，2020）；BYOL、SimSiam 和 DINO 通过架构或优化不对称性，在没有显式负样本对时学习（Grill 等，2020；Chen 与 He，2021；Caron 等，2021）。掩码预测与自蒸馏提供其他路径（He 等，2022；Zhou 等，2022；Assran 等，2022；Baevski 等，2022；Oquab 等，2023）。这些方法的预测目标和防坍塌机制不同；SADL 则研究上游供给如何沿表征路径传输，以满足端点需求。

端点几何与统计正则化。对齐与均匀性、白化、Barlow Twins、VICReg 和 VICRegL 通过分布或矩统计量约束嵌入几何（Wang 与 Isola，2020；Ermolov 等，2021；Zbontar 等，2021；Bardes 等，2022a;b；Balestriero 与 LeCun，2025；Zhao 等，2026）。VISReg 形状项使用切片 W_2^2 匹配，QQWorld 分位数目标近似一维 W_2^2（Wu 等，2026；Yu 等，2026）。在平衡设置中，对具有度量底层代价的归一化离散测度，推土机距离对应 W_1（Peyr´e 与 Cuturi，2019；Rubner 等，2000）。上述基于传输的目标约束端点几何；SADL 保留所选损失，却将其需求与上游激发、跨端点关联比较，不优化传输耦合。

投影接口与中间正则化。DirectCLR 与 Guillotine Regularization 表明投影和读出选择会影响维度坍塌与下游效用（Jing 等，2022；Bordes 等，2023）。在特定投影器条件下，端点方差–协方差控制可诱导投影前独立性；VCReg 与 Seq-VCR 也约束中间表征（Mialon 等，2022；Zhu 等，2023；Arefin 等，2024）。非对比式孪生学习分析研究绕开坍塌的架构与优化路径（Zhang 等，2022；Li 等，2022）；对比–非对比对偶性则讨论给定假设下损失族之间的关系（Garrido 等，2023）。与固定中间约束不同，SADL 根据所选端点目标的需求决定校正。

3 方法

对于 N 个对齐样本，编码器 f_θ 与投影器 g_ϕ 产生 $H = f_\theta(X) \in \mathbb{R}^{N\times d_h}$ 和 $Z = g_\phi(H) \in \mathbb{R}^{N\times d_z}$。我们保留任务损失 $\mathcal{L}_{\text{task}}$ 与端点约束 $\mathcal{D}(Z) \geq 0$；值越低表示端点满足程度越高，且 $0 < \mathcal{D}(0) < \infty$。若 $\mathcal{D}$ 基于传输，SADL 用其停止梯度的标量值校准 q_R；桥接项不求解任何传输问题（Peyr´e 与 Cuturi，2019；Wu 等，2026；Yu 等，2026）。SADL 读取停止梯度的端点需求 q，测量可用路径供给 k，仅在 $k < q$ 时校正 H。图 2 总结需求、供给分支及梯度路径。

3.1 端点约束与表征路径容量

对于批次级投影器 $Z = \mathcal{G}_\phi(H)$，反向传播给出 $\nabla_H \mathcal{D}(Z) = J_{\mathcal{G}_\phi}(H)^\top \nabla_Z \mathcal{D}(Z)$。因此，对任意扰动 ΔH，

$$|\langle \nabla_H \mathcal{D}(Z), \Delta H\rangle_F| \leq \|\nabla_Z \mathcal{D}(Z)\|_F \left\|J_{\mathcal{G}_\phi}(H)[\Delta H]\right\|_F . \tag{1}$$

沿上游方向的校正信号受投影器局部响应限制，并在其雅可比零空间中消失。这推动我们直接约束 H，同时与 Z 形成的结构协调。我们用衡量样本相关变化的编码器激发，以及衡量其

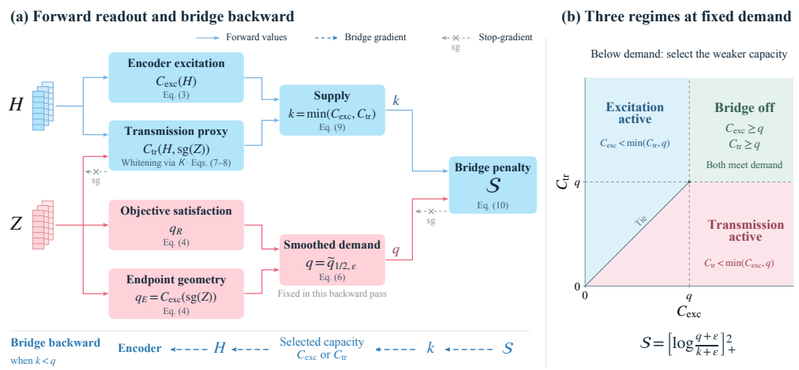


Figure 2: Demand–supply mechanism. Detached endpoint readouts determine demand q, while the weaker of encoder excitation and cross-endpoint transmission determines supply k. The $\tilde{q}_{1/2,\epsilon}$ diagram depicts the smoothed symmetric demand used in training, as defined in equation 6. The bridge acts only for $k < q$ and updates the encoder-side representation; ordinary task and endpoint gradients remain in place.

linear association with Z. This association score operationalizes path support but does not estimate the projector Jacobian.

3.2 Normalized Geometry and Endpoint Demand

For $U \in \mathbb{R}^{N \times d_U}$ and $\epsilon > 0$, define

$$\mu_U = \frac{1}{N} U^\top \mathbf{1}, \quad \bar{U} = U - \mathbf{1}\mu_U^\top,$$
$$\Sigma_U = \frac{1}{N} \bar{U}^\top \bar{U}, \quad e_U = \max\left\{ \frac{\|U\|_F^2}{N d_U}, \epsilon \right\}. \tag{2}$$

The same normalized excitation score is read at Z to calibrate demand and at H to measure encoder-side supply:

$$C_{\mathrm{exc}}(U) = \frac{\log\det(I + \gamma \Sigma_U / e_U)}{d_U \log(1+\gamma)}, \qquad \gamma > 0. \tag{3}$$

This score lies in $[0, 1]$, is orthogonally invariant, and is scale invariant whenever the energy floor remains inactive. Normalization by uncentered energy makes it sensitive to common-mean energy and spectral concentration rather than uniform feature amplification.

The endpoint supplies an objective-based reference and a realized-geometry reference. With $\mathcal{D}_0 = \mathcal{D}(0)$, define

$$q_R := \mathrm{clip}\left(1 - \frac{\mathcal{D}(Z)}{\mathcal{D}_0}, 0, 1\right), \qquad q_E = C_{\mathrm{exc}}(\mathrm{sg}(Z)). \tag{4}$$

Here q_R increases as the endpoint objective is satisfied, while q_E measures current endpoint excitation. The calibration is invariant to common positive scaling of $\mathcal{D}(Z)$ and $\mathcal{D}_0$, but depends on the prescribed loss zero point. For positive references, the ideal log-space interpolation is

$$q_\alpha^\star = q_R^{1-\alpha} q_E^\alpha, \qquad \alpha \in [0, 1]. \tag{5}$$

The implementation evaluates its numerically smoothed counterpart

$$\widetilde{q}_{\alpha,\epsilon} := \exp\big((1-\alpha)\log(q_R + \epsilon) + \alpha \log(q_E + \epsilon)\big). \tag{6}$$

We use the symmetric default $q = \widetilde{q}_{1/2,\epsilon}$; $q_0 = q_R$ and $q_1 = q_E$ denote separately defined boundary methods. The ideal demand extends continuously to zero when either positive-weight reference is zero, while implementation retains an ϵ-scale numerical floor. Adaptation follows the evolving endpoint statistics without a learned coefficient.

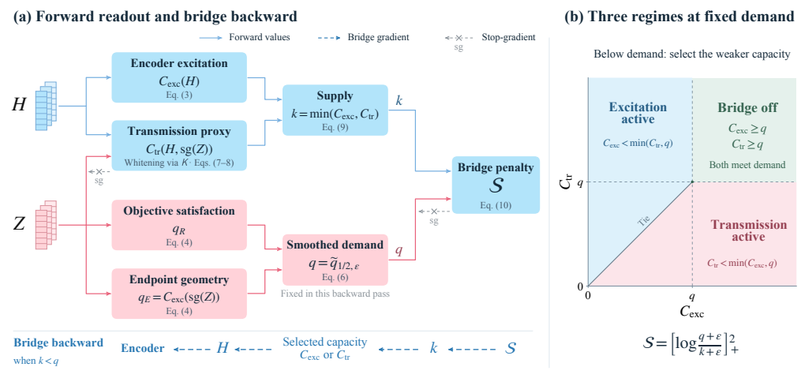


图 2：需求–供给机制。停止梯度的端点读数决定需求 q；编码器激发与跨端点传输中较弱者决定供给 k。$\tilde{q}_{1/2,\ \epsilon}$ 图展示式 6 定义、训练使用的平滑对称需求。桥接只在 $k < q$ 时激活，更新编码器侧表征；常规任务与端点梯度仍然保留。

与 Z 的线性关联。该关联分数操作化路径支持，但不估计投影器雅可比。

3.2 归一化几何与端点需求

对于 $U \in \mathbb{R}^{N \times d_U}$、$\epsilon > 0$，定义

$$\mu_U = \frac{1}{N} U^\top \mathbf{1}, \quad \bar{U} = U - \mathbf{1}\mu_U^\top,$$
$$\Sigma_U = \frac{1}{N} \bar{U}^\top \bar{U}, \quad e_U = \max\left\{ \frac{\|U\|_F^2}{N d_U}, \epsilon \right\}. \tag{2}$$

同一个归一化激发分数在 Z 处读取以校准需求，在 H 处读取以测量编码器侧供给：

$$C_{\mathrm{exc}}(U) = \frac{\log\det(I + \gamma \Sigma_U / e_U)}{d_U \log(1+\gamma)}, \qquad \gamma > 0. \tag{3}$$

该分数位于 $[0, 1]$, ，具有正交不变性；能量下限未激活时还具有尺度不变性。用未中心化能量归一化使其对公共均值能量和谱集中敏感，而不是对统一特征放大敏感。

端点提供基于目标的参考与已实现几何参考。对于 $\mathcal{D}_0 = \mathcal{D}(0)$，定义

$$q_R := \mathrm{clip}\left(1 - \frac{\mathcal{D}(Z)}{\mathcal{D}_0}, 0, 1\right), \qquad q_E = C_{\mathrm{exc}}(\mathrm{sg}(Z)). \tag{4}$$

这里 q_R 随端点目标被满足而增加， q_E 衡量当前端点激发。校准对 $\mathcal{D}(Z)$ 与 $\mathcal{D}_0$ 同时正缩放不变，但依赖规定的损失零点。正参考下，理想对数空间插值为

$$q_\alpha^\star = q_R^{1-\alpha} q_E^\alpha, \qquad \alpha \in [0, 1]. \tag{5}$$

实现计算其数值平滑形式

$$\widetilde{q}_{\alpha,\epsilon} := \exp\big((1-\alpha)\log(q_R + \epsilon) + \alpha \log(q_E + \epsilon)\big). \tag{6}$$

我们使用对称默认值 $q = \widetilde{q}_{1/2}$、ϵ； $q_0 = q_R$、 $q_1 = q_E$ 表示另行定义的边界方法。当任一具有正权重的参考为零时，理想需求连续延拓为零；实现则保留 ϵ 尺度数值下限。自适应跟随变化的端点统计，不使用学习系数。

3.3 Excitation, Transmission, and Path Supply

Let $\Sigma_{HZ} = N^{-1}\bar{H}^\top \bar{Z}$. Using a regularized form of canonical correlation (Hotelling, 1936), define

$$A_H = \Sigma_H + \rho\tau_H I, \qquad A_Z = \Sigma_Z + \rho\tau_Z I,$$
$$\tau_U = \max\left\{\frac{\mathrm{tr}(\Sigma_U)}{d_U}, \epsilon\right\}, \quad K = A_H^{-1/2}\Sigma_{HZ}A_Z^{-1/2}, \tag{7}$$

where $\rho > 0$. The endpoint entering this score is $\mathrm{sg}(Z)$, whereas the relative ridge scale on the H side remains differentiable. This preserves the backward, as well as forward, invariance of the channel score to global rescaling of H whenever the floor is inactive. We measure transmission by

$$C_{\mathrm{tr}}(H, \mathrm{sg}(Z)) = \frac{\log\det(I + \gamma K^\top K)}{r\log(1+\gamma)}, \qquad r = \min(d_h, d_z). \tag{8}$$

Normalization by all potential interface dimensions distinguishes broad association from strong correlation in only a few directions. This second-order association proxy does not capture general nonlinear dependence; at zero cross-covariance, both its value and first-order gradient vanish.

The path supply is the limiting capacity:

$$k = \min\{C_{\mathrm{exc}}(H), C_{\mathrm{tr}}(H, \mathrm{sg}(Z))\}. \tag{9}$$

Hence $k \geq q$ requires both scores to reach q. In this two-stage path-capacity model, the weaker of excitation and transmission is the active cut.

3.4 One-Sided Demand–Supply Matching

We penalize insufficient path supply through the one-sided supply-shortfall correction

$$\mathcal{S}\big(H, \mathrm{sg}(Z); \mathrm{sg}(q)\big) = \left[\log\frac{q+\epsilon}{k+\epsilon}\right]_+^2, \qquad [x]_+ = \max(x, 0). \tag{10}$$

This supply-side constraint covers both excitation and cross-endpoint transmission. In log-capacity coordinates, let $E_D = \log(q+\epsilon)$ and $E_j = \log(k_j+\epsilon)$ for $j \in \{\mathrm{exc}, \mathrm{tr}\}$. The same correction has the energy representation

$$\mathcal{S} = \big[E_D - \min(E_{\mathrm{exc}}, E_{\mathrm{tr}})\big]_+^2 = \max_{j\in\{\mathrm{exc},\mathrm{tr}\}}\big[E_D - E_j\big]_+^2. \tag{11}$$

Geometrically, this is the squared ℓ_∞ distance to the joint support region in smoothed log-capacity coordinates. The correction is zero by construction whenever supply satisfies demand.

The complete objective separates the task loss from the two weighted parts of **SADL**:

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{task}} + \mathcal{L}_{\mathrm{SADL}},$$
$$\mathcal{L}_{\mathrm{SADL}} = \lambda_D\mathcal{D}(Z) + \lambda_S\mathcal{S}\big(H, \mathrm{sg}(Z); \mathrm{sg}(q)\big). \tag{12}$$

Here λ_D weights the chosen endpoint constraint, λ_S weights the supply-shortfall loss, and α remains solely the demand-interpolation coefficient. Setting $\lambda_S = 0$ removes the supply correction and recovers the selected endpoint-only objective. Detached demand and endpoint references direct the correction toward upstream supply; the task and endpoint terms retain their normal gradients. The ridge scale derived from H is not detached. The ϵ-smoothed demand is used in the correction, and the ideal construction is recovered as $\epsilon \to 0$ away from the zero boundary. Because the correction remains active whenever a nonzero demand exceeds the available path supply, it can also repair an already degraded checkpoint, rather than only prevent collapse during initial training. We evaluate this post-collapse setting in Appendix D.6.

3.5 Gradient Action of Conditional Correction

Let $G_k = \nabla_H^{\mathrm{AD}} k$ denote the supply gradient under the stated stop-gradient conventions. Where the selected branch is differentiable,

$$\nabla_H^{\mathrm{AD}}\big(\lambda_S\mathcal{S}\big) = -a(H, Z)G_k,$$
$$a(H, Z) = \frac{2\lambda_S}{k+\epsilon}\left[\log\frac{q+\epsilon}{k+\epsilon}\right]_+. \tag{13}$$

3.3 激发、传输与路径供给

设 $\Sigma_{HZ} = N^{-1}\bar{H}^\top \bar{Z}$。使用正则化的典型相关形式（Hotelling，1936），定义

$$A_H = \Sigma_H + \rho\tau_H I, \qquad A_Z = \Sigma_Z + \rho\tau_Z I,$$
$$\tau_U = \max\left\{\frac{\mathrm{tr}(\Sigma_U)}{d_U}, \epsilon\right\}, \quad K = A_H^{-1/2}\Sigma_{HZ}A_Z^{-1/2}, \tag{7}$$

其中 $\rho > 0$。该分数的端点为 $\mathrm{sg}(Z)$，而 H 侧相对岭尺度保持可微。这保证在下限未激活时，通道分数对 H 的全局缩放具有前向及反向不变性。传输衡量为

$$C_{\mathrm{tr}}(H, \mathrm{sg}(Z)) = \frac{\log\det(I + \gamma K^\top K)}{r\log(1+\gamma)}, \qquad r = \min(d_h, d_z). \tag{8}$$

以全部潜在接口维度归一化，可区分广泛关联与仅少数方向上的强相关。这个二阶关联代理不能捕捉一般非线性依赖；交叉协方差为零时，其值和一阶梯度均为零。

路径供给为限制容量：

$$k = \min\{C_{\mathrm{exc}}(H), C_{\mathrm{tr}}(H, \mathrm{sg}(Z))\}. \tag{9}$$

因此 $k \geq q$ 要求两个分数均达到 q。该两阶段路径容量模型中，激发与传输较弱者为活跃截断。

3.4 单侧需求–供给匹配

通过单侧供给短缺校正惩罚不足的路径供给

$$\mathcal{S}\big(H, \mathrm{sg}(Z); \mathrm{sg}(q)\big) = \left[\log\frac{q+\epsilon}{k+\epsilon}\right]_+^2, \qquad [x]_+ = \max(x, 0). \tag{10}$$

这一供给侧约束覆盖激发和跨端点传输。在对数容量坐标中，设 $E_D = \log(q+\epsilon)$，以及对于 $j \in \{\mathrm{exc,tr}\}$ 的 $E_j = \log(k_j+\epsilon)$。同一校正可表示为能量

$$\mathcal{S} = \big[E_D - \min(E_{\mathrm{exc}}, E_{\mathrm{tr}})\big]_+^2 = \max_{j\in\{\mathrm{exc},\mathrm{tr}\}}\big[E_D - E_j\big]_+^2. \tag{11}$$

几何上，这是平滑对数容量坐标中到联合支持区域的平方 ℓ_∞ 距离。供给满足需求时，校正按构造为零。

完整目标将任务损失与 SADL 的两个加权部分分开：

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{task}} + \mathcal{L}_{\mathrm{SADL}},$$
$$\mathcal{L}_{\mathrm{SADL}} = \lambda_D\mathcal{D}(Z) + \lambda_S\mathcal{S}\big(H, \mathrm{sg}(Z); \mathrm{sg}(q)\big). \tag{12}$$

这里 λ_D 加权所选端点约束，λ_S 加权供给短缺损失，α 仅作为需求插值系数。设置 $\lambda_S = 0$ 可移除供给校正，恢复所选的仅端点目标。停止梯度的需求和端点参考将校正指向上游供给；任务与端点项保留正常梯度。由 H 得到的岭尺度不停止梯度。校正使用 ϵ 平滑需求，在远离零边界处，当 $\epsilon \to 0$ 时恢复理想构造。只要非零需求超过可用路径供给，校正就仍然激活，因此它也能修复已经退化的检查点，而不只是预防初始训练坍塌。我们在附录 D.6 评估这一坍塌后设置。

3.5 条件校正的梯度作用

设 $G_k = \nabla_H^{\mathrm{AD}} k$ 为既定停止梯度约定下的供给梯度。在所选分支可微处，

$$\nabla_H^{\mathrm{AD}}\big(\lambda_S\mathcal{S}\big) = -a(H, Z)G_k,$$
$$a(H, Z) = \frac{2\lambda_S}{k+\epsilon}\left[\log\frac{q+\epsilon}{k+\epsilon}\right]_+. \tag{13}$$

The deficit sets the correction strength a, and the active bottleneck selects its direction: excitation or cross-endpoint association. Once $k \geq q$, $a = 0$. The encoder gradient is $-aJ_{f_\theta}^\top G_k$, bypassing the projector Jacobian while retaining dependence on the encoder Jacobian and capacity gradient. The correction has no direct gradient on separate projector parameters.

To characterize the excitation branch, treat H as a directly optimized variable and define $r_i = \lambda_i(\Sigma_H)/e_H$.

Proposition 1 (Spectral equalization under excitation limitation)**.** *Assume $\lambda_S > 0$, an inactive energy floor, and $0 < C_{\text{exc}}(H) < \min\{q, C_{\text{tr}}(H, \text{sg}(Z))\}$ throughout the interval considered. The correction-only feature flow $\dot{H} = a\nabla_H C_{\text{exc}}$, with $a > 0$, preserves total feature energy, makes the common-mean norm non-increasing, and satisfies*

$$\frac{\mathrm{d}}{\mathrm{d}t} \log \frac{r_i}{r_j} = \frac{4a\gamma}{Nd_h e_H \log(1+\gamma)} \left(\frac{1}{1+\gamma r_i} - \frac{1}{1+\gamma r_j} \right) \leq 0 \tag{14}$$

for positive relative eigenvalues $r_i \geq r_j > 0$.

Thus, weaker positive directions receive larger relative growth rates, making the positive-spectrum condition number non-increasing while preserving total energy. Exact zero modes remain zero. Under an active excitation bottleneck, **SADL** preferentially strengthens weak but nonzero spectral directions, counteracting incipient collapse before sample variation is completely lost; Appendix D.6 shows that the same deficit-triggered correction can also reopen several near-collapsed checkpoints. This result describes the correction-only feature flow; joint training also incorporates task and endpoint updates.

Sequence features are aligned in batch–time order. Centering, covariance, whitening, and log-determinant calculations use FP32 within each task's local minibatch before shared-parameter gradients are averaged across tasks. Excitation energy and covariance-relative whitening scales retain their gradients, while $\mathcal{D}_0$ is cached from a shape-matched zero input. The architecture and endpoint computation paths remain unchanged.

4 Experiments

Our experiments test whether **SADL** prevents path collapse, accelerates convergence, improves representation quality, and generalizes across endpoint objectives and tasks. We report E1 control and fixed probes, then examine mechanism, endpoint geometries, and image classification; extended benchmarks and per-task results appear in the appendix.

4.1 Experimental setup

We evaluate INTACT and LeWorldModel (LeWM) (Maes et al., 2026; Sun et al., 2026) on PushT (Chi et al., 2023), Cube, Reacher, and TwoRoom, both separately and under joint four-task training, with three endpoint objectives: VISReg, SIGReg, and QQWorld. In the primary comparison, joint models share a ViT-Tiny encoder, projector, and BatchNorm; the BN configuration used for the independent Flash collapse diagnostic is specified in Table 2. Unless noted otherwise, **SADL** uses $\alpha = 0.5$ and $\lambda_S = 0.01$, and control models use BF16. INTACT uses search-free Direct control, whereas LeWM uses Pure-CEM. All control scores below are four-task macro success on the Official benchmark. The primary performance comparison uses three training seeds per configuration and evaluates 300 episodes per task at each checkpoint; we report the mean and sample standard deviation across training seeds. The independent collapse diagnostic uses ten training seeds, with its protocol specified in the table caption.

Table 2(b) holds the encoder and predictor backends, endpoint objective, and coefficient fixed between baseline and **+SADL** (SIGReg 0.03, VISReg 1.5, QQWorld 3.5). Table 1 retains the original single-task protocol; its backend details, additional benchmarks, and multi-epoch results are provided in Appendix E.

4.2 Convergence speed and representation quality

We first test whether path correction makes control-relevant representations available early in training. Each of PushT, Cube, Reacher, and TwoRoom is trained separately, and Official success is

缺口决定校正强度 a，活跃瓶颈决定方向：激发或跨端点关联。一旦 $k \geq q$， $a = 0$。编码器梯度为 $-aJ_{f_\theta}^\top G_k$，绕过投影器雅可比，仍依赖编码器雅可比与容量梯度。校正不直接向独立投影器参数传递梯度。

为刻画激发分支，将 H 视为直接优化变量，定义 $r_i = \lambda_i(\Sigma_H)/e_H$。

命题 *1*（激发限制下的谱均衡化）。假设所考虑区间内 $\lambda_S > 0$、能量下限未激活，且 $0 < C_{\text{exc}}(H) < \min\{q, C_{\text{tr}}(H, \text{sg}(Z))\}$。仅校正特征流 $\dot{H} = a\nabla_H C_{\text{exc}}$ 在 $a > 0$ 下保留总特征能量，使公共均值范数不增，并满足

$$\frac{\mathrm{d}}{\mathrm{d}t} \log \frac{r_i}{r_j} = \frac{4a\gamma}{Nd_h e_H \log(1+\gamma)} \left(\frac{1}{1+\gamma r_i} - \frac{1}{1+\gamma r_j} \right) \leq 0 \tag{14}$$

对于正相对特征值 $r_i \geq r_j > 0$。

因此，较弱正方向获得更大的相对增长率，在保留总能量的同时使正谱条件数不增。精确零模式仍为零。当激发是活跃瓶颈时，SADL 优先加强弱但非零的谱方向，在样本变化完全丢失前抵抗初发坍塌；附录 D.6 表明同一缺口触发校正也能重新展开一些近坍塌检查点。该结果描述仅校正的特征流；联合训练还包含任务及端点更新。

序列特征按批次–时间顺序对齐。中心化、协方差、白化、对数行列式计算在各任务局部小批次内以 FP32 执行，再跨任务平均共享参数梯度。激发能量和协方差相对的白化尺度保留梯度； $\mathcal{D}_0$ 从形状匹配的零输入缓存。架构与端点计算路径不变。

4 实验

实验检验 SADL 是否防止路径坍塌、加速收敛、改善表征质量，以及跨端点目标和任务泛化。我们报告 E1 控制与固定探针，再检验机制、端点几何和图像分类；扩展基准及逐任务结果见附录。

4.1 实验设置

我们在 PushT（Chi 等，2023）、Cube、Reacher 和 TwoRoom 上分别以及联合四任务训练评估 INTACT 与 LeWorldModel（LeWM）（Maes 等，2026；Sun 等，2026），采用 VISReg、SIGReg、QQWorld 三种端点目标。主要比较中，联合模型共享 ViT-Tiny 编码器、投影器和 BatchNorm；独立 Flash 坍塌诊断的 BN 配置见表 2。除另有说明，SADL 使用 $\alpha = 0.5$ 与 $\lambda_S = 0.01$，控制模型使用 BF16。INTACT 使用无搜索 Direct 控制，LeWM 使用 Pure-CEM。下面的控制得分均为官方基准的四任务宏平均成功率。主要性能比较每种配置使用三个训练种子，每个检查点每任务评估 300 回合；报告跨训练种子的均值和样本标准差。独立坍塌诊断使用十个训练种子，协议见表注。

表 2(b) 在基线与 +SADL 间固定编码器、预测器后端、端点目标及系数（SIGReg 0.03、VISReg 1.5、QQWorld 3.5）。表 1 保留原单任务协议；后端细节、额外基准及多轮结果见附录 E。

4.2 收敛速度与表征质量

我们首先检验路径校正能否在训练早期提供控制相关表征。PushT、Cube、Reacher、TwoRoom 分别独立训练，官方成功率为

Table 1: Single-task E1 Official macro success rate (SR, %; mean ± SD over three seeds) and matched-stage fixed probes. SR_0/SR_A denote baseline/+**SADL**; r is effective rank, c is mean off-diagonal sample cosine, and subscripts h/z indicate representation stage. Probe entries show baseline → +**SADL**.

Model	Endpoint	SR_0	SR_A	r_h	c_h	r_z	c_z	L_{pred}
LeWM	SIGReg	45.25 ± 6.50	**72.19 ± 0.97**	15.08 → **51.90**	.963 → **.013**	22.97 → **46.32**	.421 → **.007**	3.247 → **.021**
	VISReg	55.67 ± 5.46	**76.31 ± 1.23**	13.30 → **45.60**	.943 → **.013**	21.36 → **34.59**	.214 → **.007**	.586 → **.011**
	QQWorld	51.19 ± 6.98	**75.28 ± 1.73**	15.33 → **51.49**	.956 → **.013**	26.25 → **47.54**	.287 → **.007**	.869 → **.018**
INTACT	SIGReg	58.64 ± 8.14	**95.33 ± 0.44**	15.28 → **51.96**	.976 → **.011**	25.86 → **47.95**	.363 → **.007**	3.160 → **.021**
	VISReg	71.97 ± 12.79	**95.75 ± 0.08**	13.74 → **46.07**	.963 → **.014**	23.77 → **37.39**	.264 → **.007**	.620 → **.012**
	QQWorld	61.56 ± 8.61	**95.47 ± 0.21**	15.16 → **51.26**	.970 → **.013**	27.35 → **49.88**	.350 → **.006**	2.279 → **.019**

macro-averaged across the four tasks. After one epoch, across all six pairs, **SADL** adds 20.6–36.7 points of SR, raises h/z effective rank by 3.35–3.44×/1.57–2.02×, and lowers prediction loss by 48–155× (Table 1).

Joint four-task training tests one shared encoder and projector across all tasks. Table 2 pairs six low-diversity Flash E1 runs with same-seed +**SADL** runs and reports the three-seed comparison. Across the six Flash-matched model–objective pairs in Table 2(b), **SADL** raises E1 Official SR by 23.9–67.4 percentage points: LeWM reaches 63.0–71.6% and INTACT 95.4–95.6%. Effective rank at h/z rises 3.4–6.4/1.6–5.5 times, mean sample cosine falls 57–71/41–93 times, and prediction loss falls 20–303 times. **Both single-task and joint four-task training yield more useful control policies and more discriminative representations after only one epoch.** Per-task results are in Appendix E.

Table 2: Joint four-task E1 Official control and fixed-probe comparison. (a) Six matched single-seed low-diversity Flash examples. (b) Three-seed Flash-matched comparison. SR/SR_{SADL} are without/with **SADL**; arrows show the corresponding probe changes. In (b), SR is mean ± SD and probes are three-seed means.

Model	Endpoint	SR	SR_{SADL}	r_h	c_h	r_z	c_z	L_{pred}
(a) Collapse comparison								
LeWM	SIGReg s42/E1	31.3	**61.8**	17.4 → **54.0**	.926 → **.012**	23.5 → **50.4**	.601 → **.007**	.912 → **.017**
	SIGReg s3072/E1	46.7	**62.9**	12.3 → **53.8**	.960 → **.014**	24.9 → **51.1**	.103 → **.008**	.301 → **.018**
	QQWorld s3072/E1	23.9	**64.0**	14.1 → **55.2**	.992 → **.014**	13.0 → **53.5**	.864 → **.009**	3.752 → **.018**
INTACT	SIGReg s0/E1	26.9	**95.8**	5.4 → **53.5**	.995 → **.013**	9.4 → **53.3**	.795 → **.007**	4.430 → **.017**
	SIGReg s3072/E1	22.5	**95.6**	8.1 → **52.8**	.991 → **.015**	7.2 → **53.4**	.885 → **.009**	10.041 → **.018**
	QQWorld s3072/E1	24.2	**96.0**	4.6 → **53.7**	1.000 → **.015**	7.9 → **54.9**	.833 → **.009**	4.495 → **.019**
(b) Normal comparison								
LeWM	SIGReg	37.2 ± 8.3	**63.0 ± 1.3**	15.7 → **54.0**	.937 → **.013**	24.4 → **50.9**	.419 → **.008**	.651 → **.018**
	VISReg	47.7 ± 5.1	**71.6 ± 3.0**	13.7 → **48.0**	.853 → **.014**	22.7 → **37.7**	.376 → **.009**	.267 → **.014**
	QQWorld	32.9 ± 8.3	**68.4 ± 4.2**	15.8 → **54.5**	.944 → **.013**	21.7 → **52.7**	.619 → **.008**	1.643 → **.017**
INTACT	SIGReg	28.2 ± 6.4	**95.6 ± .2**	8.3 → **53.1**	.994 → **.014**	9.7 → **53.1**	.744 → **.008**	5.300 → **.017**
	VISReg	54.4 ± 2.9	**95.4 ± .6**	13.9 → **48.2**	.912 → **.016**	21.8 → **39.8**	.492 → **.010**	.526 → **.016**
	QQWorld	38.6 ± 12.8	**95.6 ± .3**	12.9 → **53.6**	.952 → **.014**	18.2 → **54.4**	.661 → **.008**	2.044 → **.019**

4.3 Collapse prevention and mechanistic ablations

Endpoint regularization alone does not ensure a healthy representation path. Across six low-diversity E1 Flash checkpoints spanning LeWM, INTACT, SIGReg, and QQWorld, matched **SADL** training restores control and representation diversity (Table 2(a)). In a ten-seed Flash+SIGReg audit, relative path-degradation events fall from 5/10 to 0/10 with full **SADL**. Formal definitions and broader VISReg evidence are in Appendix D.

In four-task LeWM, excitation drives most of the observed gain, so excitation-only approaches Full **SADL** in Table 3(b); transmission remains necessary when the cross-endpoint channel is the limiting capacity, as the controlled optimization and forward replay in Appendix Figure 5 demonstrate.

表 1：单任务 E1 官方宏平均成功率（SR，%；三种子的均值 ± 标准差）与匹配阶段固定探针。SR_0/SR_A 表示基线/+SADL；r 为有效秩，c 为平均非对角样本余弦，下标 h/z 表示表征阶段。探针数值展示基线 → +SADL。

模型	端点	成功率₀	成功率ₐ	r_h	c_h	r_z	c_z	L_{pred}
LeWM	SIGReg	45.25 ± 6.50	**72.19 ± 0.97**	15.08 → **51.90**	.963 → **.013**	22.97 → **46.32**	.421 → **.007**	3.247 → **.021**
	VISReg	55.67 ± 5.46	**76.31 ± 1.23**	13.30 → **45.60**	.943 → **.013**	21.36 → **34.59**	.214 → **.007**	.586 → **.011**
	QQWorld	51.19 ± 6.98	**75.28 ± 1.73**	15.33 → **51.49**	.956 → **.013**	26.25 → **47.54**	.287 → **.007**	.869 → **.018**
INTACT	SIGReg	58.64 ± 8.14	**95.33 ± 0.44**	15.28 → **51.96**	.976 → **.011**	25.86 → **47.95**	.363 → **.007**	3.160 → **.021**
	VISReg	71.97 ± 12.79	**95.75 ± 0.08**	13.74 → **46.07**	.963 → **.014**	23.77 → **37.39**	.264 → **.007**	.620 → **.012**
	QQWorld	61.56 ± 8.61	**95.47 ± 0.21**	15.16 → **51.26**	.970 → **.013**	27.35 → **49.88**	.350 → **.006**	2.279 → **.019**

对四任务取宏平均。训练一轮后，全部六组配对中，SADL 增加成功率 20.6–36.7 个百分点，使 h/z 有效秩增加 3.35–3.44×/1.57–2.02×，预测损失降低 48–155×（表 1）。

联合四任务训练在所有任务上检验一个共享编码器与投影器。表 2 将六个低多样性 Flash E1 运行与相同种子的 +SADL 运行配对，并报告三种子比较。在表 2(b) 六组 Flash 匹配的模型–目标对中，SADL 将 E1 官方成功率提高 23.9–67.4 个百分点：LeWM 达到 63.0–71.6%，INTACT 达到 95.4–95.6%。h/z 处有效秩增加 3.4–6.4/1.6–5.5 倍，平均样本余弦降低 57–71/41–93 倍，预测损失降低 20–303 倍。单任务与联合四任务仅训练一轮，均得到更有用的控制策略和更具区分性的表征。逐任务结果见附录 E。

表 2：联合四任务 E1 官方控制与固定探针比较。(a) 六组匹配单种子低多样性 Flash 示例。(b) 三种子 Flash 匹配比较。SR/SR_{SADL} 为不使用/使用 SADL；箭头显示相应探针变化。(b) 中成功率为均值 ±标准差，探针为三种子均值。

模型	端点	SR	成功率 SADL	r_h	c_h	r_z	c_z	L_{pred}
(a) 坍塌比较								
LeWM	SIGReg s42/E1	31.3	**61.8**	17.4 → **54.0**	.926 → **.012**	23.5 → **50.4**	.601 → **.007**	.912 → **.017**
	SIGReg s3072/E1	46.7	**62.9**	12.3 → **53.8**	.960 → **.014**	24.9 → **51.1**	.103 → **.008**	.301 → **.018**
	QQWorld s3072/E1	23.9	**64.0**	14.1 → **55.2**	.992 → **.014**	13.0 → **53.5**	.864 → **.009**	3.752 → **.018**
INTACT	SIGReg s0/E1	26.9	**95.8**	5.4 → **53.5**	.995 → **.013**	9.4 → **53.3**	.795 → **.007**	4.430 → **.017**
	SIGReg s3072/E1	22.5	**95.6**	8.1 → **52.8**	.991 → **.015**	7.2 → **53.4**	.885 → **.009**	10.041 → **.018**
	QQWorld s3072/E1	24.2	**96.0**	4.6 → **53.7**	1.000 → **.015**	7.9 → **54.9**	.833 → **.009**	4.495 → **.019**
(b) 正常比较								
LeWM	SIGReg	37.2 ± 8.3	**63.0 ± 1.3**	15.7 → **54.0**	.937 → **.013**	24.4 → **50.9**	.419 → **.008**	.651 → **.018**
	VISReg	47.7 ± 5.1	**71.6 ± 3.0**	13.7 → **48.0**	.853 → **.014**	22.7 → **37.7**	.376 → **.009**	.267 → **.014**
	QQWorld	32.9 ± 8.3	**68.4 ± 4.2**	15.8 → **54.5**	.944 → **.013**	21.7 → **52.7**	.619 → **.008**	1.643 → **.017**
INTACT	SIGReg	28.2 ± 6.4	**95.6 ± .2**	8.3 → **53.1**	.994 → **.014**	9.7 → **53.1**	.744 → **.008**	5.300 → **.017**
	VISReg	54.4 ± 2.9	**95.4 ± .6**	13.9 → **48.2**	.912 → **.016**	21.8 → **39.8**	.492 → **.010**	.526 → **.016**
	QQWorld	38.6 ± 12.8	**95.6 ± .3**	12.9 → **53.6**	.952 → **.014**	18.2 → **54.4**	.661 → **.008**	2.044 → **.019**

4.3 坍塌预防与机制消融

仅端点正则化不能保证健康表征路径。在涵盖 LeWM、INTACT、SIGReg、QQWorld 的六个低多样性 E1 Flash 检查点中，匹配 SADL 训练恢复控制和表征多样性（表 2(a)）。十种子 Flash+SIGReg 审计中，完整 SADL 将相对路径退化事件从 5/10 降至 0/10。正式定义与更广 VISReg 证据见附录 D。

四任务 LeWM 中，激发驱动大部分观测增益，因此表 3(b) 的仅激发接近完整 SADL；当跨端点通道是限制容量时，传输仍然必要，附录图 5 的受控优化与前向重放证实这一点。

Table 3: Path collapse and mechanistic ablations. (a) Collapse incidence for four-task LeWM at E1: fraction of observed events among ten seeds; C0 serves only as the matched reference, and ranks and losses are at E1. (b) Independent SIGReg four-task E2 Official SR (%): mean ± SD over three seeds.

(a) Matched collapse events and path metrics

Cell	Incidence	S_{path}	r_h/r_z	L_{pred}
C0 MATH+SIG	reference	.031	5.9/27.4	.7767
C1 Flash	100%	.005	1.3/1.1	27.0896
C2 C1+SIG	50%	.032	6.2/21.5	1.8586
C3 C2+Exc	0%	.370	113.2/71.0	.0174
C4 +**SADL**	0%	.373	113.1/71.6	.0173

C4 adds **SADL** to C2. C1–C4 are judged against seed-matched C0; C0 has no self-incidence. 0% means no event was observed in these ten seeds; C3 is excitation-only.

(b) SIGReg internal ablation: three-seed Official E2

Cell	LeWM CEM	INTACT Direct
Excitation-only	68.86 ± .38	96.25 ± .30
Transmission-only	46.17 ± 22.13	48.67 ± 39.49
Full **SADL**	**69.06 ± 1.16**	**96.56 ± .17**

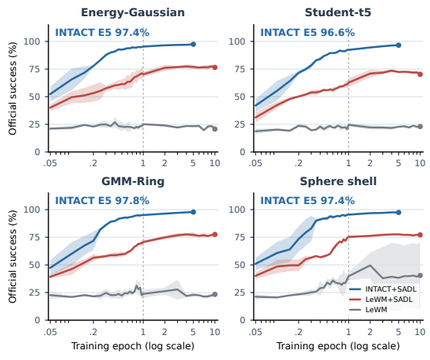


Figure 3: Four-task Official success across four endpoint objectives: INTACT+**SADL** (blue), LeWM+**SADL** (red), and endpoint-only LeWM (gray). Endpoint-only training is collapse-prone and unstable across seeds; with **SADL**, diverse endpoint geometries yield stable, high-quality representations.

4.4 Generality across endpoint objectives

We test **SADL** with eight endpoint objectives rather than assuming a single Gaussian target. After one epoch, all eight give INTACT over 92% search-free Direct success (seven reach 95.03%–95.81%), while LeWM+**SADL** Pure-CEM reaches 62.81%–75.33% (Table 4). Against matched endpoint-only LeWM at 23.81%–47.69%, **SADL** gains 23.89–46.94 points (Appendix Table 33). Student-t_5, GMM-Ring, Sphere shell, and VIC moments further cover non-Gaussian or density-unspecified constraints. **SADL therefore adapts across endpoint objectives without implying that all geometries are equivalent.**

Table 4: Eight-endpoint E1 Official joint control and LeJEPA-related diagnostics. Entries are three-seed means (Macro: mean ± SD); LeWM uses Pure-CEM 300×30, INTACT Direct, and $\lambda_S = .01$. $\widehat{J}_{\text{proj}}$ and b_{1024} are projected Fisher and finite-sample kNN-bias proxies; full definitions, uncertainty, and evaluation protocol appear in Appendix G.

Endpoint	λ_D	PushT	Cube	Reacher	TwoRoom	Macro	$\widehat{J}_{\text{proj}}$	b_{1024}
LeWM – Gaussian targets								
SIGReg	0.0114	56.00	67.11	82.33	80.78	71.56 ± 1.27	1.289	.1772
VISReg	1.5000	47.33	68.44	81.00	89.56	71.58 ± 4.67	1.184	.1717
QQWorld	1.2729	60.89	68.00	80.33	88.22	74.36 ± 2.90	1.179	.1699
Energy-Gaussian	2.3383	49.44	61.56	79.44	90.56	70.25 ± 2.35	1.161	.1712
LeWM – non-Gaussian or density-free targets								
VIC moments	1.9130	35.56	62.00	79.67	92.89	67.53 ± 6.09	1.308	.2228
Sphere	1.1318	73.11	59.56	81.89	86.78	75.33 ± 1.36	1.189	.1660
Student-t_5	1.2104	30.11	69.56	70.44	81.11	62.81 ± 3.99	1.591	.2034
GMM-Ring	2.3345	51.00	63.33	79.78	88.89	70.75 ± 1.23	1.232	.1756
INTACT – Gaussian targets								
SIGReg	0.0117	85.44	100.00	96.89	100.00	95.58 ± 0.00	1.287	.1762
VISReg	1.5000	83.00	99.89	97.44	100.00	95.08 ± 0.17	1.250	.1716
QQWorld	1.2746	85.89	100.00	97.44	99.89	95.81 ± 0.13	1.213	.1745
Energy-Gaussian	2.2074	86.33	100.00	95.11	99.89	95.33 ± 0.63	1.259	.1683

表 3：路径坍塌与机制消融。(a) 四任务 LeWM E1 坍塌发生率：十种子中的观测事件比例；C0 仅作为匹配参考，秩与损失在 E1 测量。(b) 独立 SIGReg 四任务 E2 官方成功率（%）：三种子的均值 ± 标准差。

(a) 匹配坍塌事件与路径指标

Cell	发生率	S_{path}	r_h/r_z	L_{pred}
C0 MATH+SIG	参考	.031	5.9/27.4	.7767
C1 Flash	100%	.005	1.3/1.1	27.0896
C2 C1+SIG	50%	.032	6.2/21.5	1.8586
C3 C2+Exc	0%	.370	113.2/71.0	.0174
C4 +**SADL**	0%	.373	113.1/71.6	.0173

C4 在 C2 上加入 SADL。C1–C4 相对于种子匹配的 C0 判定；C0 没有自身事件发生率。0% 表示这些实验中未观测到事件十个种子；C3 为仅激发。

(b) SIGReg 内部消融：三种子官方 E2

Cell	LeWM CEM	INTACT Direct
仅激发	68.86 ± .38	96.25 ± .30
仅传输	46.17 ± 22.13	48.67 ± 39.49
完整 SADL	**69.06 ± 1.16**	**96.56 ± .17**

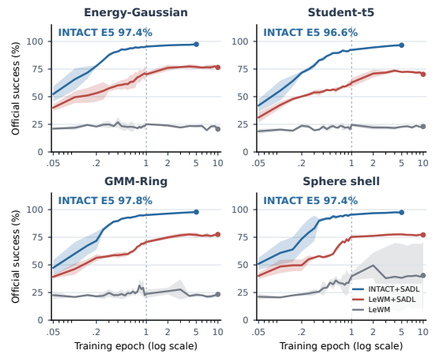


图 3：四种端点目标的四任务官方成功率：INTACT+SADL（蓝）、LeWM+SADL（红）与仅端点 LeWM（灰）。仅端点训练易坍塌且跨种子不稳定；加入 SADL 后，多种端点几何均产生稳定、高质量表征。

4.4 跨端点目标的通用性

我们以八种端点目标检验 SADL，而不预设单一高斯目标。训练一轮后，八种目标的 INTACT 无搜索 Direct 成功率均超过 92%（其中七种为 95.03%–95.81%），LeWM+SADL Pure-CEM 达到 62.81%–75.33%（表 4）。相对于 23.81%–47.69% 的匹配仅端点 LeWM，SADL 增益为 23.89–46.94 个百分点（附录表 33）。Student-t_5、GMM-Ring、球壳及 VIC 矩进一步覆盖非高斯或未规定密度的约束。因此 SADL 可适应不同端点目标，但这并不意味着所有几何结构等价。

表 4：八端点 E1 官方联合控制与 LeJEPA 相关诊断。数值为三种子均值（宏平均：均值 ± 标准差）；LeWM 用 Pure-CEM 300×30，INTACT 用 Direct，且 $\lambda_S = .01$。$\widehat{J}_{\text{proj}}$ 与 b_{1024} 为投影 Fisher 和有限样本 kNN 偏差代理；完整定义、不确定性及评估协议见附录 G。

端点	λ_D	PushT	Cube	Reacher	TwoRoom	宏平均	$\widehat{J}_{\text{proj}}$	b_{1024}
LeWM – 高斯目标								
SIGReg	0.0114	56.00	67.11	82.33	80.78	71.56 ± 1.27	1.289	.1772
VISReg	1.5000	47.33	68.44	81.00	89.56	71.58 ± 4.67	1.184	.1717
QQWorld	1.2729	60.89	68.00	80.33	88.22	74.36 ± 2.90	1.179	.1699
能量–高斯	2.3383	49.44	61.56	79.44	90.56	70.25 ± 2.35	1.161	.1712
LeWM – 非高斯或不规定密度的目标								
VIC 矩	1.9130	35.56	62.00	79.67	92.89	67.53 ± 6.09	1.308	.2228
球面	1.1318	73.11	59.56	81.89	86.78	75.33 ± 1.36	1.189	.1660
Student-t_5	1.2104	30.11	69.56	70.44	81.11	62.81 ± 3.99	1.591	.2034
GMM-Ring	2.3345	51.00	63.33	79.78	88.89	70.75 ± 1.23	1.232	.1756
INTACT – 高斯目标								
SIGReg	0.0117	85.44	100.00	96.89	100.00	95.58 ± 0.00	1.287	.1762
VISReg	1.5000	83.00	99.89	97.44	100.00	95.08 ± 0.17	1.250	.1716
QQWorld	1.2746	85.89	100.00	97.44	99.89	95.81 ± 0.13	1.213	.1745
能量–高斯	2.2074	86.33	100.00	95.11	99.89	95.33 ± 0.63	1.259	.1683

Table 4 continued

Endpoint	λ_D	PushT	Cube	Reacher	TwoRoom	Macro	$\widehat{J}_{\text{proj}}$	b_{1024}
INTACT – non-Gaussian or density-free targets								
VIC moments	1.7339	83.78	100.00	96.33	100.00	95.03 ± 0.99	1.330	.2287
Sphere	1.0725	86.33	100.00	95.89	99.56	95.44 ± 0.82	1.251	.1590
Student-t_5	1.0662	76.67	99.78	93.00	99.89	92.33 ± 1.04	1.660	.2052
GMM-Ring	2.1996	84.89	100.00	95.67	99.89	95.11 ± 0.35	1.220	.1661

4.5 Image representation learning

Beyond control, **SADL** improves ImageNet-LT (Liu et al., 2019), Galaxy10 (Leung & Bovy, 2024), CIFAR-10, and Imagenette under matched budgets and endpoint objectives (Figure 4; Appendix Figure 7, Table 34). It adds no trainable parameters and shows no systematic step-time overhead in a separate control-model audit (Appendix I). Early paired gains span 1.56–8.91 points; under a LeJEPA-style ViT-S/8 SIGReg recipe, all seven Imagenette seeds improve at E5, E20, and E100 (+8.91 points at E20).

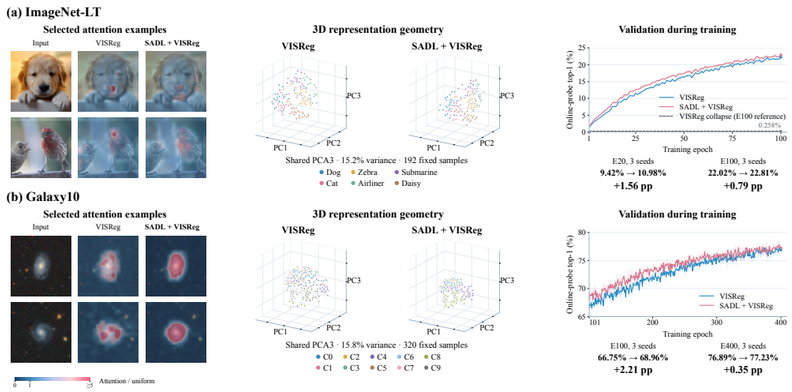


Figure 4: ImageNet-LT (top) and Galaxy10 (bottom): representative attention maps, shared three-dimensional feature geometry, and paired online-validation curves. The baseline remains vulnerable to path degradation; **SADL** converges faster and more stably.

5 Conclusion

We introduce **SADL**, an adaptive one-sided bridge that matches endpoint demand to encoder excitation and cross-endpoint transmission capacity. Across JEPA-style world models and endpoint regularization functions, it accelerates learning and yields more stable, useful representations. Ablations distinguish excitation-driven expansion from correction of transmission bottlenecks; image experiments show benefits beyond control. Together, these results support demand–supply matching as a low-overhead principle for coordinating endpoint objectives with the representation path that must realize them.

Future work. The present transmission score captures a linear, regularized association between hidden and endpoint representations. A natural next step is to generalize path capacity to nonlinear, multimodal, and temporally structured transmission operators. We also aim to develop online diagnostics that anticipate emerging collapse modes and trigger state-adaptive repair, extending **SADL** from early prevention and checkpoint rescue toward prediction and automatic recovery for a broader class of representation failures.

表 4 续

端点	λ_D	PushT	Cube	Reacher	TwoRoom	宏平均	$\widehat{J}_{\text{proj}}$	b_{1024}
INTACT – 非高斯或不规定密度的目标								
VIC 矩	1.7339	83.78	100.00	96.33	100.00	95.03 ± 0.99	1.330	.2287
球面	1.0725	86.33	100.00	95.89	99.56	95.44 ± 0.82	1.251	.1590
Student-t_5	1.0662	76.67	99.78	93.00	99.89	92.33 ± 1.04	1.660	.2052
GMM-Ring	2.1996	84.89	100.00	95.67	99.89	95.11 ± 0.35	1.220	.1661

4.5 图像表征学习

除控制外，SADL 在匹配预算和端点目标下改善 ImageNet-LT（Liu 等，2019）、Galaxy10（Leung 与 Bovy，2024）、CIFAR-10 与 Imagenette（图 4；附录图 7、表 34）。它不增加可训练参数，在独立控制模型审计中也未显示系统性的单步时间开销（附录 I）。早期配对增益为 1.56–8.91 个百分点；在 LeJEPA 式 ViT-S/8 SIGReg 配方下，Imagenette 全部七个种子在 E5 、 E20 、 E100 均改善（ E20 增加 +8.91 个百分点）。

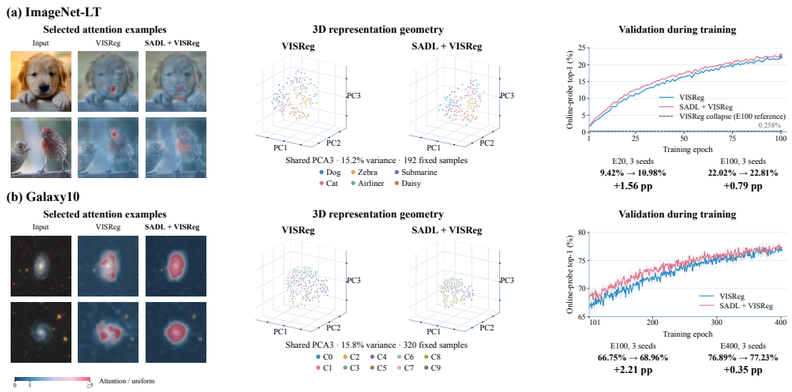


图 4：ImageNet-LT（上）与 Galaxy10（下）的代表性注意力图、共享三维特征几何及配对在线验证曲线。基线仍易发生路径退化；SADL 收敛更快、更稳定。

5 结论

我们提出 SADL，一种将端点需求与编码器激发、跨端点传输容量匹配的自适应单侧桥接机制。在 JEPA 式世界模型与多种端点正则化函数中，它加速学习，产生更稳定、有用的表征。消融区分激发驱动展开与传输瓶颈校正；图像实验展示控制之外的收益。综合结果支持需求–供给匹配作为低开销原则，协调端点目标与实现它所需的表征路径。

未来工作。目前的传输分数捕捉隐表征与端点表征间线性、正则化的关联。自然的下一步是将路径容量推广至非线性、多模态及具有时间结构的传输算子。我们也计划开发在线诊断，提前发现正在形成的坍塌模式并触发状态自适应修复，将 SADL 从早期预防和检查点修复扩展到更广表征失效的预测与自动恢复。

REPRODUCIBILITY STATEMENT

Section 3 and Appendices A–B specify the **SADL** objective, its stop-gradient paths, numerical stabilization, and gradient computation. Appendix C documents model configurations, training and evaluation protocols, success criteria, and aggregation across seeds. Appendices E–H report the complete control comparisons, mechanism ablations, endpoint objectives, and image-classification protocols; the corresponding tables state evaluation budgets and sample counts.

ETHICS STATEMENT

We conduct this work in accordance with the ICLR Code of Ethics. The reported experiments use established image datasets and simulated-control benchmarks and do not involve human participants. Real-world deployment of learned control policies requires task-specific safety evaluation.

AI USE STATEMENT

Generative AI tools were used to aid and polish manuscript writing, refine LaTeX formatting and page layout, and provide suggestions on coding style and implementation. The authors reviewed and verified all resulting text and code and take full responsibility for the final submission.

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可复现性声明

第 3 节及附录 A–B 规定 SADL 目标、停止梯度路径、数值稳定化和梯度计算。附录 C 记录模型配置、训练与评估协议、成功标准及跨种子汇总。附录 E–H 报告完整控制比较、机制消融、端点目标和图像分类协议；各表标注评估预算与样本数量。

伦理声明

本工作遵循 ICLR 伦理准则。实验使用既有图像数据集及仿真控制基准，不涉及人类受试者。学习控制策略的现实部署需要针对任务的安全评估。

AI 使用声明

生成式 AI 工具用于辅助及润色稿件、改进 LaTeX 格式和页面布局，并提供编码风格与实现建议。作者审查、验证全部文本和代码，对最终投稿承担全部责任。

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APPENDIX

This appendix gives the theoretical derivations, numerical implementation, training and evaluation protocols, collapse diagnosis, complete control results, mechanism ablations, alternative endpoint geometries, image experiments, and computational costs. Unless stated otherwise, control comparisons use three training seeds; each task is evaluated with three evaluation seeds and 100 episodes per seed. Other sample counts are stated with their respective results.

A THEORETICAL DERIVATIONS AND IMPLEMENTATION DETAILS

This appendix provides the proofs, gradient expressions, and numerical details for Section 3.

Table 5: Notation and experimental labels used throughout the appendix.

Symbol	Meaning
N, d_h, d_z	Aligned sample count and encoder/projector dimensions.
$\mathbf{1}, P$	All-ones vector $\mathbf{1} \in \mathbb{R}^N$ and sample-centering matrix $P = I_N - \mathbf{1}\mathbf{1}^\top/N$.
H, Z; h_i, z_i	Aligned representation matrices and their sample rows.
$\hat{Z}$	Endpoint representation predicted by the forward model.
$\bar{U}, \Sigma_U, \Sigma_{HZ}, \mu_U, e_U, v_U, \gamma$	Centered features, sample covariance, cross-covariance, mean, normalized energy, centered-energy fraction, and spectral sensitivity for U.
$\mathcal{D}, \mathcal{D}_0$	Endpoint loss $\mathcal{D}(Z)$ and its zero-representation reference $\mathcal{D}_0 = \mathcal{D}(0)$.
$\mathcal{L}_{\text{task}}, \mathcal{L}_{\text{SADL}}, \mathcal{S}$	Task loss, combined SADL regularizer, and one-sided supply-shortfall loss. The latter takes $(H, \text{sg}(Z); \text{sg}(q))$ and also uses q_{img} in image experiments.
$q_R, q_E, q^\star_\alpha, \widetilde{q}_{\alpha,\epsilon}, q, q_{\text{img}}$	Calibrated, realized, ideal interpolated, numerically smoothed, default, and image-capped demands.
$C_{\text{exc}}, C_{\text{tr}}$	Encoder excitation and cross-endpoint transmission capacities.
$\tau_U, \rho, A_H, A_Z, K, r$	Relative ridge scale and coefficient, regularized endpoint covariances, whitened cross-covariance, and interface dimension $r = \min(d_h, d_z)$.
k	Available path capacity $k = \min\{C_{\text{exc}}(H), C_{\text{tr}}(H, \text{sg}(Z))\}$.
sg, ϵ	Stop-gradient and numerical floor/smoothing parameter.
$\alpha, \lambda_D, \lambda_S$	Log-space demand interpolation, endpoint weight, and bridge weight.
$r_{\text{eff}}, \eta_1, \bar{\kappa}$	Effective covariance rank, leading-eigenvalue share, and mean absolute standardized excess kurtosis.
$\widehat{J}_{\text{proj}}, J(p), n_{\text{nbr}}$	Smoothed projected Fisher proxy, full Fisher functional, and kNN neighbor count; the first two are not interchangeable.
$p, \mathcal{F}, \mathcal{P}_T$	Target endpoint density, downstream task family, and endpoints attainable within training budget T.
E1/E5/E10, Official	Epoch-1/5/10 checkpoints and the control-evaluation protocol defined below.
Endpoint-only, Full SADL	Endpoint regularization alone versus the same endpoint plus the complete SADL bridge. Excitation-only and Transmission-only retain one supply branch each.
MATH, Flash	Explicit versus fused memory-efficient encoder-attention backends, not distinct regularization objectives.

A.1 NOTATION AND BASIC SETUP

Let $H \in \mathbb{R}^{N \times d_h}$ and $Z \in \mathbb{R}^{N \times d_z}$ contain encoder and projected representations of the same samples in the same row order. Define the centering matrix

$$P = I_N - \frac{1}{N}\mathbf{1}\mathbf{1}^\top.$$

For any representation matrix $U \in \mathbb{R}^{N \times d_U}$, write

$$\mu_U = \frac{1}{N}U^\top\mathbf{1}, \qquad \Sigma_U = \frac{1}{N}U^\top PU,$$

$$e_U^{\text{raw}} = \frac{\|U\|_F^2}{Nd_U}, \qquad e_U = \max\{e_U^{\text{raw}}, \epsilon\},$$

附录

本附录给出理论推导、数值实现、训练与评估协议、坍塌诊断、完整控制结果、机制消融、其他端点几何、图像实验及计算成本。除另有说明，控制比较使用三个训练种子；每任务以三个评估种子评估，各 100 回合。其他样本数量随各结果标注。

A 理论推导与实现细节

本附录给出第 3 节的证明、梯度表达式与数值细节 3.

表 5：附录使用的符号与实验标签。

符号	含义
N, d_h, d_z	对齐样本数及编码器/投影器维度。
$\mathbf{1}, P$	全一向量 $\mathbf{1} \in \mathbb{R}^N$ 与样本中心化矩阵 $P = I_N - \mathbf{1}\mathbf{1}^\top/N$。
H, Z; h_i, z_i	对齐表征矩阵及其样本行。
$\hat{Z}$	前向模型预测的端点表征。
$\bar{U}, \Sigma_U, \Sigma_{HZ}, \mu_U, e_U, v_U, \gamma$	中心化特征、样本协方差、交叉协方差、均值、归一化能量、中心化能量比例与 U 的谱敏感性。
$\mathcal{D}, \mathcal{D}_0$	端点损失 $\mathcal{D}(Z)$ 与零表征参考 $\mathcal{D}_0 = \mathcal{D}(0)$。
$\mathcal{L}_{\text{task}}, \mathcal{L}_{\text{SADL}}, \mathcal{S}$	任务损失、组合 SADL 正则项与单侧供给短缺损失。后者输入为 $(H, \text{sg}(Z); \text{sg}(q))$，图像实验还使用 q_{img}。
$q_R, q_E, q^\star_\alpha, \widetilde{q}_{\alpha,\epsilon}, q, q_{\text{img}}$	校准、已实现、理想插值、数值平滑、默认，以及图像容量封顶需求。
$C_{\text{exc}}, C_{\text{tr}}$	编码器激发容量与跨端点传输容量。
$\tau_U, \rho, A_H, A_Z, K, r$	相对岭尺度与系数、正则化端点协方差、白化交叉协方差，以及接口维度 $r = \min(d_h, d_z)$。
k	可用路径容量 $k = \min\{C_{\text{exc}}(H), C_{\text{tr}}(H, \text{sg}(Z))\}$。
sg, ϵ	停止梯度与数值下限/平滑参数。
$\alpha, \lambda_D, \lambda_S$	对数空间需求插值、端点权重与桥接权重。
$r_{\text{eff}}, \eta_1, \bar{\kappa}$	有效协方差秩、首特征值占比及平均绝对标准化超额峰度。
$\widehat{J}_{\text{proj}}, J(p), n_{\text{nbr}}$	平滑投影 Fisher 代理、完整 Fisher 泛函与 kNN 邻居数量；前两者不能互换。
$p, \mathcal{F}, \mathcal{P}_T$	目标端点密度、下游任务族与可达到的端点在训练预算 T 内。
E1/E5/E10, Official	第 1/5/10 轮检查点与下文定义的控制评估协议。
仅端点，完整 SADL	仅端点正则化与相同端点加完整 SADL 桥接。 仅激发与仅传输保留一种供给各一个分支。
MATH, Flash	显式与融合显存高效编码器注意力后端，而非不同的正则化目标。

A.1 符号与基本设置

设 $H \in \mathbb{R}^{N \times d_h}$、$Z \in \mathbb{R}^{N \times d_z}$ 包含相同样本、相同行顺序的编码器与投影表征。定义中心化矩阵

$$P = I_N - \frac{1}{N}\mathbf{1}\mathbf{1}^\top.$$

对任意表征矩阵 $U \in \mathbb{R}^{N \times d_U}$，记

$$\mu_U = \frac{1}{N}U^\top\mathbf{1}, \qquad \Sigma_U = \frac{1}{N}U^\top PU,$$

$$e_U^{\text{raw}} = \frac{\|U\|_F^2}{Nd_U}, \qquad e_U = \max\{e_U^{\text{raw}}, \epsilon\},$$

where $\epsilon > 0$ is a numerical floor. The total energy decomposes as

$$d_U e_U^{\text{raw}} = \text{tr}(\Sigma_U) + \|\mu_U\|_2^2.$$

Consequently, the centered-energy fraction

$$v_U = \frac{\text{tr}(\Sigma_U)}{d_U e_U}$$

satisfies $0 \le v_U \le 1$. When the energy floor is inactive,

$$v_U = 1 - \frac{\|\mu_U\|_2^2}{d_U e_U}.$$

All results below concern finite-batch empirical statistics. They require neither Gaussian representations nor independence between samples within the batch.

A.2 Properties of Excitation Capacity

Recall that

$$C_{\text{exc}}(U) = \frac{\log\det\left(I + \gamma \Sigma_U / e_U\right)}{d_U \log(1+\gamma)}, \qquad \gamma > 0.$$

Let $r_i = \lambda_i(\Sigma_U)/e_U$. Then

$$C_{\text{exc}}(U) = \frac{\sum_{i=1}^{d_U} \log(1+\gamma r_i)}{d_U \log(1+\gamma)}, \qquad \frac{1}{d_U}\sum_{i=1}^{d_U} r_i = v_U.$$

Range and equality conditions. Each logarithmic term is nonnegative because $r_i \ge 0$. By the strict concavity of $\log(1+\gamma r)$,

$$\begin{aligned}\frac{1}{d_U}\sum_{i=1}^{d_U} \log(1+\gamma r_i) &\le \log\left(1 + \frac{\gamma}{d_U}\sum_{i=1}^{d_U} r_i\right) \\ &= \log(1+\gamma v_U) \\ &\le \log(1+\gamma).\end{aligned}$$

Therefore,

$$0 \le C_{\text{exc}}(U) \le 1.$$

The lower bound is attained if and only if $\Sigma_U = 0$. Equality at the upper bound requires every relative eigenvalue to equal one, equivalently

$$\mu_U = 0, \qquad \Sigma_U = e_U I, \qquad e_U^{\text{raw}} = e_U.$$

Since $\text{rank}(\Sigma_U) \le N - 1$, attaining the upper bound also requires $N - 1 \ge d_U$.

Coordinate and scale properties. For any orthogonal matrix Q, the transformation $U \mapsto UQ$ preserves total energy and orthogonally conjugates the covariance. Hence

$$C_{\text{exc}}(UQ) = C_{\text{exc}}(U).$$

Now consider $U \mapsto cUQ$, where $c \ne 0$. If the energy floor is inactive before and after the transformation, then

$$\Sigma_{cUQ} = c^2 Q^\top \Sigma_U Q, \qquad e_{cUQ} = c^2 e_U.$$

The normalized covariance changes only by orthogonal conjugation, which gives

$$C_{\text{exc}}(cUQ) = C_{\text{exc}}(U).$$

The normalization retains sensitivity to the common feature mean. At fixed centered covariance, increasing the mean energy increases e_U and decreases the relative covariance spectrum.

其中 $\epsilon > 0$ 为数值下限。总能量分解为

$$d_U e_U^{\text{raw}} = \text{tr}(\Sigma_U) + \|\mu_U\|_2^2.$$

因此，中心化能量比例

$$v_U = \frac{\text{tr}(\Sigma_U)}{d_U e_U}$$

满足 $0 \le v_U \le 1$。能量下限未激活时，

$$v_U = 1 - \frac{\|\mu_U\|_2^2}{d_U e_U}.$$

下列结果均针对有限批次经验统计，不要求高斯表征，也不要求批次样本独立。

A.2 激发容量的性质

回顾

$$C_{\text{exc}}(U) = \frac{\log\det\left(I + \gamma \Sigma_U / e_U\right)}{d_U \log(1+\gamma)}, \qquad \gamma > 0.$$

设 $r_i = \lambda_i(\Sigma_U)/e_U$。则

$$C_{\text{exc}}(U) = \frac{\sum_{i=1}^{d_U} \log(1+\gamma r_i)}{d_U \log(1+\gamma)}, \qquad \frac{1}{d_U}\sum_{i=1}^{d_U} r_i = v_U.$$

范围与等号条件。因 $r_i \ge 0$，各对数项非负。由 $\log(1+\gamma r)$ 的严格凹性，

$$\begin{aligned}\frac{1}{d_U}\sum_{i=1}^{d_U} \log(1+\gamma r_i) &\le \log\left(1 + \frac{\gamma}{d_U}\sum_{i=1}^{d_U} r_i\right) \\ &= \log(1+\gamma v_U) \\ &\le \log(1+\gamma).\end{aligned}$$

因此，

$$0 \le C_{\text{exc}}(U) \le 1.$$

下界取等当且仅当 $\Sigma_U = 0$。上界取等要求每个相对特征值均为 1，等价于

$$\mu_U = 0, \qquad \Sigma_U = e_U I, \qquad e_U^{\text{raw}} = e_U.$$

因 $\text{rank}(\Sigma_U) \le N - 1$，达到上界还要求 $N - 1 \ge d_U$。

坐标与尺度性质。对于任意正交矩阵 Q，变换 $U \mapsto UQ$ 保留总能量并对协方差作正交共轭。因此

$$C_{\text{exc}}(UQ) = C_{\text{exc}}(U).$$

现在考虑 $U \mapsto cUQ$，其中 $c \ne 0$。若变换前后能量下限均未激活，则

$$\Sigma_{cUQ} = c^2 Q^\top \Sigma_U Q, \qquad e_{cUQ} = c^2 e_U.$$

归一化协方差仅作正交共轭变化，得到

$$C_{\text{exc}}(cUQ) = C_{\text{exc}}(U).$$

归一化保留对公共特征均值的敏感性。在固定中心化协方差 ，
增加均值能量会增大 e_U，减小相对协方差谱。

Rank–energy bound. Let $k_U = \operatorname{rank}(\Sigma_U) > 0$. Only k_U relative eigenvalues are nonzero, and their sum is $d_U v_U$. Applying Jensen's inequality to these nonzero terms gives

$$\sum_{i=1}^{k_U} \log(1+\gamma r_i) \le k_U \log\left(1 + \frac{\gamma d_U v_U}{k_U}\right).$$

Thus,

$$C_{\mathrm{exc}}(U) \le \frac{k_U}{d_U} \frac{\log\left(1+\gamma d_U v_U / k_U\right)}{\log(1+\gamma)}. \tag{15}$$

Equality holds if and only if all positive covariance eigenvalues are equal. If $k_U = 0$, the excitation capacity is zero.

This bound captures both the number of available representation directions and the fraction of total energy devoted to sample-dependent variation. At fixed rank and centered energy, a uniform nonzero spectrum maximizes excitation capacity.

A.3 Transmission Capacity and the Path-Supply Bound

Define

$$\Sigma_{HZ} = \frac{1}{N} H^\top P Z,$$

$$\tau_H = \max\left\{\frac{\operatorname{tr}(\Sigma_H)}{d_h}, \epsilon\right\}, \qquad \tau_Z = \max\left\{\frac{\operatorname{tr}(\Sigma_Z)}{d_z}, \epsilon\right\},$$

and

$$A_H = \Sigma_H + \rho\tau_H I, \qquad A_Z = \Sigma_Z + \rho\tau_Z I, \qquad \rho > 0.$$

The regularized whitened cross-covariance is

$$K = A_H^{-1/2} \Sigma_{HZ} A_Z^{-1/2}.$$

Let $r = \min(d_h, d_z)$ and let $\sigma_1, \ldots, \sigma_r$ denote the singular values of K. The transmission capacity can be written as

$$C_{\mathrm{tr}}(H, \mathrm{sg}(Z)) = \frac{\sum_{i=1}^{r} \log(1+\gamma\sigma_i^2)}{r\log(1+\gamma)}.$$

Singular-value bound. The joint empirical covariance is positive semidefinite:

$$\begin{pmatrix} \Sigma_H & \Sigma_{HZ} \\ \Sigma_{HZ}^\top & \Sigma_Z \end{pmatrix} = \frac{1}{N} \begin{pmatrix} H^\top \\ Z^\top \end{pmatrix} P \begin{pmatrix} H & Z \end{pmatrix} \succeq 0.$$

Adding the two positive ridge terms yields

$$\begin{pmatrix} A_H & \Sigma_{HZ} \\ \Sigma_{HZ}^\top & A_Z \end{pmatrix} \succ 0.$$

Its Schur complement satisfies

$$A_Z - \Sigma_{HZ}^\top A_H^{-1} \Sigma_{HZ} \succ 0.$$

Congruence by $A_Z^{-1/2}$ then gives

$$I - K^\top K \succ 0.$$

Therefore, $0 \le \sigma_i < 1$ for every i.

Let $k_t = \operatorname{rank}(\Sigma_{HZ})$. Both whitening matrices are invertible, so K has exactly k_t nonzero singular values. It follows that

$$0 \le C_{\mathrm{tr}}(H, \mathrm{sg}(Z)) \le \frac{k_t}{r} \le 1. \tag{16}$$

For $\rho > 0$ and $k_t > 0$, the middle inequality is strict. Transmission capacity is zero if and only if the cross-covariance is zero.

The denominator counts all potential interface modes. Strong transmission within a small active subspace therefore retains a capacity deficit associated with the missing directions.

秩-能量界。令 $k_U = \operatorname{rank}(\Sigma_U) > 0$。只有 k_U 个相对特征值非零，其和为 $d_U v_U$。对这些非零项应用 Jensen 不等式得到

$$\sum_{i=1}^{k_U} \log(1+\gamma r_i) \le k_U \log\left(1 + \frac{\gamma d_U v_U}{k_U}\right).$$

于是，

$$C_{\mathrm{exc}}(U) \le \frac{k_U}{d_U} \frac{\log\left(1+\gamma d_U v_U / k_U\right)}{\log(1+\gamma)}. \tag{15}$$

等号当且仅当全部正协方差特征值相等时成立。若 $k_U = 0$，激发容量为零。

该界同时捕捉可用表征方向数，以及用于样本相关变化的总能量比例。在固定秩和中心化能量下，均匀非零谱最大化激发容量。

A.3 传输容量与路径供给界

定义

$$\Sigma_{HZ} = \frac{1}{N} H^\top P Z,$$

$$\tau_H = \max\left\{\frac{\operatorname{tr}(\Sigma_H)}{d_h}, \epsilon\right\}, \qquad \tau_Z = \max\left\{\frac{\operatorname{tr}(\Sigma_Z)}{d_z}, \epsilon\right\},$$

及

$$A_H = \Sigma_H + \rho\tau_H I, \qquad A_Z = \Sigma_Z + \rho\tau_Z I, \qquad \rho > 0.$$

正则化白化交叉协方差为

$$K = A_H^{-1/2} \Sigma_{HZ} A_Z^{-1/2}.$$

设 $r = \min(d_h, d_z)$，令 $\sigma_1, \cdots, \sigma_r$ 为 K 的奇异值。传输容量可写为

$$C_{\mathrm{tr}}(H, \mathrm{sg}(Z)) = \frac{\sum_{i=1}^{r} \log(1+\gamma\sigma_i^2)}{r\log(1+\gamma)}.$$

奇异值界。联合经验协方差为半正定：

$$\begin{pmatrix} \Sigma_H & \Sigma_{HZ} \\ \Sigma_{HZ}^\top & \Sigma_Z \end{pmatrix} = \frac{1}{N} \begin{pmatrix} H^\top \\ Z^\top \end{pmatrix} P \begin{pmatrix} H & Z \end{pmatrix} \succeq 0.$$

加入两个正岭项得到

$$\begin{pmatrix} A_H & \Sigma_{HZ} \\ \Sigma_{HZ}^\top & A_Z \end{pmatrix} \succ 0.$$

其 Schur 补满足

$$A_Z - \Sigma_{HZ}^\top A_H^{-1} \Sigma_{HZ} \succ 0.$$

再通过 $A_Z^{-1/2}$ 合同变换得到

$$I - K^\top K \succ 0.$$

因此，对每个 i 均有 $0 \le \sigma_i < 1$。

设 $k_t = \operatorname{rank}(\Sigma_{HZ})$。两个白化矩阵均可逆，因此 K 恰有 k_t 个非零奇异值。于是

$$0 \le C_{\mathrm{tr}}(H, \mathrm{sg}(Z)) \le \frac{k_t}{r} \le 1. \tag{16}$$

对于 $\rho > 0$ 且 $k_t > 0$ 时，中间不等式严格成立。传输容量为零当且仅当交叉协方差为零。

分母计数全部潜在接口模式。因此，小型活跃子空间内的强传输仍保留与缺失方向有关的容量不足。

Finite ridge and invariance. Transmission capacity is invariant to translations and independent orthogonal transformations at the two endpoints. When the corresponding scale floors are inactive, it is also invariant to independent nonzero global rescalings.

A finite ridge contracts the whitened correlation strengths. For example, if $H = Z$, the covariance is isotropic, and the scale floor is inactive, then

$$K = \frac{1}{1+\rho} I$$

and

$$C_{\mathrm{tr}} = \frac{\log\left(1+\gamma/(1+\rho)^2\right)}{\log(1+\gamma)} < 1.$$

The normalization thus refers to ideal transmission in the vanishing-ridge limit.

The ridge is relative to the current covariance scale and remains on the automatic-differentiation path. In particular, under $H \mapsto aH$ with an inactive floor, $\Sigma_H \mapsto a^2\Sigma_H$, $\tau_H \mapsto a^2\tau_H$, and $\Sigma_{HZ} \mapsto a\Sigma_{HZ}$, so the channel score and its radial directional derivative are invariant. Detaching τ_H would leave the forward value unchanged at the current point but would break this backward invariance, creating a spurious radial gradient whenever transmission is the active bottleneck.

The score measures second-order cross-endpoint transmission. Zero cross-covariance does not exclude nonlinear dependence between the representations.

Supply-bound derivation. By definition,

$$k(H, \mathrm{sg}(Z)) = \min\left\{C_{\mathrm{exc}}(H), C_{\mathrm{tr}}(H, \mathrm{sg}(Z))\right\}.$$

Combining Equation 15 with Equation 16 gives

$$k(H, \mathrm{sg}(Z)) \le \min\left\{\frac{k_h}{d_h}\frac{\log\left(1+\gamma d_h v_h/k_h\right)}{\log(1+\gamma)}, \frac{k_t}{r}\right\}.$$

If $k_h = 0$, then $C_{\mathrm{exc}}(H) = 0$ and hence $k(H, \mathrm{sg}(Z)) = 0$.

The minimum corresponds to simultaneous support by both capacity scores:

$$k \ge q \iff C_{\mathrm{exc}}(H) \ge q \text{ and } C_{\mathrm{tr}}(H, \mathrm{sg}(Z)) \ge q.$$

This is the joint support criterion used in the construction. □

A.4 Demand Calibration and Log-Space Interpolation

Assume that $\mathcal{D}(Z) \ge 0$ and that the zero-representation reference is finite and positive:

$$\mathcal{D}_0 = \mathcal{D}(0) > 0.$$

Reference calibration. Consider an affine calibration of endpoint energy satisfying

$$\mathcal{D}(Z) = 0 \mapsto q_R = 1, \qquad \mathcal{D}(Z) = \mathcal{D}_0 \mapsto q_R = 0.$$

These conditions uniquely determine the affine map $1 - \mathcal{D}(Z)/\mathcal{D}_0$. Restricting its range to the unit interval gives

$$q_R = \mathrm{clip}\left(1 - \frac{\mathcal{D}(Z)}{\mathcal{D}_0}, 0, 1\right).$$

For any constant $c > 0$,

$$1 - \frac{c\mathcal{D}(Z)}{c\mathcal{D}_0} = 1 - \frac{\mathcal{D}(Z)}{\mathcal{D}_0}.$$

The calibrated demand is therefore invariant to the overall units of the endpoint energy.

The second demand readout is

$$q_E = C_{\mathrm{exc}}(\mathrm{sg}(Z)).$$

It uses the relative covariance structure already formed at the endpoint to calibrate external demand. Excitation and transmission have different spectral normalizations, so their values need not obey a fixed ordering.

有限岭与不变性。传输容量对两端点的平移、独立正交变换不变。对应尺度下限未激活时，也对独立非零全局缩放不变。

有限岭使白化相关强度收缩。例如，若 $H = Z$、协方差各向同性且尺度下限未激活，则

$$K = \frac{1}{1+\rho} I$$

及

$$C_{\mathrm{tr}} = \frac{\log\left(1+\gamma/(1+\rho)^2\right)}{\log(1+\gamma)} < 1.$$

因此，归一化参考的是岭趋零极限下的理想传输。

岭尺度相对于当前协方差尺度确定，并保留在自动微分路径中。尤其在 $H \mapsto aH$ 且下限未激活时， $\Sigma_H \mapsto a^2\Sigma_H$、 $\tau_H \mapsto a^2\tau_H$、$\Sigma_{HZ} \mapsto a\Sigma_{HZ}$，因此通道分数及其径向方向导数不变。对 τ_H 停止梯度会保持当前点前向值，却破坏这一反向不变性：传输为活跃瓶颈时会产生虚假径向梯度。

该分数衡量二阶跨端点传输。零交叉协方差不能排除表征之间的非线性依赖。

供给界推导。根据定义，

$$k(H, \mathrm{sg}(Z)) = \min\left\{C_{\mathrm{exc}}(H), C_{\mathrm{tr}}(H, \mathrm{sg}(Z))\right\}.$$

结合式 15 与式 16 得到

$$k(H, \mathrm{sg}(Z)) \le \min\left\{\frac{k_h}{d_h}\frac{\log\left(1+\gamma d_h v_h/k_h\right)}{\log(1+\gamma)}, \frac{k_t}{r}\right\}.$$

若 $k_h = 0$，则 $C_{\mathrm{exc}}(H) = 0$，因此 $k(H, \mathrm{sg}(Z)) = 0$。

取最小值对应两种容量分数同时提供支持：

$$k \ge q \iff C_{\mathrm{exc}}(H) \ge q \text{ and } C_{\mathrm{tr}}(H, \mathrm{sg}(Z)) \ge q.$$

这是构造中使用的联合支持标准。 □

A.4 需求校准与对数空间插值

假设 $\mathcal{D}(Z) \ge 0$，零表征参考有限且为正：

$$\mathcal{D}_0 = \mathcal{D}(0) > 0.$$

参考校准。考虑满足以下条件的端点能量仿射校准

$$\mathcal{D}(Z) = 0 \mapsto q_R = 1, \qquad \mathcal{D}(Z) = \mathcal{D}_0 \mapsto q_R = 0.$$

这些条件唯一确定仿射映射 $1 - \mathcal{D}(Z)/\mathcal{D}_0$。将其值域限制于单位区间得到

$$q_R = \mathrm{clip}\left(1 - \frac{\mathcal{D}(Z)}{\mathcal{D}_0}, 0, 1\right).$$

对任意常量 $c > 0$,

$$1 - \frac{c\mathcal{D}(Z)}{c\mathcal{D}_0} = 1 - \frac{\mathcal{D}(Z)}{\mathcal{D}_0}.$$

因此，校准需求对端点能量的总体单位不变。

第二个需求读数为

$$q_E = C_{\mathrm{exc}}(\mathrm{sg}(Z)).$$

它用端点已形成的相对协方差结构校准外部需求。激发和传输具有不同谱归一化，因此数值不必具有固定次序。

Log-space barycenter. For $q_R, q_E > 0$, consider

$$F_\alpha(q) = (1-\alpha)(\log q - \log q_R)^2 + \alpha(\log q - \log q_E)^2, \qquad \alpha \in [0,1].$$

Let $y = \log q$, $a = \log q_R$, and $b = \log q_E$. Completing the square gives

$$F_\alpha(q) = (y - [(1-\alpha)a + \alpha b])^2 + \alpha(1-\alpha)(a-b)^2.$$

The second term is independent of q. The unique minimizer therefore satisfies

$$\log q_\alpha^\star = (1-\alpha)\log q_R + \alpha \log q_E,$$

or equivalently,

$$q_\alpha^\star = q_R^{1-\alpha} q_E^\alpha.$$

Under the metric $\mathrm{d}\ell^2 = (\mathrm{d}q/q)^2$ on the positive half-line, this interpolation is a geodesic between the two demand readouts. It also satisfies

$$\min(q_R, q_E) \le q_\alpha^\star \le \max(q_R, q_E).$$

The external and realized-geometry demands are recovered at $\alpha = 0$ and $\alpha = 1$, respectively.

Minimax interpretation of the symmetric choice. For any $y \in \mathbb{R}$,

$$|a-b| \le |a-y| + |y-b| \le 2\max\{|a-y|, |y-b|\}.$$

The largest deviation is thus at least $|a-b|/2$, with equality only at $y = (a+b)/2$. Consequently,

$$\sqrt{q_R q_E} = \operatorname*{argmin}_{q>0} \max\left\{\left|\log\frac{q}{q_R}\right|, \left|\log\frac{q}{q_E}\right|\right\}.$$

The choice $\alpha = \frac{1}{2}$ is therefore the symmetric center of the two readouts in log space. Adaptation follows from changes in the endpoint energy and geometry, without learning the interpolation weight.

Zero-demand boundaries. For fixed $0 < \alpha < 1$, the geometric demand has a continuous extension to the nonnegative quadrant:

$$q_\alpha^\star = 0 \qquad \text{whenever} \qquad q_R q_E = 0.$$

The two boundary methods are defined separately by $q_0 = q_R$ and $q_1 = q_E$.

For numerical evaluation at the boundary, the implementation uses Equation 6. Thus $\widetilde{q}_{\alpha,\epsilon} > 0$ for $\epsilon > 0$, while $\widetilde{q}_{\alpha,\epsilon} \to q_\alpha^\star$ as $\epsilon \to 0$ wherever the ideal demand is positive. This distinguishes the geometric object from its finite-precision realization. For the symmetric default, $\epsilon \le q \le 1+\epsilon$: $q_R = q_E = 1$ gives $q = 1+\epsilon$, $q_R = 0$ and $q_E = c > 0$ give $q = \sqrt{\epsilon(c+\epsilon)} = O(\sqrt{\epsilon})$, and $q_R = q_E = 0$ gives $q = \epsilon$. At this last corner, $k = 0$ yields $\mathcal{S} = (\log 2)^2$; an exactly constant H still has zero first-order capacity gradient, as shown below.

A.5 Geometric Derivation of the One-Sided Bridge

Fix a demand $q \ge 0$ and write

$$c_1 = C_{\mathrm{exc}}(H), \qquad c_2 = C_{\mathrm{tr}}(H, \mathrm{sg}(Z)), \qquad k = \min(c_1, c_2).$$

Introduce log-supply coordinates

$$u_i = \log(c_i + \epsilon), \qquad b_q = \log(q+\epsilon).$$

The two support conditions define the region

$$\mathcal{C}_q = [b_q, \infty)^2.$$

For any $v \in \mathcal{C}_q$,

$$\|u - v\|_\infty \ge \max_{i=1,2}[b_q - u_i]_+.$$

对数空间重心。对于 q_R、$q_E > 0$，**考虑**

$$F_\alpha(q) = (1-\alpha)(\log q - \log q_R)^2 + \alpha(\log q - \log q_E)^2, \qquad \alpha \in [0,1].$$

设 $y = \log q$、$a = \log q_R$、$b = \log q_E$。配方得到

$$F_\alpha(q) = (y - [(1-\alpha)a + \alpha b])^2 + \alpha(1-\alpha)(a-b)^2.$$

第二项与 q 无关。因此唯一最小值点满足

$$\log q_\alpha^\star = (1-\alpha)\log q_R + \alpha \log q_E,$$

或等价地，

$$q_\alpha^\star = q_R^{1-\alpha} q_E^\alpha.$$

在正半轴的度量 $\mathrm{d}\ell^2 = (\mathrm{d}q/q)^2$ 下，该插值是两个需求读数之间的测地线，还满足

$$\min(q_R, q_E) \le q_\alpha^\star \le \max(q_R, q_E).$$

外部需求与已实现几何需求分别在 $\alpha = 0$、$\alpha = 1$ 恢复。

对称选择的极小极大解释。对任意 $y \in \mathbb{R}$**，**

$$|a-b| \le |a-y| + |y-b| \le 2\max\{|a-y|, |y-b|\}.$$

最大偏差因此至少为 $|a-b|/2$，仅在 $y = (a+b)/2$ 时取等。因此，

$$\sqrt{q_R q_E} = \operatorname*{argmin}_{q>0} \max\left\{\left|\log\frac{q}{q_R}\right|, \left|\log\frac{q}{q_E}\right|\right\}.$$

因此，$\alpha = \frac{1}{2}$ 的选择是两个读数在对数空间的对称中心。自适应来自端点能量与几何变化，无需学习插值权重。

零需求边界。对于固定 $0 < \alpha < 1$，几何需求可连续延拓到非负象限：

$$q_\alpha^\star = 0 \qquad \text{whenever} \qquad q_R q_E = 0.$$

两个边界方法分别定义为 $q_0 = q_R$ 与 $q_1 = q_E$。

在边界进行数值计算时，实现使用式 6。因此，当 $\epsilon > 0$ 时有 $\widetilde{q}_{\alpha、\epsilon} > 0$；在理想需求为正处，当 $\epsilon \to 0$ 时有 $\widetilde{q}_{\alpha、\epsilon} \to q_\alpha^\star$。这区分了几何对象与有限精度实现。对于对称默认设置，$\epsilon \le q \le 1+\epsilon$：$q_R = q_E = 1$ 给出 $q = 1+\epsilon$，$q_R = 0$ 和 $q_E = c > 0$ 给出 $q = \sqrt{\epsilon(c+\epsilon)} = O(\sqrt{\epsilon})$，$q_R = q_E = 0$ 给出 $q = \epsilon$。在最后这个角点，$k = 0$ 得到 $\mathcal{S} = (\log 2)^2$；严格常量的 H 仍具有零一阶容量梯度，如下所示。

A.5 单侧桥接的几何推导

固定需求 $q \ge 0$，记

$$c_1 = C_{\mathrm{exc}}(H), \qquad c_2 = C_{\mathrm{tr}}(H, \mathrm{sg}(Z)), \qquad k = \min(c_1, c_2).$$

引入对数供给坐标

$$u_i = \log(c_i + \epsilon), \qquad b_q = \log(q+\epsilon).$$

两个支持条件定义区域

$$\mathcal{C}_q = [b_q, \infty)^2.$$

对任意 $v \in \mathcal{C}_q$，

$$\|u - v\|_\infty \ge \max_{i=1,2}[b_q - u_i]_+.$$

The choice $v_i = \max(u_i, b_q)$ attains this lower bound. Therefore,

$$\begin{aligned}\mathrm{dist}_\infty^2(u, \mathcal{C}_q) &= \max_{i=1,2}[b_q - u_i]_+^2 \\ &= [b_q - \min(u_1, u_2)]_+^2 \\ &= \left[\log \frac{q+\epsilon}{k+\epsilon}\right]_+^2.\end{aligned}$$

This proves the distance representation in Equation 10.

The bridge equals the largest squared log-violation of the two supply constraints and satisfies

$$\mathcal{S} = 0 \quad \Longleftrightarrow \quad c_1 \geq q \text{ and } c_2 \geq q.$$

It exerts no additional geometric pressure once both support conditions hold. Otherwise, its correction is determined by the current bottleneck.

The distance is defined in capacity coordinates. The network parameterization determines how these coordinates change under parameter updates.

A.6 Gradient Paths and Boundary Behavior

Response to insufficient supply. For fixed demand, let

$$\Delta(k) = \left[\log \frac{q+\epsilon}{k+\epsilon}\right]_+.$$

Then

$$\frac{\partial \mathcal{S}}{\partial k} = -\frac{2\Delta(k)}{k+\epsilon}.$$

Within the active region $k < q$,

$$\frac{\partial^2 \mathcal{S}}{\partial k^2} = \frac{2\left(1 + \log \frac{q+\epsilon}{k+\epsilon}\right)}{(k+\epsilon)^2} > 0.$$

At fixed demand, the magnitude of the negative scalar derivative therefore increases as supply decreases. The first derivative is zero for $k \geq q$ and is continuous at $k = q$.

When the two capacities differ,

$$\nabla_H \mathcal{S} = -\frac{2\Delta(k)}{k+\epsilon} \begin{cases} \nabla_H C_{\mathrm{exc}}, & C_{\mathrm{exc}} < C_{\mathrm{tr}}, \\ \nabla_H C_{\mathrm{tr}}, & C_{\mathrm{tr}} < C_{\mathrm{exc}}. \end{cases}$$

At a tie, a generalized gradient is selected from the convex hull of the active branch gradients.

Detached references and endpoint roles. The bridge uses a stopped endpoint representation and demand:

$$\mathcal{S}\big(H, \mathrm{sg}(Z); \mathrm{sg}(q)\big) = \left[\log \frac{q+\epsilon}{k+\epsilon}\right]_+^2.$$

The endpoint tensor and demand are fixed references during the current backward pass. The covariance-relative ridge scale derived from H remains differentiable; the Z side receives no gradient because the endpoint tensor has already been stopped.

With separate encoder and projector parameters θ and ϕ, the bridge's automatic-differentiation gradients satisfy

$$\nabla_\phi^{\mathrm{AD}} \mathcal{S} = 0, \qquad \nabla_\theta^{\mathrm{AD}} \mathcal{S} = J_{f_\theta}^\top \nabla_H \mathcal{S}.$$

The same loss decomposition as Equation 12 remains valid in automatic differentiation:

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{task}} + \mathcal{L}_{\mathrm{SADL}}, \qquad \mathcal{L}_{\mathrm{SADL}} = \lambda_D \mathcal{D}(Z) + \lambda_S \mathcal{S}\big(H, \mathrm{sg}(Z); \mathrm{sg}(q)\big),$$

The task objective and endpoint regularizer retain their normal gradients. Stopping the references directs the bridge toward current upstream supply, rather than directly reducing demand to eliminate the mismatch.

选择 $v_i = \max(u_i, b_q)$ 达到该下界。因此，

$$\begin{aligned}\mathrm{dist}_\infty^2(u, \mathcal{C}_q) &= \max_{i=1,2}[b_q - u_i]_+^2 \\ &= [b_q - \min(u_1, u_2)]_+^2 \\ &= \left[\log \frac{q+\epsilon}{k+\epsilon}\right]_+^2.\end{aligned}$$

这证明了式 10 的距离表示。

桥接等于两个供给约束中最大的平方对数违反量，并满足

$$\mathcal{S} = 0 \quad \Longleftrightarrow \quad c_1 \geq q \text{ and } c_2 \geq q.$$

一旦两个支持条件成立，就不施加额外几何压力；否则校正由当前瓶颈决定。

距离在容量坐标中定义。网络参数化决定这些坐标如何随参数更新变化。

A.6 梯度路径与边界行为

对供给不足的响应。固定需求，设

$$\Delta(k) = \left[\log \frac{q+\epsilon}{k+\epsilon}\right]_+.$$

则

$$\frac{\partial \mathcal{S}}{\partial k} = -\frac{2\Delta(k)}{k+\epsilon}.$$

活跃区域 $k < q$ 内，

$$\frac{\partial^2 \mathcal{S}}{\partial k^2} = \frac{2\left(1 + \log \frac{q+\epsilon}{k+\epsilon}\right)}{(k+\epsilon)^2} > 0.$$

固定需求下，负标量导数的绝对值随供给降低而增大。当 $k \geq q$ 时一阶导数为零，且在 $k = q$ 处连续。

两种容量不同时，

$$\nabla_H \mathcal{S} = -\frac{2\Delta(k)}{k+\epsilon} \begin{cases} \nabla_H C_{\mathrm{exc}}, & C_{\mathrm{exc}} < C_{\mathrm{tr}}, \\ \nabla_H C_{\mathrm{tr}}, & C_{\mathrm{tr}} < C_{\mathrm{exc}}. \end{cases}$$

相等时，从活跃分支梯度的凸包选择广义梯度。

停止梯度的参考与端点角色。桥接使用停止梯度的端点表征及需求：

$$\mathcal{S}\big(H, \mathrm{sg}(Z); \mathrm{sg}(q)\big) = \left[\log \frac{q+\epsilon}{k+\epsilon}\right]_+^2.$$

在当前反向传播中，端点张量与需求是固定参考。由 H 得到的协方差相对岭尺度保持可微； Z 侧不接收梯度，因为端点张量已停止梯度。

编码器与投影器参数分别为 θ 、 ϕ 时，桥接自动微分梯度满足

$$\nabla_\phi^{\mathrm{AD}} \mathcal{S} = 0, \qquad \nabla_\theta^{\mathrm{AD}} \mathcal{S} = J_{f_\theta}^\top \nabla_H \mathcal{S}.$$

与式 12 相同的损失分解在自动微分中仍成立：

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{task}} + \mathcal{L}_{\mathrm{SADL}}, \qquad \mathcal{L}_{\mathrm{SADL}} = \lambda_D \mathcal{D}(Z) + \lambda_S \mathcal{S}\big(H, \mathrm{sg}(Z); \mathrm{sg}(q)\big),$$

任务目标和端点正则项保留正常梯度。停止参考的梯度使桥接指向当前上游供给，而非直接降低需求来消除失配。

Feature gradient of excitation capacity. Assume that the energy floor is inactive, with $e_H^{\text{raw}} > \epsilon$. Define

$$M = I + \gamma \Sigma_H / e_H, \qquad a_H = \frac{1}{d_h} \operatorname{tr}\left(\frac{\Sigma_H}{e_H} M^{-1}\right).$$

Differentiating excitation capacity gives

$$\mathrm{d}C_{\text{exc}} = \frac{\gamma}{d_h \log(1+\gamma)} \left[\frac{\operatorname{tr}(M^{-1}\mathrm{d}\Sigma_H)}{e_H} - \frac{\operatorname{tr}(M^{-1}\Sigma_H)}{e_H^2}\,\mathrm{d}e_H\right].$$

Substituting the covariance and energy differentials yields

$$\nabla_H C_{\text{exc}} = \frac{2\gamma}{N d_h e_H \log(1+\gamma)} \left(PHM^{-1} - a_H H\right). \tag{17}$$

Along a relative covariance eigenmode r_i, the spectral response of M^{-1} is

$$\frac{1}{1+\gamma r_i}.$$

Weaker directions therefore receive larger spectral response weights. The second term compensates for changes along the global scaling direction, giving

$$\langle H, \nabla_H C_{\text{exc}} \rangle_F = 0.$$

This identity describes the local scale property of excitation capacity itself in representation space.

Zero-demand limit and exact degeneracy. The ideal and implemented demands are, respectively,

$$q^\star = q_R^{1-\alpha} q_E^{\alpha}, \qquad q = \exp((1-\alpha)\log(q_R+\epsilon) + \alpha \log(q_E+\epsilon)),$$

and the bridge uses

$$\mathcal{S} = [\log(q+\epsilon) - \log(k+\epsilon)]_+^2.$$

If either reference is zero, then $q^\star = 0$ for an interior interpolation weight, but the implemented q remains at an ϵ-controlled positive scale. Exact bridge deactivation is therefore determined by $k \geq q$ in the smoothed implementation, not by substituting the ideal boundary value into the code.

An exactly constant encoder representation has a separate boundary behavior. If $H = \mathbf{1}h_0^\top$, then

$$PH = 0, \qquad \Sigma_H = 0, \qquad \Sigma_{HZ} = 0.$$

Both capacity scores have zero first-order gradients with respect to H at this point. The scalar supply response and the resulting feature gradient must therefore be distinguished. The original task objective and endpoint regularizer remain in the complete objective, while the bridge responds to the nonzero demand and structure formed during training.

激发容量的特征梯度。假设能量下限未激活，且 $e_H^{\text{raw}} > \epsilon$。定义

$$M = I + \gamma \Sigma_H / e_H, \qquad a_H = \frac{1}{d_h} \operatorname{tr}\left(\frac{\Sigma_H}{e_H} M^{-1}\right).$$

对激发容量求导得到

$$\mathrm{d}C_{\text{exc}} = \frac{\gamma}{d_h \log(1+\gamma)} \left[\frac{\operatorname{tr}(M^{-1}\mathrm{d}\Sigma_H)}{e_H} - \frac{\operatorname{tr}(M^{-1}\Sigma_H)}{e_H^2}\,\mathrm{d}e_H\right].$$

代入协方差与能量的微分得到

$$\nabla_H C_{\text{exc}} = \frac{2\gamma}{N d_h e_H \log(1+\gamma)} \left(PHM^{-1} - a_H H\right). \tag{17}$$

沿相对协方差特征模式 r_i，M^{-1} 的谱响应为

$$\frac{1}{1+\gamma r_i}.$$

因此，较弱方向获得更大谱响应权重。第二项补偿全局缩放方向的变化，得到

$$\langle H, \nabla_H C_{\text{exc}} \rangle_F = 0.$$

该恒等式描述表征空间中激发容量本身的局部尺度性质。

零需求极限与精确退化。理想需求和实现需求分别为

$$q^\star = q_R^{1-\alpha} q_E^{\alpha}, \qquad q = \exp((1-\alpha)\log(q_R+\epsilon) + \alpha \log(q_E+\epsilon)),$$

且桥接使用

$$\mathcal{S} = [\log(q+\epsilon) - \log(k+\epsilon)]_+^2.$$

任一参考为零时，内部插值权重下有 $q^\star = 0$，但实现的 q 仍保持由 ϵ 控制的正尺度。因此，平滑实现中桥接是否精确关闭由 $k \geq q$ 决定，不能把理想边界值直接代入代码。

严格常量编码器表征具有独立边界行为。若 $H = \mathbf{1}h_0^\top$，则

$$PH = 0, \qquad \Sigma_H = 0, \qquad \Sigma_{HZ} = 0.$$

此点两种容量分数对 H 的一阶梯度均为零。因此必须区分标量供给响应与所得特征梯度。完整目标仍保留原任务目标及端点正则项，桥接响应训练过程中形成的非零需求与结构。

B NUMERICAL IMPLEMENTATION

Sample alignment and statistical precision. Sequence representations are flattened in the same batch–time order before computing cross-covariance. If the encoder output has shape $B \times S \times d_h$ and the projected output has shape $S \times B \times d_z$, the first two projected dimensions are exchanged before both tensors are flattened into $N = BS$ samples.

Centering, covariance estimation, whitening, and log-determinant operations within the bridge are performed in FP32 with autocast disabled. Endpoint regularizers retain their respective computation paths.

Cholesky whitening. The implementation uses Cholesky factorizations and triangular solves instead of explicitly forming inverse square roots. Let

$$A = L_H L_H^\top, \qquad B = L_Z L_Z^\top,$$

and compute

$$\widetilde{K} = L_H^{-1} \Sigma_{HZ} L_Z^{-\top}.$$

The matrices $\widetilde{K}$ and K differ only by orthogonal transformations on the left and right, and therefore have the same singular values.

Using

$$\det\left(I + \gamma \widetilde{K}^\top \widetilde{K}\right) = \det\left(I + \gamma \widetilde{K} \widetilde{K}^\top\right),$$

the log-determinant can be computed from the smaller Gram matrix, whose dimension is $r = \min(d_h, d_z)$.

The ridge scales are computed from differentiable mean covariance energies:

$$\tau_U = \max\left\{\frac{\mathrm{tr}(\Sigma_U)}{d_U}, \epsilon\right\}.$$

Both this relative ridge scale and the total energy in excitation capacity retain their gradients, subject to their numerical floors.

Gradient-semantic audit. The release implementation keeps the relative ridge scale differentiable. A historical control runner detached this scalar, which preserves the forward score but can create a radial gradient when transmission is the active bottleneck. Across 25,396 audited capacity records, excitation was the bottleneck in every case. A matched E1 replay further produced identical values for all 204 common diagnostics through step 5,650 of 5,695, excluding wall time. Thus the audited headline configurations are unchanged, while the released implementation follows the scale-invariant gradient semantics stated above.

Reference energy and smoothing. The reference energy $\mathcal{D}_0$ is evaluated on an all-zero tensor with the same shape as the endpoint representation and cached for the current configuration. The evaluation does not update model parameters. Its random-number state is saved and restored so that reference calibration does not alter the subsequent training random sequence.

The energy floors and logarithmic smoothing serve different numerical roles. The former define stable capacity normalizations, while the latter keep the comparison finite at zero supply. Equation 6 also places ϵ inside the implemented interpolation. This is a numerical regularization of the ideal demand, not a change to the ideal log-geodesic definition.

Configuration in the robotic control experiments. Table 6 lists the bridge settings used in the robotic control experiments. All three endpoint regularizers share the same SADL parameters.

B 数值实现

样本对齐与统计精度。计算交叉协方差前，序列表征按相同批次–时间顺序展平。若编码器输出形状为 $B \times S \times d_h$，投影输出为 $S \times B \times d_z$，则先交换投影输出前两个维度，再将两张量均展平为 $N = BS$ 个样本。

桥接内中心化、协方差估计、白化和对数行列式以 FP32 执行，并禁用 autocast。各端点正则项保留自身计算路径。

Cholesky 白化。实现采用 Cholesky 分解与三角求解，不显式形成逆平方根。设

$$A = L_H L_H^\top, \qquad B = L_Z L_Z^\top,$$

并计算

$$\widetilde{K} = L_H^{-1} \Sigma_{HZ} L_Z^{-\top}.$$

~矩阵 K 与 K 仅相差左右正交变换，因此奇异值相同。

使用

$$\det\left(I + \gamma \widetilde{K}^\top \widetilde{K}\right) = \det\left(I + \gamma \widetilde{K} \widetilde{K}^\top\right),$$

可从较小的 Gram 矩阵计算对数行列式，其维度为 r =min(d_h, d_z)。

岭尺度由可微的平均协方差能量计算：

$$\tau_U = \max\left\{\frac{\mathrm{tr}(\Sigma_U)}{d_U}, \epsilon\right\}.$$

相对岭尺度与激发容量中的总能量均保留梯度，受各自数值下限约束。

梯度语义审计。发布实现保持相对岭尺度可微。历史控制运行器对该标量停止梯度，保留前向分数，却在传输为活跃瓶颈时可能产生径向梯度。25,396 条审计容量记录中，每条的瓶颈均为激发。匹配 E1 重放进一步在 5,695 步中的前 5,650 步，对全部 204 个共有诊断得到相同数值，墙钟时间除外。因此，审计的主要配置不受影响，而发布实现遵循上述尺度不变梯度语义。

参考能量与平滑。参考能量 $\mathcal{D}_0$ 在与端点表征同形状的全零张量上计算，并按当前配置缓存。评估不更新模型参数。其随机数状态保存并恢复，使参考校准不改变后续训练随机序列。

能量下限与对数平滑承担不同数值作用。前者定义稳定容量归一化，后者保证零供给时比较有限。式 6 还将 ϵ 放入实现的插值中。这是理想需求的数值正则化，而非修改理想对数测地线定义。

机器人控制实验配置。表 6 列出机器人控制实验的桥接设置。三种端点正则项共享相同 SADL 参数。

Algorithm 1 Single-batch computation of SADL

Require: Aligned representations H, Z; task loss $\mathcal{L}_{\text{task}}$; endpoint regularizer $\mathcal{D}$
Require: Reference $\mathcal{D}_0$; weights λ_D, λ_S; smoothing ϵ
Ensure: Total objective $\mathcal{L}_{\text{total}}$
Evaluate $\mathcal{D}(Z)$ with its endpoint gradient retained
$q_R \leftarrow \text{clip}(1 - \mathcal{D}(Z)/\mathcal{D}_0, 0, 1)$
$q_E \leftarrow C_{\text{exc}}(\text{sg}(Z))$
$q \leftarrow \text{sg}(\exp\{[\log(q_R + \epsilon) + \log(q_E + \epsilon)]/2\})$
$k \leftarrow \min\{C_{\text{exc}}(H), C_{\text{tr}}(H, \text{sg}(Z))\}$
$\mathcal{S} \leftarrow [\log(q + \epsilon) - \log(k + \epsilon)]_+^2$
$\mathcal{L}_{\text{SADL}} \leftarrow \lambda_D \mathcal{D}(Z) + \lambda_S \mathcal{S}$
return $\mathcal{L}_{\text{task}} + \mathcal{L}_{\text{SADL}}$
Capacity statistics and matrix factorizations use FP32 with autocast disabled.

Table 6: SADL configuration in the robotic control experiments. Endpoint weights multiply the corresponding raw regularizer implementations.

Parameter	Value
Demand interpolation weight α	0.5
Capacity parameter γ	9
Ridge coefficient ρ	10^{-3}
Numerical floor and log smoothing ϵ	10^{-6}
Bridge weight λ_S	0.01
SIGReg endpoint weight	0.03
VISReg endpoint weight	1.5
QQ endpoint weight	3.5

算法 1：SADL 的单批次计算

输入：对齐表征 H、Z；任务损失 $\mathcal{L}_{\text{task}}$；端点正则项 $\mathcal{D}$
输入：参考 $\mathcal{D}_0$；权重 λ_D、λ_S；平滑 ϵ
输出：总目标 $\mathcal{L}_{\text{total}}$
计算 $\mathcal{D}(Z)$，保留其端点梯度
$q_R \leftarrow \text{clip}(1 - \mathcal{D}(Z)/\mathcal{D}_0, 0, 1)$
$q_E \leftarrow C_{\text{exc}}(\text{sg}(Z))$
$q \leftarrow \text{sg}(\exp\{[\log(q_R + \epsilon) + \log(q_E + \epsilon)]/2\})$
$k \leftarrow \min\{C_{\text{exc}}(H), C_{\text{tr}}(H, \text{sg}(Z))\}$
$\mathcal{S} \leftarrow [\log(q + \epsilon) - \log(k + \epsilon)]_+^2$
$\mathcal{L}_{\text{SADL}} \leftarrow \lambda_D \mathcal{D}(Z) + \lambda_S \mathcal{S}$
返回 $\mathcal{L}_{\text{task}} + \mathcal{L}_{\text{SADL}}$
容量统计与矩阵分解使用 *FP32*，并禁用 *autocast*。

表 6：机器人控制实验的 SADL 配置。端点权重乘对应的原始正则项实现。

参数	值
需求插值权重 α	0.5
容量参数 γ	9
岭系数 ρ	10^{-3}
数值下限与对数平滑 ϵ	10^{-6}
桥接权重 λ_S	0.01
SIGReg 端点权重	0.03
VISReg 端点权重	1.5
QQ 端点权重	3.5

C Experimental Setup and Statistical Protocols

C.1 World-Model Training

Single-task experiments train one model per task. Joint experiments share the encoder and projector while retaining task-specific prediction and control modules. BatchNorm running buffers and shared-parameter gradients are averaged across task workers.

Capacity statistics are computed from each task's local batch before gradient aggregation. The bridge therefore acts on within-task statistics, rather than a covariance pooled across tasks.

Table 7: Core world-model training configuration.

Configuration	Setting
Tasks	PushT, Cube, Reacher, TwoRoom
Training seeds	0, 42, 3072
Encoder	ViT-Tiny, trained from scratch
Encoder representation dimension	192
Input image size	224×224
Patch size	14×14
Projector	MLP with BatchNorm; hidden dimension 2,048
Projected dimension	192
Predictor	Six transformer blocks
Training clip length	Eight frames
Conditioning history	Three steps
Numerical precision	BF16 mixed precision
Batch size per task	256 clips
Joint batch size	1,024 clips across four workers
Optimizer	AdamW
Learning rate	5×10^{-4}
Weight decay	10^{-3}
Endpoint-only encoder backend	MATH
SADL and structural-ablation encoder backend	Flash
Predictor backend	MATH

INTACT retains its forward-prediction, inverse-action, and goal-displacement objectives with weights 1.0, 0.1, and 0.05, respectively. LeWM uses forward prediction without the INTACT inverse-action and goal-action objectives. The shared bridge parameters are given in Table 6.

Joint endpoint-only runs extend to ten epochs. Full SADL and excitation-only runs extend to five epochs. The transmission-only, bilateral, one-sided-linear, and bridge-only comparisons use two-epoch runs. Single-task comparisons use the fixed first-epoch checkpoint.

C.2 Control Interfaces and Success Criteria

Control execution. INTACT uses Direct control with $H = 5/RH = 5$. Each replanning step executes five distinct predicted actions. The initial context contains the actual actions preceding the evaluation start, with zero padding only before a true episode boundary. Subsequent context entries contain executed actions.

LeWM uses Pure-CEM with 300 candidate sequences per iteration, 30 iterations, and 30 retained candidates. Both horizons are five, and no actor warm start is used.

CLEAR-LeWM. Moderate and Strict evaluation use a goal offset of 25 primitive steps and an evaluation budget of 50 primitive actions. Sampling is episode-balanced over the `all` split, and initially successful pairs are excluded. These evaluations are not held-out-episode tests. TwoRoom additionally uses cross-room pairs, clean source windows, and swept-collision handling.

C 实验设置与统计协议

C.1 世界模型训练

单任务实验每任务训练一个模型。联合实验共享编码器与投影器，保留任务专用预测、控制模块。BatchNorm 运行缓冲与共享参数梯度在任务工作进程间平均。

容量统计在梯度汇总前由各任务局部批次计算。因此桥接作用于任务内统计，而非跨任务汇集的协方差。

表 7：世界模型核心训练配置。

配置	设置
任务	PushT，Cube，Reacher，TwoRoom
训练种子	0, 42, 3072
编码器	ViT-Tiny，从头训练
编码器表征维度	192
输入图像尺寸	224×224
图块尺寸	14×14
投影器	含 BatchNorm 的 MLP；隐藏维度 2,048
投影维度	192
预测器	六个 Transformer 块
训练片段长度	八帧
条件历史	三步
数值精度	BF16 混合精度
每任务批量大小	256 个片段
联合批量大小	四个工作进程中共 1,024 个片段
优化器	AdamW
学习率	5×10^{-4}
权重衰减	10^{-3}
仅端点编码器后端	MATH
SADL 与结构消融的编码器后端：Flash	
预测器后端	MATH

INTACT 保留前向预测、逆向动作和目标位移目标，权重分别为 1.0、0.1、0.05。LeWM 使用前向预测，不含 INTACT 逆向动作和目标动作目标。共享桥接参数见表 6。

联合仅端点运行延长至十轮；完整 SADL 与仅激发延长至五轮；仅传输、双侧、单侧线性、仅桥接比较使用两轮运行。单任务比较使用固定第一轮检查点。

C.2 控制接口与成功标准

控制执行。INTACT 使用 Direct 控制，配置 $H = 5/RH = 5$。每次重新规划执行五个不同预测动作。初始上下文含评估起点前的实际动作，只在真实回合边界前零填充；后续上下文包含已执行动作。

LeWM 使用 Pure-CEM，每轮 300 条候选序列、30 轮迭代、保留 30 个候选。两个时域均为五，不使用 actor 热启动。

CLEAR-LeWM。Moderate、Strict 评估使用 25 个原始步的目标偏移与 50 个原始动作的评估预算。在 all 划分上按回合均衡采样，排除初始已成功配对。这些评估不是保留回合测试。TwoRoom 还采用跨房间配对、清洁源窗口与扫掠碰撞处理。

Table 8: Task-level CLEAR-LeWM success predicates. "Hold" gives the number of consecutive primitive steps satisfying the predicate. Thresholds use the respective environment coordinate systems.

Task	Moderate	Strict
PushT	Joint agent-and-block position error < 20 native pixel units; block angle error $< 20°$; hold one step.	Block position error < 10; block angle error $< 10°$; hold three steps.
Cube	Position error ≤ 0.04 m; no orientation constraint; hold one step.	Position error ≤ 0.03 m and cube-symmetry-aware orientation error $\leq 15°$; hold three steps.
Reacher	Maximum joint error < 0.05 rad; periodic shoulder and unwrapped elbow error; hold one step.	End-effector distance ≤ 0.01 m; hold two steps.
TwoRoom	Goal distance < 16 native units; hold one step.	Goal distance < 8, valid collision-free route, doorway crossing, and correct goal room; hold one step.

Official. Official retains the upstream evaluator's row-uniform sampling, initially successful pairs, and first-hit success semantics. Its base thresholds are 20 position units and 20° for PushT, 0.04 m without an orientation condition for Cube, 0.05 rad using raw joint errors for Reacher, and 16 native units for TwoRoom. The configured goal offset and evaluation budget are 25 and 50, respectively. The evaluator retains its historical action-loop compatibility convention. Official and CLEAR results are reported separately.

C.3 Aggregation and Pairing

For each training seed and task, evaluation uses seeds 0, 1, and 42, with 100 episodes per evaluation seed. Let $s_{k,t,e}$ denote the corresponding success percentage. We compute

$$s_{k,t} = \frac{1}{3}\sum_{e=1}^{3} s_{k,t,e}, \qquad m_k = \frac{1}{4}\sum_{t=1}^{4} s_{k,t},$$

and report

$$\bar{m} = \frac{1}{3}\sum_{k=1}^{3} m_k, \qquad \mathrm{SD}(m) = \sqrt{\frac{1}{2}\sum_{k=1}^{3}(m_k - \bar{m})^2}.$$

Task-level standard deviations use the same training-seed aggregation. Evaluation episodes are not treated as independent training replicates.

For paired comparisons, differences are computed before aggregation:

$$\Delta_k = m_k^A - m_k^B.$$

We report the mean and sample standard deviation of Δ_k. Rounding is applied only after aggregation.

A complete four-task result contains all 36 task–training-seed–evaluation-seed records. Checkpoint identity, training epoch, control interface, and success criterion are checked before aggregation.

表 8：任务级 CLEAR-LeWM 成功判定。"Hold"表示连续满足判定的原始步数。阈值使用各环境坐标系。

Task	Moderate	Strict
PushT	智能体与块的联合位置误差 < 20 原生像素单位；块角度误差 $< 20°$；保持一步。	块位置误差 < 10；块角度误差 $< 10°$；保持三步。
Cube	位置误差 ≤ 0.04 米；不要求姿态约束；保持一步。	位置误差 ≤ 0.03 米及考虑立方体对称性的姿态误差 $\leq 15°$；保持三步。
Reacher	最大关节误差 < 0.05 弧度；周期性肩关节与未包裹肘关节误差；保持一步。	末端执行器距离 ≤ 0.01 米；保持两步。
TwoRoom	目标距离 < 16 原生单位；保持一步。	目标距离 < 8，有效无碰撞路径，穿过门口且目标房间正确；保持一步。

官方协议。官方保留上游评估器的行均匀采样、初始已成功配对及首次命中成功语义。基础阈值为 PushT 位置 20 单位及 20°，Cube 为 0.04 米且不要求姿态，Reacher 以原始关节误差判定 0.05 弧度，TwoRoom 为 16 原生单位。目标偏移和评估预算分别为 25、50。评估器保留历史动作循环兼容约定。官方与 CLEAR 结果分别报告。

C.3 汇总与配对

每个训练种子和任务使用评估种子 0、1、42，每种子 100 回合。设 $s_{k,t,e}$ 为对应成功百分比，计算

$$s_{k,t} = \frac{1}{3}\sum_{e=1}^{3} s_{k,t,e}, \qquad m_k = \frac{1}{4}\sum_{t=1}^{4} s_{k,t},$$

并报告

$$\bar{m} = \frac{1}{3}\sum_{k=1}^{3} m_k, \qquad \mathrm{SD}(m) = \sqrt{\frac{1}{2}\sum_{k=1}^{3}(m_k - \bar{m})^2}.$$

任务级标准差使用相同训练种子汇总；评估回合不视为独立训练重复。

配对比较先计算差异，再汇总：

$$\Delta_k = m_k^A - m_k^B.$$

报告 Δ_k 的均值和样本标准差；仅在汇总后舍入。

完整四任务结果包含全部 36 条任务–训练种子–评估种子记录。汇总前检查检查点身份、训练轮次、控制接口与成功标准。

Table 9: Paired 100-step component swaps from identical initialization, input order, random state, and optimizer. Seeds: 3072, 3073, 3074. The first divergence is a numerical trajectory event, not itself a collapse event.

Visual encoder	Latent predictor	Bitwise match	First divergence	Max. parameter gap at step 100
MATH	Flash	3/3	Not observed	0
Flash	MATH	0/3	Step 1, 3/3	.0556–.0671

D Collapse Phenomenology and Path Diagnosis

We proceed from the observed failure through controlled component and constraint-placement tests to an operational definition and a path-capacity explanation. Fixed probes identify where variation is lost; a paired ten-seed study estimates relative path-degradation events.

D.1 Observed failure: endpoint constraints do not certify the path

In one jointly trained four-task INTACT–VISReg run, Official Direct success decreases from 55.67% at E1 to 24.75% at E5. At the same checkpoints, $r_{\text{eff}}(H)$ decreases from 13.74 to 6.64, while the mean inter-sample cosine c_H increases from .910 to .997. The control failure coincides with increasingly indistinguishable encoder outputs; endpoint regularization loss alone does not establish representation utility.

The layer-wise audit fixes 256 trajectories per task and eight frames per trajectory. At matched observation indices we probe successive visual-encoder layers, $H = h_{\text{cls}}$, projected Z, and predicted $\hat{Z}$, centering within each task before comparison. Across 37 runs, the cross-run correlation between effective ranks of H and Z is .993. In all 13 observed severe low-rank runs, the earliest abnormal boundary is already in the visual encoder; we find no case in which H remains healthy while Z or the predictor alone first becomes low-rank.

D.2 Controlled localization of the first numerical divergence

Component swap. We clone each initialization, input sequence, random state, and optimizer, changing only the attention backend of the visual encoder or latent predictor. Keeping predictor Flash while switching the encoder to MATH gives bitwise agreement for all three seeds over 100 steps; the reverse swap diverges on step one in all three. In one seed, the first detected difference is an encoder backward-gradient perturbation of 1.49×10^{-8}, which is amplified over subsequent parameter updates. This locates the first measurable numerical perturbation; long-run training separately establishes the representation outcome.

Visual-encoder layer position. With predictor MATH and endpoint SIGReg weight .03, we vary which encoder layers use MATH. In independent run A, protecting the final six layers preserves substantially more variation in $H/Z/\hat{Z}$ than protecting the first six. In run B, MATH in layers 10–12 approaches the all-MATH reference, whereas changing only layers 7–9 does not recover it. Late attention is thus a sensitive location under this architecture and protocol.

Constraint location. Taking the MATH encoder as a healthy reference, we compare endpoint-only SIGReg, a fixed variance/spectral barrier on H alone, and their direct sum with Flash encoder and MATH predictor. The H-only barrier raises $r_{\text{eff}}(H)$ to 27.90, but $Z/\hat{Z}$ reach only $9.67/2.96$; SIGReg(Z) alone gives $4.65/7.65/7.74$. Directly stacking the two constraints does not automatically repair the entire path. Neither *expanding H alone* nor *matching the marginal distribution of Z alone* guarantees a usable readout.

D.3 Numerical trigger versus representational bottleneck

The component swap identifies the first perturbation in visual-encoder Flash-SDPA backward. Shared four-task optimization, BF16, and a relatively large learning rate can amplify this small

表 9：相同初始化、输入顺序、随机状态和优化器出发的配对 100 步组件交换。种子为 3072、3073、3074。首次分歧是数值轨迹事件，本身不是坍塌事件。

视觉编码器	潜变量预测器	逐位匹配	首次分歧	第 100 步的最大参数差距
MATH	Flash	3/3	未观测到	0
Flash	MATH	0/3	第 1 步，3/3	.0556–.0671

D 坍塌现象与路径诊断

我们从观测失效出发，经由受控组件与约束位置测试，给出操作性定义与路径容量解释。固定探针定位变化丢失处；配对十种子研究估计相对路径退化事件。

D.1 观测失效：端点约束不能保证路径健康

一次联合四任务 INTACT–VISReg 运行中，官方 Direct 成功率从 E1 的 55.67% 降至 E5 的 24.75%。相同检查点的 $r_{\text{eff}}(H)$ 从 13.74 降至 6.64，平均样本间余弦 c_H 从 .910 升至 .997。控制失效与编码器输出越来越不可区分同时出现；仅端点正则化损失不能证明表征有用。

逐层审计固定每任务 256 条轨迹、每轨迹八帧。在匹配观测索引处，依次探测视觉编码器各层、$H = h_{\text{cls}}$、投影后的 Z 和预测的 $\hat{Z}$；比较前在各任务内中心化。37 次运行中，H 与 Z 有效秩的跨运行相关系数为 .993。13 次观测到的严重低秩运行中，最早异常边界均已在视觉编码器内；未发现 H 仍健康而 Z 或预测器单独首先变低秩的情况。

D.2 首次数值分歧的受控定位

组件交换。我们克隆每个初始化、输入序列、随机状态和优化器，只修改视觉编码器或潜变量预测器的注意力后端。保持预测器 Flash、将编码器改为 MATH，三个种子在 100 步内均逐位一致；反向交换则全部在第一步分歧。一个种子中最先检测到的差异为编码器反向梯度扰动 1.49×10^{-8}，随后参数更新将其放大。这定位了第一个可测数值扰动；长期训练另行确定表征结果。

视觉编码器层位置。在预测器 MATH、端点 SIGReg 权重 .03 下，改变使用 MATH 的编码器层。独立运行 A 中，保护最后六层比保护前六层在 $H/Z/\hat{Z}$ 中保留更多变化。运行 B 中，第 10–12 层使用 MATH 接近全 MATH 参考，而只改变第 7–9 层不能恢复。因此，该架构和协议下的后期注意力是敏感位置。

约束位置。以 MATH 编码器为健康参考，我们比较仅端点 SIGReg、仅作用于 H 的固定方差/谱屏障，以及在 Flash 编码器、MATH 预测器上的两者直接相加。仅 H 屏障使 $r_{\text{eff}}(H)$ 达到 27.90，但 $Z/\hat{Z}$ 仅达到 $9.67/2.96$；仅 SIGReg(Z) 给出 $4.65/7.65/7.74$。直接叠加约束并不会自动修复整个路径。单独展开 H 或单独匹配 Z 的边缘分布，都不保证有用的读出。

D.3 数值触发与表征瓶颈

组件交换将首次扰动定位到视觉编码器 Flash-SDPA 反向传播。共享四任务优化、BF16 和较大学习率可将这一微小

Table 10: Encoder-layer backend ablation. Runs A and B are independently trained four-task INTACT models with seed 3072, SIGReg(Z) weight .03, MATH predictor, and the same fixed PushT probe.

Run	Encoder MATH layers	$r_{\text{eff}}(H)$	$r_{\text{eff}}(Z)$	$r_{\text{eff}}(\hat{Z})$
A	All 12	21.13	50.18	49.14
A	None (all Flash)	4.65	7.65	7.74
A	1–6	10.58	20.72	23.00
A	7–12	20.62	51.22	51.32
B	All 12	26.84	66.63	64.78
B	None (all Flash)	13.80	14.86	4.73
B	7–9	7.53	12.08	2.64
B	10–12	27.09	64.61	64.39

Table 11: Fixed-threshold H barrier and endpoint SIGReg placement. Four-task INTACT, seed 3072, E2 fixed PushT probe; predictor MATH throughout. The H barrier penalizes mean standard deviation below .10 or leading eigenvalue share above .35.

Encoder	Constraint	$r_{\text{eff}}(H)$	$r_{\text{eff}}(Z)$	$r_{\text{eff}}(\hat{Z})$
MATH	SIGReg(Z) .03	21.13	50.18	49.14
Flash	SIGReg(Z) .03	4.65	7.65	7.74
Flash	H barrier .01, no Z constraint	27.90	9.67	2.96
Flash	H barrier .01 + SIGReg(Z) .03	2.12	6.14	3.33

difference into distinct non-convex training trajectories. Predictor Flash alone remains bitwise reproducible. Conversely, enabling deterministic algorithms while retaining Flash still yields a low-rank phase in a single-seed run (mean four-task E1 rank 7.36, versus 25.42 with MATH). Short-run divergence and long-run rank loss together connect a numerical trigger to representation degradation.

Once the encoder has erased distinctions between samples, a deterministic downstream map cannot restore them from identical inputs. Moreover, an endpoint gradient reaches H through the projector Jacobian,

$$\nabla_H \mathcal{D}(Z) = J_P(H)^\top \nabla_Z \mathcal{D}(Z).$$

An endpoint objective may therefore produce a gradient at Z without sufficiently exciting or transmitting useful differences in H. Conversely, expanding H alone does not ensure a useful endpoint or predictive readout.

D.4 Operational definition of collapse and degradation

For a fixed probe $U \in \mathbb{R}^{N\times d_U}$, define the centered sample covariance $\Sigma_U = (PU)^\top(PU)/(N-1)$. An exactly constant representation has $\operatorname{rank}(\Sigma_U) = 0$. Training more often produces a concentrated spectrum, nearly vanishing scale, or deteriorated predictive readout, rather than exact rank zero. We therefore report effective rank, mean inter-sample cosine, scale, and prediction error. The 13 severe low-rank runs above were screened by effective rank below half the healthy same-epoch reference.

For the fixed-probe path dimension, let

$$S_{\text{path}} = \min\left\{\frac{r_{\text{eff}}(H)}{\min(N-1, d_h)}, \frac{r_{\text{eff}}(Z)}{\min(N-1, d_z)}\right\}.$$

This is an external diagnostic, distinct from the training-time supply k. In the formal paired ten-seed audit, each candidate is compared with a same-seed, same-epoch MATH reference. We bootstrap trajectories 10,000 times and record a *relative path-degradation event* only when the upper endpoint of the one-sided 95% confidence interval of $\log(S^{\text{cand}}_{\text{path}}/S^{\text{ref}}_{\text{path}})$ is below zero. This event is distinct from exact rank-zero collapse and task success loss.

表 10：编码器层后端消融。A、B 为独立训练的四任务 INTACT，种子 3072，SIGReg(Z) 权重 .03，MATH 预测器，使用相同固定 PushT 探针。

Run	编码器 MATH 层	$r_{\text{eff}}(H)$	$r_{\text{eff}}(Z)$	$r_{\text{eff}}(\hat{Z})$
A	全部 12 层	21.13	50.18	49.14
A	无（全部 Flash）	4.65	7.65	7.74
A	1–6	10.58	20.72	23.00
A	7–12	20.62	51.22	51.32
B	全部 12 层	26.84	66.63	64.78
B	无（全部 Flash）	13.80	14.86	4.73
B	7–9	7.53	12.08	2.64
B	10–12	27.09	64.61	64.39

表 11：固定阈值 H 屏障与端点 SIGReg 位置。四任务 INTACT，种子 3072，E2 固定 PushT 探针；预测器始终 MATH。H 屏障惩罚低于 .10 的平均标准差或高于 .35 的首特征值占比。

编码器	约束	$r_{\text{eff}}(H)$	$r_{\text{eff}}(Z)$	$r_{\text{eff}}(\hat{Z})$
MATH	SIGReg(Z) .03	21.13	50.18	49.14
Flash	SIGReg(Z) .03	4.65	7.65	7.74
Flash	H 屏障 .01 ，无 Z 约束	27.90	9.67	2.96
Flash	H 屏障 .01 +SIGReg(Z) .03	2.12	6.14	3.33

差异转化为不同非凸训练轨迹。仅预测器 Flash 仍可逐位复现。相反，保留 Flash 而启用确定性算法时，单种子运行仍进入低秩阶段（四任务 E1 平均秩 7.36，MATH 为 25.42）。短期分歧与长期秩下降共同将数值触发与表征退化联系起来。

一旦编码器抹去样本差异，确定性下游映射就无法从相同输入恢复它们。此外，端点梯度通过投影器雅可比到达 H，

$$\nabla_H \mathcal{D}(Z) = J_P(H)^\top \nabla_Z \mathcal{D}(Z).$$

因此，端点目标可能在 Z 产生梯度，却不足以激发或传输 H 的有用差异。相反，单独展开 H 不保证有用的端点或预测读出。

D.4 坍塌与退化的操作性定义

对固定探针 $U \in \mathbb{R}^{N\times d_U}$，定义中心化样本协方差 $\Sigma_U = (PU)^\top(PU)/(N-1)$。严格常量表征满足 $\operatorname{rank}(\Sigma_U) = 0$。训练更常产生集中的谱、近乎消失的尺度或退化的预测读出，而非精确零秩。因此我们报告有效秩、平均样本间余弦、尺度与预测误差。上述 13 次严重低秩运行按有效秩低于同轮健康参考一半筛选。

对于固定探针路径维度，设

$$S_{\text{path}} = \min\left\{\frac{r_{\text{eff}}(H)}{\min(N-1, d_h)}, \frac{r_{\text{eff}}(Z)}{\min(N-1, d_z)}\right\}.$$

这是外部诊断，区别于训练时供给 k。正式配对十种子审计中，各候选与相同种子、相同轮次的 MATH 参考比较。我们对轨迹 bootstrap 10,000 次，仅当 $\log(S^{\text{cand}}_{\text{path}}/S^{\text{ref}}_{\text{path}})$ 单侧 95% 置信区间上界小于零时，记录相对路径退化事件。该事件不同于精确零秩坍塌或任务成功率下降。

Table 12: Paired ten-training-seed E1 four-task LeWM audit. A degradation event is defined relative to the same-seed MATH reference by the fixed-probe bootstrap criterion above; it is not an exact rank-zero event. Success is Official Pure-CEM macro SR (%).

Training condition	Relative events	S_{path}	Prediction loss	Official SR
MATH + SIGReg (reference)	–	.031	.7767	32.02
Flash, no endpoint constraint	$10/10$	.005	27.0896	24.11
Flash + SIGReg	$5/10$	.032	1.8586	31.92
Flash + SIGReg + excitation-only SADL	$0/10$	.370	.0174	66.67
Flash + SIGReg + full SADL	$0/10$	.373	.0173	64.97

D.5 Matching endpoint demand to path supply

The central remedy is not to force latents onto a fixed manifold, but to ensure that sample excitation in the hidden states and transmission across the projection interface can realize the endpoint's requested representation demand. The endpoint function $\mathcal{D}(Z)$ contributes demand q. The path must both preserve differences in H (excitation capacity C_{exc}) and transmit them to Z (transmission capacity C_{tr}), giving bottleneck supply $k = \min(C_{\text{exc}}, C_{\text{tr}})$. If $q > k$, the request remains unsupported; if $k \geq q$, an auxiliary bridge should not continue forcing H to expand. Excess supply is not itself noise: task readout determines which directions are relevant.

SADL implements this principle with a bridge active only at a deficit,

$$\mathcal{S} = \left[\log(q+\epsilon) - \log(k+\epsilon)\right]_+^2.$$

Unlike an endpoint-only objective, it addresses the upstream path; unlike a fixed H-expansion term, it switches off when supply suffices. In an independent paired ten-seed four-task LeWM audit, Flash+SIGReg has $5/10$ relative degradation events, versus $0/10$ after adding full SADL. S_{path} rises from .032 to .373, prediction loss falls from 1.8586 to .0173, and Official Pure-CEM macro success rises from 31.92% to 64.97%.

Excitation alone also reaches $0/10$ in this training regime, consistent with an excitation-limited path; a separate intervention that makes transmission limiting is given in Section F.3. The paired audit shows that SADL reduces relative path degradation under Flash from $5/10$ to $0/10$, retaining the faster and lower-memory fused attention implementation rather than reverting to MATH (FlashAttention). The bridge adds no model parameters; per-step overhead is reported in Table 35. **SADL also accelerates representation formation and improves E1 path dimension, predictive quality, and control success.**

D.6 SADL Repairs Near-Collapsed Checkpoints

We next ask whether SADL can repair an already degraded path, rather than only prevent degradation from the beginning of training. Starting from fixed near-collapsed checkpoints, we resume training with either the original endpoint objective or the same objective augmented by SADL. Each reported control value averages four tasks and evaluation seeds $\{0, 1, 42\}$, with 100 episodes per task–seed cell. LeWM uses Official Pure-CEM (300×30, $H = R_H = 5$), whereas INTACT uses Official Direct control ($H = R_H = 5$). Within each row, the checkpoint, continuation protocol, training host, and evaluation seeds are matched; rows from different training hosts are not pooled.

The native bridge rescues three of four LeWM checkpoints only weakly or strongly, but fails on seed 42. For that same LeWM state, moderate increases to .03 or .10 make the macro improvement positive, whereas larger or intermediate doses can again fail. INTACT is similarly checkpoint-dependent: seed 0 is not robustly rescued, while seeds 3072 and 42 improve substantially. For seed 42, the Pro6000D matched group is strongest at the native weight and every larger tested dose weakens recovery. The H200 replication confirms a non-monotone response—.15 rebounds above native, while .10 and .20 largely erase the gain—so absolute scores are not combined across hosts.

SADL therefore has genuine post-collapse repair capability, but recovery depends on the checkpoint state and bridge strength; increasing λ_S is not monotonically beneficial. This is more informative than a binary "can rescue" claim: a deficit-triggered correction can reopen a dam-

表 12：配对十训练种子 E1 四任务 LeWM 审计。退化事件由上述固定探针 bootstrap 标准相对于同种子 MATH 参考定义，不是精确零秩事件。成功率为官方 Pure-CEM 宏平均（%）。

训练条件	相对事件	S_{path}	预测损失	官方成功率
MATH + SIGReg（参考）	–	.031	.7767	32.02
Flash，无端点约束	$10/10$	.005	27.0896	24.11
Flash + SIGReg	$5/10$	.032	1.8586	31.92
Flash + SIGReg + 仅激发 SADL	$0/10$	.370	.0174	66.67
Flash + SIGReg + 完整 SADL	$0/10$	.373	.0173	64.97

D.5 将端点需求与路径供给匹配

核心补救不是把潜变量强制放到固定流形，而是保证隐状态的样本激发和跨投影接口传输能够实现端点请求的表征需求。端点函数 $\mathcal{D}(Z)$ 贡献需求 q。路径必须保留 H 的差异（激发容量 C_{exc}），并传输至 Z（传输容量 C_{tr}），得到瓶颈供给 $k = \min(C_{\text{exc}}, C_{\text{tr}})$。若 $q > k$，请求仍缺乏支持；若 $k \geq q$，辅助桥接不应继续强迫 H 展开。多余供给本身并非噪声：任务读出决定哪些方向相关。

SADL 通过仅在缺口处激活的桥接实现这一原则，

$$\mathcal{S} = \left[\log(q+\epsilon) - \log(k+\epsilon)\right]_+^2.$$

与仅端点目标不同，它处理上游路径；与固定 H 展开项不同，供给足够时它关闭。在独立配对十种子四任务 LeWM 审计中，Flash+SIGReg 有 $5/10$ 次相对退化事件，加入完整 SADL 后为 $0/10$。S_{path} 从 .032 升至 .373，预测损失从 1.8586 降至 .0173，官方 Pure-CEM 宏平均成功率从 31.92% 升至 64.97%。

在该训练设置中，仅激发也达到 $0/10$，符合路径受激发限制的解释；令传输成为限制因素的独立干预见 F.3 节。配对审计显示，SADL 将 Flash 下相对路径退化从 $5/10$ 降至 $0/10$，保留更快、显存更低的融合注意力实现，而不退回 MATH（FlashAttention）。桥接不增加模型参数；单步开销见表 35。SADL 还加速表征形成，改善 E1 路径维度、预测质量和控制成功率。

D.6 SADL 修复近坍塌检查点

接下来，我们检验 SADL 能否修复已经退化的路径，而非仅从训练开始防止退化。从固定的近坍塌检查点出发，分别以原端点目标或加入 SADL 的同一目标恢复训练。每个控制值对四个任务和评估种子 $\{0, 1, 42\}$ 求平均，每个任务–种子单元 100 回合。LeWM 使用官方 Pure-CEM（300×30, $H = R_H = 5$），INTACT 使用官方 Direct 控制（$H = R_H = 5$）。每行内部检查点、继续训练协议、训练主机和评估种子匹配；不同训练主机的行不合并。

原生桥接对四个 LeWM 检查点中的三个产生较弱或较强修复，但在种子 42 上失败。对相同 LeWM 状态，将权重适度增至 .03 或 .10 使宏平均改善转正，而更大或中间剂量仍可能失败。INTACT 同样依赖检查点：种子 0 无法稳健修复，3072 和 42 则明显改善。对于种子 42，Pro6000D 匹配组在原生权重下最强，测试的所有更大剂量均削弱恢复。H200 复现确认非单调响应：.15 反弹至高于原生，而 .10、.20 大幅消除增益。因此不跨主机合并绝对分数。

因此 SADL 具有坍塌后修复能力，但恢复取决于检查点状态和桥接强度；增加 λ_S 并非单调有益。这比二元的"能否修复"说法更有信息：缺口触发校正可以重新展开受损

Table 13: Post-collapse rescue and bridge-strength sensitivity. Panel (a) describes the two seed-42 source checkpoints before continuation using task-averaged fixed probes; r_H, r_Z are effective ranks, c_H, c_Z are mean off-diagonal cosines, and $\mathcal{L}_{\text{pred}}$ is latent prediction loss. Both states retain nonzero numerical rank and are therefore termed *near-collapsed*, not exact rank-zero. Panel (b) reports matched Official macro success rates. Native denotes $\lambda_S = .01$. Exploratory doses are reported to expose non-monotone rescue rather than to select a best result.

(a) Pre-rescue checkpoint diagnostics

Family	Seed / state	r_H	c_H	r_Z	c_Z	$\mathcal{L}_{\text{pred}}$	Diagnosis
LeWM	42 / C2 E2	15.01	.972	33.65	.128	.346	low-rank, highly aligned H
INTACT	42 / E1	18.14	.945	23.62	.599	1.126	low-rank path and poor readout

(b) Matched continuation and dose response

Family	Train seed / start	Continuation group	λ_S	Control SR	SADL SR	Δ (pp)	Outcome
LeWM	21 / E2→E4	matched native	.01	25.250	46.917	+21.667	clear rescue
LeWM	7 / E2→E4	matched native	.01	61.000	72.167	+11.167	clear rescue
LeWM	3072 / E3→E5	matched native	.01	21.500	23.083	+1.583	small positive
LeWM	42 / E2→E4	matched native	.01	24.167	22.500	-1.667	not rescued
LeWM	42 / E2→E4	same source	.03	24.167	28.333	+4.167	rescued
LeWM	42 / E2→E4	post-hoc dose	.05	24.167	23.167	-1.000	not rescued
LeWM	42 / E2→E4	post-hoc dose	.07	24.167	25.667	+1.500	marginal
LeWM	42 / E2→E4	same source	.10	24.167	29.083	+4.917	rescued
LeWM	42 / E2→E4	same source	.15	24.167	21.917	-2.250	not rescued
LeWM	42 / E2→E4	same source	.20	24.167	24.167	+0.000	neutral
LeWM	42 / E2→E4	hardware replicate	.30	24.167	22.250	-1.917	not rescued
LeWM	42 / E2→E4	hardware replicate	1.00	24.167	20.250	-3.917	not rescued
INTACT	0 / E1→E2	matched native	.01	28.917	26.500	-2.417	not rescued
INTACT	3072 / E1→E2	matched native	.01	24.417	53.917	+29.500	clear rescue
INTACT	42 / E1→E2	Pro6000D native	.01	18.917	68.333	+49.417	clear rescue
INTACT	42 / E1→E2	Pro6000D matched	.03	18.917	39.500	+20.583	weakened
INTACT	42 / E1→E2	Pro6000D matched	.10	18.917	29.750	+10.833	weakened
INTACT	42 / E1→E2	Pro6000D matched	.15	18.917	27.000	+8.083	weakened
INTACT	42 / E1→E2	Pro6000D matched	.20	18.917	27.667	+8.750	weakened
INTACT	42 / E1→E2	H200 replicate	.01	23.083	69.083	+46.000	clear rescue
INTACT	42 / E1→E2	H200 replicate	.03	23.083	55.833	+32.750	weakened
INTACT	42 / E1→E2	H200 replicate	.10	23.083	28.917	+5.834	weakened
INTACT	42 / E1→E2	H200 replicate	.15	23.083	74.583	+51.500	rebound
INTACT	42 / E1→E2	H200 replicate	.20	23.083	27.583	+4.500	weakened

aged representation path, yet the intervention must remain commensurate with the path's attainable capacity.

表 13：坍塌后修复与桥接强度敏感性。(a) 以任务平均固定探针描述继续训练前的两个种子 42 源检查点；r_H、r_Z 为有效秩，c_H、c_Z 为平均非对角余弦，$\mathcal{L}_{\text{pred}}$ 为潜变量预测损失。两状态仍保留非零数值秩，因此称近坍塌，而非精确零秩。(b) 报告匹配官方宏平均成功率。原生表示 $\lambda_S = .01$。报告探索剂量是为展示非单调修复，而非挑选最佳结果。

(a) 修复前检查点诊断

族	种子 / 状态	r_H	c_H	r_Z	c_Z	$\mathcal{L}_{\text{pred}}$	诊断
LeWM	42 / C2 E2	15.01	.972	33.65	.128	.346	低秩、高度对齐的 H
INTACT	42 / E1	18.14	.945	23.62	.599	1.126	低秩路径与较差读出

(b) 匹配继续训练与剂量响应

族	训练种子 / 起点	继续训练组	λ_S	控制成功率	SADL 成功率	Δ（百分点）	结果
LeWM	21 / E2→E4	匹配原生	.01	25.250	46.917	+21.667	明显修复
LeWM	7 / E2→E4	匹配原生	.01	61.000	72.167	+11.167	明显修复
LeWM	3072 / E3→E5	匹配原生	.01	21.500	23.083	+1.583	小幅正增益
LeWM	42 / E2→E4	匹配原生	.01	24.167	22.500	-1.667	未修复
LeWM	42 / E2→E4	相同源	.03	24.167	28.333	+4.167	已修复
LeWM	42 / E2→E4	事后剂量	.05	24.167	23.167	-1.000	未修复
LeWM	42 / E2→E4	事后剂量	.07	24.167	25.667	+1.500	边际
LeWM	42 / E2→E4	相同源	.10	24.167	29.083	+4.917	已修复
LeWM	42 / E2→E4	相同源	.15	24.167	21.917	-2.250	未修复
LeWM	42 / E2→E4	相同源	.20	24.167	24.167	+0.000	中性
LeWM	42 / E2→E4	硬件重复实验	.30	24.167	22.250	-1.917	未修复
LeWM	42 / E2→E4	硬件重复实验	1.00	24.167	20.250	-3.917	未修复
INTACT	0 / E1→E2	匹配原生	.01	28.917	26.500	-2.417	未修复
INTACT	3072 / E1→E2	匹配原生	.01	24.417	53.917	+29.500	明显修复
INTACT	42 / E1→E2	Pro6000D 原生	.01	18.917	68.333	+49.417	明显修复
INTACT	42 / E1→E2	Pro6000D 匹配	.03	18.917	39.500	+20.583	减弱
INTACT	42 / E1→E2	Pro6000D 匹配	.10	18.917	29.750	+10.833	减弱
INTACT	42 / E1→E2	Pro6000D 匹配	.15	18.917	27.000	+8.083	减弱
INTACT	42 / E1→E2	Pro6000D 匹配	.20	18.917	27.667	+8.750	减弱
INTACT	42 / E1→E2	H200 重复实验	.01	23.083	69.083	+46.000	明显修复
INTACT	42 / E1→E2	H200 重复实验	.03	23.083	55.833	+32.750	减弱
INTACT	42 / E1→E2	H200 重复实验	.10	23.083	28.917	+5.834	减弱
INTACT	42 / E1→E2	H200 重复实验	.15	23.083	74.583	+51.500	反弹
INTACT	42 / E1→E2	H200 重复实验	.20	23.083	27.583	+4.500	减弱

的表征路径，但干预必须与路径可达到的容量。

E ADDITIONAL CONTROL RESULTS

We separate Official from CLEAR-LeWM v0.8 so that success rates are compared only within the same evaluation protocol. The tables report single-task and jointly trained four-task results for LeWM and INTACT under SIGReg, VISReg, and QQWorld. Each task value is the mean ± sample SD across three training seeds after averaging the complete three-evaluation-seed triplet; the macro is first averaged over the four tasks within each training seed and then summarized across training seeds. E1 and E5 are equal-budget comparisons, while endpoint-only E10 is a longer-budget reference. The additional tables here retain the original backend protocol: endpoint-only uses a MATH encoder, +SADL uses a Flash encoder, and both use MATH predictors. The joint E1 Flash-matched comparison appears in Table 2(b).

E.1 OFFICIAL

Table 14: Single-task LeWM Official success (%). All rows are evaluated after one epoch. LeWM uses Pure-CEM with $H = 5/RH = 5$. Endpoint-only uses a MATH encoder; +SADL uses a Flash encoder; both predictors use MATH. Values are mean ± sample SD over three training seeds.

Endpoint	Method	Epoch	PushT	Cube	Reacher	TwoRoom	Macro
SIGReg	Endpoint-only	E1	67.00 ± 13.02	61.78 ± 1.64	26.44 ± 12.70	25.78 ± 8.15	45.25 ± 6.50
SIGReg	+SADL	E1	65.22 ± 3.86	58.89 ± 1.84	79.44 ± 1.71	85.22 ± 4.82	**72.19±0.97**
VISReg	Endpoint-only	E1	74.11 ± 4.62	63.22 ± 6.45	32.56 ± 2.71	52.78 ± 23.40	55.67 ± 5.46
VISReg	+SADL	E1	76.22 ± 2.17	58.22 ± 0.51	80.89 ± 2.04	89.89 ± 2.55	**76.31±1.23**
QQWorld	Endpoint-only	E1	72.22 ± 5.32	64.56 ± 4.86	34.33 ± 16.50	33.67 ± 5.36	51.19 ± 6.98
QQWorld	+SADL	E1	71.78 ± 2.17	61.00 ± 4.63	81.44 ± 1.17	86.89 ± 1.26	**75.28±1.73**

Table 15: Single-task INTACT Official success (%). All rows are evaluated after one epoch. INTACT uses search-free Direct control. Endpoint-only uses a MATH encoder; +SADL uses a Flash encoder; both predictors use MATH. Values are mean ± sample SD over three training seeds.

Endpoint	Method	Epoch	PushT	Cube	Reacher	TwoRoom	Macro
SIGReg	Endpoint-only	E1	80.89 ± 4.35	97.67 ± 2.03	28.78 ± 22.52	27.22 ± 8.98	58.64 ± 8.14
SIGReg	+SADL	E1	85.67 ± 1.45	100.00 ± 0.00	96.89 ± 2.14	98.78 ± 1.07	**95.33±0.44**
VISReg	Endpoint-only	E1	84.22 ± 2.50	97.67 ± 1.15	51.89 ± 40.50	54.11 ± 14.99	71.97 ± 12.79
VISReg	+SADL	E1	86.89 ± 0.19	100.00 ± 0.00	97.00 ± 0.58	99.11 ± 0.19	**95.75±0.08**
QQWorld	Endpoint-only	E1	82.67 ± 1.20	97.22 ± 3.10	37.33 ± 30.55	29.00 ± 6.17	61.56 ± 8.61
QQWorld	+SADL	E1	87.11 ± 1.39	100.00 ± 0.00	95.22 ± 1.58	99.56 ± 0.19	**95.47±0.21**

E 补充控制结果

我们区分官方协议与 CLEAR-LeWM v0.8，只在相同评估协议内比较成功率。表格报告 SIGReg、VISReg、QQWorld 下 LeWM 和 INTACT 的单任务及联合四任务结果。每个任务数值先对完整三个评估种子求平均，再报告三个训练种子的均值 ± 样本标准差；宏平均先在每个训练种子内对四任务取平均，再跨训练种子汇总。E1 与 E5 是相同预算比较；仅端点 E10 为较长预算参考。这里补充表保留原后端协议：仅端点使用 MATH 编码器，+SADL 使用 Flash 编码器，预测器均为 MATH。联合 E1 Flash 匹配比较见表 2(b)。

E.1 官方

表 14：单任务 LeWM 官方成功率（%）。各行均在训练一轮后评估。LeWM 使用 Pure-CEM，配置 $H = 5/RH = 5$。仅端点用 MATH 编码器，+SADL 用 Flash 编码器；预测器均为 MATH。数值为三个训练种子的均值 ± 样本标准差。

端点	方法	轮次	PushT	Cube	Reacher	TwoRoom	宏平均
SIGReg	仅端点	E1	67.00 ± 13.02	61.78 ± 1.64	26.44 ± 12.70	25.78 ± 8.15	45.25 ± 6.50
SIGReg	+SADL	E1	65.22 ± 3.86	58.89 ± 1.84	79.44 ± 1.71	85.22 ± 4.82	**72.19±0.97**
VISReg	仅端点	E1	74.11 ± 4.62	63.22 ± 6.45	32.56 ± 2.71	52.78 ± 23.40	55.67 ± 5.46
VISReg	+SADL	E1	76.22 ± 2.17	58.22 ± 0.51	80.89 ± 2.04	89.89 ± 2.55	**76.31±1.23**
QQWorld	仅端点	E1	72.22 ± 5.32	64.56 ± 4.86	34.33 ± 16.50	33.67 ± 5.36	51.19 ± 6.98
QQWorld	+SADL	E1	71.78 ± 2.17	61.00 ± 4.63	81.44 ± 1.17	86.89 ± 1.26	**75.28±1.73**

表 15：单任务 INTACT 官方成功率（%）。各行均训练一轮后评估。INTACT 使用无搜索 Direct 控制。仅端点用 MATH 编码器，+SADL 用 Flash 编码器；预测器均为 MATH。数值为三个训练种子的均值 ± 样本标准差。

端点	方法	轮次	PushT	Cube	Reacher	TwoRoom	宏平均
SIGReg	仅端点	E1	80.89 ± 4.35	97.67 ± 2.03	28.78 ± 22.52	27.22 ± 8.98	58.64 ± 8.14
SIGReg	+SADL	E1	85.67 ± 1.45	100.00 ± 0.00	96.89 ± 2.14	98.78 ± 1.07	**95.33±0.44**
VISReg	仅端点	E1	84.22 ± 2.50	97.67 ± 1.15	51.89 ± 40.50	54.11 ± 14.99	71.97 ± 12.79
VISReg	+SADL	E1	86.89 ± 0.19	100.00 ± 0.00	97.00 ± 0.58	99.11 ± 0.19	**95.75±0.08**
QQWorld	仅端点	E1	82.67 ± 1.20	97.22 ± 3.10	37.33 ± 30.55	29.00 ± 6.17	61.56 ± 8.61
QQWorld	+SADL	E1	87.11 ± 1.39	100.00 ± 0.00	95.22 ± 1.58	99.56 ± 0.19	**95.47±0.21**

Table 16: Joint four-task LeWM Official success (%). E1 and E5 compare equal training budgets; endpoint-only E10 is a longer-budget reference. LeWM uses Pure-CEM with $H = 5/RH = 5$. Endpoint-only uses a MATH encoder; +SADL uses a Flash encoder; both predictors use MATH. Values are mean ± sample SD over three training seeds.

Endpoint	Method	Epoch	PushT	Cube	Reacher	TwoRoom	Macro
SIGReg	Endpoint-only	E1	13.00 ± 1.86	53.11 ± 2.22	26.22 ± 4.11	35.67 ± 7.42	32.00 ± 2.52
SIGReg	+SADL	E1	28.78 ± 5.32	70.22 ± 3.02	79.89 ± 0.77	73.00 ± 1.45	**62.97±1.25**
SIGReg	Endpoint-only	E5	75.67 ± 3.71	67.89 ± 2.22	82.00 ± 1.53	35.11 ± 14.42	65.17 ± 2.21
SIGReg	+SADL	E5	67.00 ± 1.76	67.56 ± 2.50	83.33 ± 1.67	64.67 ± 1.76	**70.64±0.77**
SIGReg	Endpoint-only	E10	78.78 ± 3.02	67.44 ± 1.95	83.11 ± 1.71	44.78 ± 8.19	68.53 ± 0.75
VISReg	Endpoint-only	E1	53.00 ± 4.84	53.67 ± 0.33	31.56 ± 8.69	55.44 ± 12.62	48.42 ± 3.76
VISReg	+SADL	E1	53.56 ± 11.64	66.44 ± 2.41	78.44 ± 0.84	88.11 ± 1.07	**71.64±2.96**
VISReg	Endpoint-only	E5	83.78 ± 1.39	60.00 ± 1.20	83.56 ± 3.10	61.44 ± 7.20	72.19 ± 1.89
VISReg	+SADL	E5	81.44 ± 2.01	66.11 ± 2.55	84.56 ± 2.27	82.22 ± 1.64	**78.58±0.98**
VISReg	Endpoint-only	E10	86.33 ± 2.08	58.78 ± 1.26	85.89 ± 0.51	65.00 ± 4.33	74.00 ± 1.08
QQWorld	Endpoint-only	E1	16.67 ± 5.13	55.33 ± 4.10	28.44 ± 3.79	41.78 ± 5.85	35.56 ± 3.97
QQWorld	+SADL	E1	45.67 ± 11.15	67.44 ± 2.87	78.00 ± 3.53	82.56 ± 0.69	**68.42±4.19**
QQWorld	Endpoint-only	E5	80.89 ± 3.86	67.11 ± 4.50	85.00 ± 3.53	52.22 ± 12.29	71.31 ± 4.18
QQWorld	+SADL	E5	76.89 ± 2.46	68.00 ± 0.33	83.67 ± 0.88	71.89 ± 2.36	**75.11±1.34**
QQWorld	Endpoint-only	E10	86.11 ± 1.17	65.78 ± 2.50	82.78 ± 1.07	56.00 ± 4.51	72.67 ± 0.65

Table 17: Joint four-task INTACT Official success (%). E1 and E5 compare equal training budgets; endpoint-only E10 is a longer-budget reference. INTACT uses search-free Direct control. Endpoint-only uses a MATH encoder; +SADL uses a Flash encoder; both predictors use MATH. Values are mean ± sample SD over three training seeds.

Endpoint	Method	Epoch	PushT	Cube	Reacher	TwoRoom	Macro
SIGReg	Endpoint-only	E1	37.67 ± 9.29	71.67 ± 6.51	13.89 ± 4.48	55.00 ± 8.99	44.56 ± 2.80
SIGReg	+SADL	E1	85.22 ± 0.77	100.00 ± 0.00	97.11 ± 0.38	100.00 ± 0.00	**95.58±0.17**
SIGReg	Endpoint-only	E5	85.78 ± 3.83	99.89 ± 0.19	95.33 ± 0.58	66.00 ± 16.29	86.75 ± 3.20
SIGReg	+SADL	E5	90.78 ± 1.68	100.00 ± 0.00	98.22 ± 0.51	99.78 ± 0.19	**97.19±0.51**
SIGReg	Endpoint-only	E10	89.44 ± 3.79	100.00 ± 0.00	96.00 ± 0.58	80.89 ± 11.19	91.58 ± 2.01
VISReg	Endpoint-only	E1	54.67 ± 3.28	81.33 ± 9.85	34.78 ± 11.65	64.11 ± 6.35	58.72 ± 4.79
VISReg	+SADL	E1	83.78 ± 2.04	100.00 ± 0.00	97.78 ± 1.17	100.00 ± 0.00	**95.39±0.65**
VISReg	Endpoint-only	E5	89.67 ± 1.76	99.33 ± 0.33	97.67 ± 0.88	76.44 ± 1.95	90.78 ± 0.38
VISReg	+SADL	E5	90.89 ± 1.50	100.00 ± 0.00	99.33 ± 0.33	99.89 ± 0.19	**97.53±0.25**
VISReg	Endpoint-only	E10	91.00 ± 0.58	100.00 ± 0.00	97.11 ± 0.19	88.56 ± 5.98	94.17 ± 1.45
QQWorld	Endpoint-only	E1	36.89 ± 11.65	74.22 ± 2.50	21.00 ± 6.77	50.78 ± 9.50	45.72 ± 4.33
QQWorld	+SADL	E1	86.00 ± 0.88	100.00 ± 0.00	96.56 ± 0.51	100.00 ± 0.00	**95.64±0.32**
QQWorld	Endpoint-only	E5	89.11 ± 1.68	99.89 ± 0.19	97.44 ± 0.84	82.56 ± 14.20	92.25 ± 3.20
QQWorld	+SADL	E5	91.67 ± 0.88	100.00 ± 0.00	98.56 ± 0.19	100.00 ± 0.00	**97.56±0.17**
QQWorld	Endpoint-only	E10	89.56 ± 2.22	100.00 ± 0.00	97.22 ± 1.54	87.11 ± 16.28	93.47 ± 4.69

表 16：联合四任务 LeWM 官方成功率（%）。E1、E5 为等训练预算比较，仅端点 E10 为长预算参考。LeWM 用 Pure-CEM，配置 $H = 5/RH = 5$。仅端点用 MATH 编码器， +SADL 用 Flash 编码器；预测器均为 MATH。数值为三个训练种子的均值 ± 样本标准差。

端点	方法	轮次	PushT	Cube	Reacher	TwoRoom	宏平均
SIGReg	仅端点	E1	13.00 ± 1.86	53.11 ± 2.22	26.22 ± 4.11	35.67 ± 7.42	32.00 ± 2.52
SIGReg	+SADL	E1	28.78 ± 5.32	70.22 ± 3.02	79.89 ± 0.77	73.00 ± 1.45	**62.97±1.25**
SIGReg	仅端点	E5	75.67 ± 3.71	67.89 ± 2.22	82.00 ± 1.53	35.11 ± 14.42	65.17 ± 2.21
SIGReg	+SADL	E5	67.00 ± 1.76	67.56 ± 2.50	83.33 ± 1.67	64.67 ± 1.76	**70.64±0.77**
SIGReg	仅端点	E10	78.78 ± 3.02	67.44 ± 1.95	83.11 ± 1.71	44.78 ± 8.19	68.53 ± 0.75
VISReg	仅端点	E1	53.00 ± 4.84	53.67 ± 0.33	31.56 ± 8.69	55.44 ± 12.62	48.42 ± 3.76
VISReg	+SADL	E1	53.56 ± 11.64	66.44 ± 2.41	78.44 ± 0.84	88.11 ± 1.07	**71.64±2.96**
VISReg	仅端点	E5	83.78 ± 1.39	60.00 ± 1.20	83.56 ± 3.10	61.44 ± 7.20	72.19 ± 1.89
VISReg	+SADL	E5	81.44 ± 2.01	66.11 ± 2.55	84.56 ± 2.27	82.22 ± 1.64	**78.58±0.98**
VISReg	仅端点	E10	86.33 ± 2.08	58.78 ± 1.26	85.89 ± 0.51	65.00 ± 4.33	74.00 ± 1.08
QQWorld	仅端点	E1	16.67 ± 5.13	55.33 ± 4.10	28.44 ± 3.79	41.78 ± 5.85	35.56 ± 3.97
QQWorld	+SADL	E1	45.67 ± 11.15	67.44 ± 2.87	78.00 ± 3.53	82.56 ± 0.69	**68.42±4.19**
QQWorld	仅端点	E5	80.89 ± 3.86	67.11 ± 4.50	85.00 ± 3.53	52.22 ± 12.29	71.31 ± 4.18
QQWorld	+SADL	E5	76.89 ± 2.46	68.00 ± 0.33	83.67 ± 0.88	71.89 ± 2.36	**75.11±1.34**
QQWorld	仅端点	E10	86.11 ± 1.17	65.78 ± 2.50	82.78 ± 1.07	56.00 ± 4.51	72.67 ± 0.65

表 17：联合四任务 INTACT 官方成功率（%）。E1、E5 为等训练预算比较，仅端点 E10 为长预算参考。INTACT 用无搜索 Direct 控制。仅端点用 MATH 编码器， +SADL 用 Flash 编码器；预测器均为 MATH。数值为三个训练种子的均值 ± 样本标准差。

端点	方法	轮次	PushT	Cube	Reacher	TwoRoom	宏平均
SIGReg	仅端点	E1	37.67 ± 9.29	71.67 ± 6.51	13.89 ± 4.48	55.00 ± 8.99	44.56 ± 2.80
SIGReg	+SADL	E1	85.22 ± 0.77	100.00 ± 0.00	97.11 ± 0.38	100.00 ± 0.00	**95.58±0.17**
SIGReg	仅端点	E5	85.78 ± 3.83	99.89 ± 0.19	95.33 ± 0.58	66.00 ± 16.29	86.75 ± 3.20
SIGReg	+SADL	E5	90.78 ± 1.68	100.00 ± 0.00	98.22 ± 0.51	99.78 ± 0.19	**97.19±0.51**
SIGReg	仅端点	E10	89.44 ± 3.79	100.00 ± 0.00	96.00 ± 0.58	80.89 ± 11.19	91.58 ± 2.01
VISReg	仅端点	E1	54.67 ± 3.28	81.33 ± 9.85	34.78 ± 11.65	64.11 ± 6.35	58.72 ± 4.79
VISReg	+SADL	E1	83.78 ± 2.04	100.00 ± 0.00	97.78 ± 1.17	100.00 ± 0.00	**95.39±0.65**
VISReg	仅端点	E5	89.67 ± 1.76	99.33 ± 0.33	97.67 ± 0.88	76.44 ± 1.95	90.78 ± 0.38
VISReg	+SADL	E5	90.89 ± 1.50	100.00 ± 0.00	99.33 ± 0.33	99.89 ± 0.19	**97.53±0.25**
VISReg	仅端点	E10	91.00 ± 0.58	100.00 ± 0.00	97.11 ± 0.19	88.56 ± 5.98	94.17 ± 1.45
QQWorld	仅端点	E1	36.89 ± 11.65	74.22 ± 2.50	21.00 ± 6.77	50.78 ± 9.50	45.72 ± 4.33
QQWorld	+SADL	E1	86.00 ± 0.88	100.00 ± 0.00	96.56 ± 0.51	100.00 ± 0.00	**95.64±0.32**
QQWorld	仅端点	E5	89.11 ± 1.68	99.89 ± 0.19	97.44 ± 0.84	82.56 ± 14.20	92.25 ± 3.20
QQWorld	+SADL	E5	91.67 ± 0.88	100.00 ± 0.00	98.56 ± 0.19	100.00 ± 0.00	**97.56±0.17**
QQWorld	仅端点	E10	89.56 ± 2.22	100.00 ± 0.00	97.22 ± 1.54	87.11 ± 16.28	93.47 ± 4.69

E.2 CLEAR-LEWM V0.8

E.2.1 SINGLE-TASK MODERATE AND STRICT

Table 18: Single-task LeWM CLEAR-LeWM v0.8 success (%) at E1 under Moderate and Strict criteria. LeWM uses Pure-CEM with $H = 5/R_H = 5$; entries are mean ± sample SD over three training seeds.

Criterion	Endpoint	Method	Epoch	PushT	Cube	Reacher	TwoRoom	Macro
Moderate	SIGReg	Endpoint-only	E1	70.22 ± 15.32	41.33 ± 3.84	14.56 ± 7.90	10.44 ± 5.80	34.14 ± 6.55
	SIGReg	+SADL	E1	67.89 ± 4.22	38.78 ± 0.51	76.89 ± 6.36	81.67 ± 14.62	$\mathbf{66.31 \pm 2.38}$
	VISReg	Endpoint-only	E1	78.67 ± 3.84	43.33 ± 7.26	18.78 ± 2.87	45.67 ± 32.30	46.61 ± 7.94
	VISReg	+SADL	E1	77.11 ± 4.19	37.67 ± 1.20	77.78 ± 1.54	83.00 ± 5.20	$\mathbf{68.89 \pm 0.61}$
	QQWorld	Endpoint-only	E1	74.33 ± 3.76	42.33 ± 3.93	27.44 ± 16.88	18.33 ± 13.17	40.61 ± 8.59
	QQWorld	+SADL	E1	71.44 ± 3.24	42.33 ± 7.23	75.78 ± 0.69	79.67 ± 3.51	$\mathbf{67.31 \pm 2.56}$
Strict	SIGReg	Endpoint-only	E1	50.67 ± 12.91	18.44 ± 2.01	18.00 ± 7.88	3.89 ± 2.36	22.75 ± 5.40
	SIGReg	+SADL	E1	51.22 ± 0.19	19.00 ± 0.58	82.33 ± 1.15	73.11 ± 9.90	$\mathbf{56.42 \pm 2.43}$
	VISReg	Endpoint-only	E1	55.11 ± 3.02	22.22 ± 5.01	26.22 ± 1.90	30.56 ± 26.34	33.53 ± 6.44
	VISReg	+SADL	E1	57.78 ± 2.71	17.67 ± 2.73	80.11 ± 2.87	73.78 ± 2.91	$\mathbf{57.33 \pm 1.01}$
	QQWorld	Endpoint-only	E1	56.22 ± 6.05	19.33 ± 6.08	36.78 ± 20.35	11.22 ± 12.28	30.89 ± 9.64
	QQWorld	+SADL	E1	58.44 ± 1.02	22.67 ± 4.36	78.78 ± 3.89	71.22 ± 3.66	$\mathbf{57.78 \pm 3.08}$

Table 19: Single-task INTACT CLEAR-LeWM v0.8 success (%) at E1 under Moderate and Strict criteria with search-free Direct control; entries are mean ± sample SD over three training seeds.

Criterion	Endpoint	Method	Epoch	PushT	Cube	Reacher	TwoRoom	Macro
Moderate	SIGReg	Endpoint-only	E1	80.89 ± 6.95	95.00 ± 5.03	25.11 ± 20.97	10.78 ± 4.67	52.94 ± 7.19
	SIGReg	+SADL	E1	86.67 ± 2.91	100.00 ± 0.00	97.00 ± 0.88	97.22 ± 2.36	$\mathbf{95.22 \pm 1.02}$
	VISReg	Endpoint-only	E1	86.56 ± 3.67	96.22 ± 2.83	47.00 ± 39.40	37.00 ± 41.01	66.69 ± 19.72
	VISReg	+SADL	E1	87.56 ± 2.01	100.00 ± 0.00	97.22 ± 1.02	98.00 ± 1.20	$\mathbf{95.69 \pm 0.97}$
	QQWorld	Endpoint-only	E1	85.00 ± 2.33	96.67 ± 4.36	35.78 ± 31.62	9.11 ± 5.32	56.64 ± 6.16
	QQWorld	+SADL	E1	89.00 ± 0.58	100.00 ± 0.00	97.11 ± 1.50	98.89 ± 1.35	$\mathbf{96.25 \pm 0.55}$
Strict	SIGReg	Endpoint-only	E1	70.67 ± 3.93	80.78 ± 6.38	40.22 ± 29.92	3.00 ± 1.73	48.67 ± 7.91
	SIGReg	+SADL	E1	76.00 ± 2.67	86.33 ± 0.67	98.67 ± 0.88	92.89 ± 1.95	$\mathbf{88.47 \pm 0.84}$
	VISReg	Endpoint-only	E1	73.56 ± 0.51	80.11 ± 5.32	58.67 ± 38.23	23.67 ± 33.50	59.00 ± 16.07
	VISReg	+SADL	E1	78.00 ± 2.73	86.56 ± 0.19	98.22 ± 0.84	93.33 ± 2.89	$\mathbf{89.03 \pm 1.25}$
	QQWorld	Endpoint-only	E1	74.22 ± 1.50	82.33 ± 6.66	51.78 ± 33.01	3.67 ± 4.93	53.00 ± 6.07
	QQWorld	+SADL	E1	79.89 ± 1.26	86.67 ± 1.76	98.11 ± 0.19	92.44 ± 3.91	$\mathbf{89.28 \pm 1.64}$

E.2 CLEAR-LEWM V0.8

E.2.1 单任务 Moderate 与 Strict

表 18：单任务 LeWM E1 的 CLEAR-LeWM v0.8 Moderate、Strict 成功率（%）。LeWM 使用 Pure-CEM，配置 $H = 5/R_H = 5$；数值为三个训练种子的均值 ± 样本标准差。

标准	端点	方法	轮次	PushT	Cube	Reacher	TwoRoom	宏平均
Moderate	SIGReg	仅端点	E1	70.22 ± 15.32	41.33 ± 3.84	14.56 ± 7.90	10.44 ± 5.80	34.14 ± 6.55
	SIGReg	+SADL	E1	67.89 ± 4.22	38.78 ± 0.51	76.89 ± 6.36	81.67 ± 14.62	$\mathbf{66.31 \pm 2.38}$
	VISReg	仅端点	E1	78.67 ± 3.84	43.33 ± 7.26	18.78 ± 2.87	45.67 ± 32.30	46.61 ± 7.94
	VISReg	+SADL	E1	77.11 ± 4.19	37.67 ± 1.20	77.78 ± 1.54	83.00 ± 5.20	$\mathbf{68.89 \pm 0.61}$
	QQWorld	仅端点	E1	74.33 ± 3.76	42.33 ± 3.93	27.44 ± 16.88	18.33 ± 13.17	40.61 ± 8.59
	QQWorld	+SADL	E1	71.44 ± 3.24	42.33 ± 7.23	75.78 ± 0.69	79.67 ± 3.51	$\mathbf{67.31 \pm 2.56}$
Strict	SIGReg	仅端点	E1	50.67 ± 12.91	18.44 ± 2.01	18.00 ± 7.88	3.89 ± 2.36	22.75 ± 5.40
	SIGReg	+SADL	E1	51.22 ± 0.19	19.00 ± 0.58	82.33 ± 1.15	73.11 ± 9.90	$\mathbf{56.42 \pm 2.43}$
	VISReg	仅端点	E1	55.11 ± 3.02	22.22 ± 5.01	26.22 ± 1.90	30.56 ± 26.34	33.53 ± 6.44
	VISReg	+SADL	E1	57.78 ± 2.71	17.67 ± 2.73	80.11 ± 2.87	73.78 ± 2.91	$\mathbf{57.33 \pm 1.01}$
	QQWorld	仅端点	E1	56.22 ± 6.05	19.33 ± 6.08	36.78 ± 20.35	11.22 ± 12.28	30.89 ± 9.64
	QQWorld	+SADL	E1	58.44 ± 1.02	22.67 ± 4.36	78.78 ± 3.89	71.22 ± 3.66	$\mathbf{57.78 \pm 3.08}$

表 19：单任务 INTACT E1 的 CLEAR-LeWM v0.8 Moderate、Strict 成功率（%），采用无搜索 Direct 控制；数值为三个训练种子的均值 ± 样本标准差。

标准	端点	方法	轮次	PushT	Cube	Reacher	TwoRoom	宏平均
Moderate	SIGReg	仅端点	E1	80.89 ± 6.95	95.00 ± 5.03	25.11 ± 20.97	10.78 ± 4.67	52.94 ± 7.19
	SIGReg	+SADL	E1	86.67 ± 2.91	100.00 ± 0.00	97.00 ± 0.88	97.22 ± 2.36	$\mathbf{95.22 \pm 1.02}$
	VISReg	仅端点	E1	86.56 ± 3.67	96.22 ± 2.83	47.00 ± 39.40	37.00 ± 41.01	66.69 ± 19.72
	VISReg	+SADL	E1	87.56 ± 2.01	100.00 ± 0.00	97.22 ± 1.02	98.00 ± 1.20	$\mathbf{95.69 \pm 0.97}$
	QQWorld	仅端点	E1	85.00 ± 2.33	96.67 ± 4.36	35.78 ± 31.62	9.11 ± 5.32	56.64 ± 6.16
	QQWorld	+SADL	E1	89.00 ± 0.58	100.00 ± 0.00	97.11 ± 1.50	98.89 ± 1.35	$\mathbf{96.25 \pm 0.55}$
Strict	SIGReg	仅端点	E1	70.67 ± 3.93	80.78 ± 6.38	40.22 ± 29.92	3.00 ± 1.73	48.67 ± 7.91
	SIGReg	+SADL	E1	76.00 ± 2.67	86.33 ± 0.67	98.67 ± 0.88	92.89 ± 1.95	$\mathbf{88.47 \pm 0.84}$
	VISReg	仅端点	E1	73.56 ± 0.51	80.11 ± 5.32	58.67 ± 38.23	23.67 ± 33.50	59.00 ± 16.07
	VISReg	+SADL	E1	78.00 ± 2.73	86.56 ± 0.19	98.22 ± 0.84	93.33 ± 2.89	$\mathbf{89.03 \pm 1.25}$
	QQWorld	仅端点	E1	74.22 ± 1.50	82.33 ± 6.66	51.78 ± 33.01	3.67 ± 4.93	53.00 ± 6.07
	QQWorld	+SADL	E1	79.89 ± 1.26	86.67 ± 1.76	98.11 ± 0.19	92.44 ± 3.91	$\mathbf{89.28 \pm 1.64}$

E.2.2 JOINT MODERATE

Table 20: Joint four-task LeWM Moderate success (%). E1 and E5 compare equal training budgets; endpoint-only E10 provides a longer-budget reference. LeWM uses Pure-CEM with $H = 5/RH = 5$. Endpoint-only uses a MATH encoder; +SADL uses a Flash encoder; both predictors use MATH. All entries are mean ± sample SD over three training seeds.

Endpoint	Method	Epoch	PushT	Cube	Reacher	TwoRoom	Macro
SIGReg	Endpoint-only	E1	12.89 ± 3.34	29.89 ± 1.35	14.67 ± 5.78	20.33 ± 9.26	19.44 ± 3.59
SIGReg	+SADL	E1	69.00 ± 2.96	49.00 ± 2.00	81.44 ± 2.01	58.11 ± 1.58	$\mathbf{64.39 \pm 1.44}$
SIGReg	Endpoint-only	E5	79.00 ± 3.48	50.44 ± 4.50	61.56 ± 16.78	22.33 ± 18.05	53.33 ± 3.63
SIGReg	+SADL	E5	68.33 ± 2.73	49.00 ± 2.00	67.67 ± 13.86	58.11 ± 1.58	$\mathbf{60.78 \pm 3.63}$
SIGReg	Endpoint-only	E10	76.67 ± 3.76	50.78 ± 1.35	70.44 ± 18.03	28.56 ± 11.03	56.61 ± 6.49
VISReg	Endpoint-only	E1	52.67 ± 8.17	32.56 ± 3.10	21.11 ± 6.30	32.22 ± 15.87	34.64 ± 3.67
VISReg	+SADL	E1	62.56 ± 21.91	45.89 ± 7.34	79.56 ± 3.29	80.00 ± 8.97	$\mathbf{67.00 \pm 3.67}$
VISReg	Endpoint-only	E5	84.78 ± 0.51	39.67 ± 3.84	79.89 ± 0.69	40.56 ± 10.21	61.22 ± 2.94
VISReg	+SADL	E5	83.22 ± 2.27	47.67 ± 3.48	82.56 ± 1.35	68.00 ± 4.06	$\mathbf{70.36 \pm 0.53}$
VISReg	Endpoint-only	E10	86.33 ± 2.19	39.33 ± 1.53	70.56 ± 20.69	47.44 ± 6.16	60.92 ± 4.79
QQWorld	Endpoint-only	E1	14.56 ± 4.86	31.89 ± 6.55	19.89 ± 3.69	26.22 ± 10.87	23.14 ± 5.32
QQWorld	+SADL	E1	71.44 ± 9.31	50.78 ± 1.35	79.44 ± 3.53	67.67 ± 14.47	$\mathbf{67.33 \pm 1.04}$
QQWorld	Endpoint-only	E5	80.67 ± 5.49	49.67 ± 3.18	68.78 ± 21.17	35.11 ± 14.26	58.56 ± 10.60
QQWorld	+SADL	E5	76.78 ± 3.67	52.11 ± 1.17	80.33 ± 0.88	58.56 ± 1.68	$\mathbf{66.94 \pm 0.97}$
QQWorld	Endpoint-only	E10	87.33 ± 1.67	49.00 ± 5.03	68.67 ± 13.89	38.44 ± 7.12	60.86 ± 3.57

Table 21: Joint four-task INTACT Moderate success (%) with search-free Direct control. E1 and E5 compare equal training budgets; endpoint-only E10 provides a longer-budget reference. Endpoint-only uses a MATH encoder; +SADL uses a Flash encoder; both predictors use MATH. All entries are mean ± sample SD over three training seeds.

Endpoint	Method	Epoch	PushT	Cube	Reacher	TwoRoom	Macro
SIGReg	Endpoint-only	E1	39.11 ± 9.14	54.11 ± 12.04	9.89 ± 4.17	41.33 ± 8.21	36.11 ± 5.28
SIGReg	+SADL	E1	86.22 ± 0.38	100.00 ± 0.00	97.00 ± 0.58	99.67 ± 0.00	$\mathbf{95.72 \pm 0.05}$
SIGReg	Endpoint-only	E5	87.56 ± 5.59	99.67 ± 0.58	97.00 ± 1.45	58.89 ± 20.42	85.78 ± 3.78
SIGReg	+SADL	E5	92.89 ± 0.38	100.00 ± 0.00	99.44 ± 0.69	99.78 ± 0.19	$\mathbf{98.03 \pm 0.13}$
SIGReg	Endpoint-only	E10	91.00 ± 3.18	100.00 ± 0.00	96.89 ± 0.69	73.89 ± 15.08	90.44 ± 3.14
VISReg	Endpoint-only	E1	55.78 ± 3.83	69.78 ± 13.62	31.11 ± 9.67	55.00 ± 11.32	52.92 ± 4.08
VISReg	+SADL	E1	86.78 ± 1.84	100.00 ± 0.00	97.56 ± 0.84	99.67 ± 0.00	$\mathbf{96.00 \pm 0.46}$
VISReg	Endpoint-only	E5	91.89 ± 1.84	98.56 ± 0.51	97.67 ± 0.58	69.22 ± 3.91	89.33 ± 0.92
VISReg	+SADL	E5	93.22 ± 0.38	100.00 ± 0.00	98.89 ± 0.77	99.78 ± 0.38	$\mathbf{97.97 \pm 0.10}$
VISReg	Endpoint-only	E10	93.78 ± 1.64	100.00 ± 0.00	97.78 ± 0.84	89.11 ± 5.30	95.17 ± 1.10
QQWorld	Endpoint-only	E1	37.78 ± 10.67	60.78 ± 3.15	16.56 ± 7.07	30.89 ± 15.12	36.50 ± 4.59
QQWorld	+SADL	E1	88.78 ± 0.84	100.00 ± 0.00	97.44 ± 0.38	99.44 ± 0.19	$\mathbf{96.42 \pm 0.22}$
QQWorld	Endpoint-only	E5	90.11 ± 1.26	99.89 ± 0.19	97.67 ± 0.58	77.11 ± 22.04	91.19 ± 5.22
QQWorld	+SADL	E5	92.00 ± 1.15	100.00 ± 0.00	99.00 ± 0.58	99.67 ± 0.33	$\mathbf{97.67 \pm 0.08}$
QQWorld	Endpoint-only	E10	92.56 ± 1.17	100.00 ± 0.00	97.33 ± 0.58	85.22 ± 22.13	93.78 ± 5.38

E.2.2 联合 Moderate

表 20：联合四任务 LeWM Moderate 成功率（%）。E1、E5 比较等训练预算，仅端点 E10 为长预算参考。LeWM 用 Pure-CEM，配置 $H = 5/RH$ =5。仅端点用 MATH 编码器， +SADL 用 Flash 编码器；预测器均为 MATH。数值为三个训练种子的均值 ± 样本标准差。

端点	方法	轮次	PushT	Cube	Reacher	TwoRoom	宏平均
SIGReg	仅端点	E1	12.89 ± 3.34	29.89 ± 1.35	14.67 ± 5.78	20.33 ± 9.26	19.44 ± 3.59
SIGReg	+SADL	E1	69.00 ± 2.96	49.00 ± 2.00	81.44 ± 2.01	58.11 ± 1.58	$\mathbf{64.39 \pm 1.44}$
SIGReg	仅端点	E5	79.00 ± 3.48	50.44 ± 4.50	61.56 ± 16.78	22.33 ± 18.05	53.33 ± 3.63
SIGReg	+SADL	E5	68.33 ± 2.73	49.00 ± 2.00	67.67 ± 13.86	58.11 ± 1.58	$\mathbf{60.78 \pm 3.63}$
SIGReg	仅端点	E10	76.67 ± 3.76	50.78 ± 1.35	70.44 ± 18.03	28.56 ± 11.03	56.61 ± 6.49
VISReg	仅端点	E1	52.67 ± 8.17	32.56 ± 3.10	21.11 ± 6.30	32.22 ± 15.87	34.64 ± 3.67
VISReg	+SADL	E1	62.56 ± 21.91	45.89 ± 7.34	79.56 ± 3.29	80.00 ± 8.97	$\mathbf{67.00 \pm 3.67}$
VISReg	仅端点	E5	84.78 ± 0.51	39.67 ± 3.84	79.89 ± 0.69	40.56 ± 10.21	61.22 ± 2.94
VISReg	+SADL	E5	83.22 ± 2.27	47.67 ± 3.48	82.56 ± 1.35	68.00 ± 4.06	$\mathbf{70.36 \pm 0.53}$
VISReg	仅端点	E10	86.33 ± 2.19	39.33 ± 1.53	70.56 ± 20.69	47.44 ± 6.16	60.92 ± 4.79
QQWorld	仅端点	E1	14.56 ± 4.86	31.89 ± 6.55	19.89 ± 3.69	26.22 ± 10.87	23.14 ± 5.32
QQWorld	+SADL	E1	71.44 ± 9.31	50.78 ± 1.35	79.44 ± 3.53	67.67 ± 14.47	$\mathbf{67.33 \pm 1.04}$
QQWorld	仅端点	E5	80.67 ± 5.49	49.67 ± 3.18	68.78 ± 21.17	35.11 ± 14.26	58.56 ± 10.60
QQWorld	+SADL	E5	76.78 ± 3.67	52.11 ± 1.17	80.33 ± 0.88	58.56 ± 1.68	$\mathbf{66.94 \pm 0.97}$
QQWorld	仅端点	E10	87.33 ± 1.67	49.00 ± 5.03	68.67 ± 13.89	38.44 ± 7.12	60.86 ± 3.57

表 21：联合四任务 INTACT 的 Moderate 成功率（%），使用无搜索 Direct 控制。E1、E5 比较等训练预算，仅端点 E10 为长预算参考。仅端点使用 MATH 编码器， +SADL 使用 Flash 编码器；预测器均为 MATH。数值为三个训练种子的均值 ± 样本标准差。

端点	方法	轮次	PushT	Cube	Reacher	TwoRoom	宏平均
SIGReg	仅端点	E1	39.11 ± 9.14	54.11 ± 12.04	9.89 ± 4.17	41.33 ± 8.21	36.11 ± 5.28
SIGReg	+SADL	E1	86.22 ± 0.38	100.00 ± 0.00	97.00 ± 0.58	99.67 ± 0.00	$\mathbf{95.72 \pm 0.05}$
SIGReg	仅端点	E5	87.56 ± 5.59	99.67 ± 0.58	97.00 ± 1.45	58.89 ± 20.42	85.78 ± 3.78
SIGReg	+SADL	E5	92.89 ± 0.38	100.00 ± 0.00	99.44 ± 0.69	99.78 ± 0.19	$\mathbf{98.03 \pm 0.13}$
SIGReg	仅端点	E10	91.00 ± 3.18	100.00 ± 0.00	96.89 ± 0.69	73.89 ± 15.08	90.44 ± 3.14
VISReg	仅端点	E1	55.78 ± 3.83	69.78 ± 13.62	31.11 ± 9.67	55.00 ± 11.32	52.92 ± 4.08
VISReg	+SADL	E1	86.78 ± 1.84	100.00 ± 0.00	97.56 ± 0.84	99.67 ± 0.00	$\mathbf{96.00 \pm 0.46}$
VISReg	仅端点	E5	91.89 ± 1.84	98.56 ± 0.51	97.67 ± 0.58	69.22 ± 3.91	89.33 ± 0.92
VISReg	+SADL	E5	93.22 ± 0.38	100.00 ± 0.00	98.89 ± 0.77	99.78 ± 0.38	$\mathbf{97.97 \pm 0.10}$
VISReg	仅端点	E10	93.78 ± 1.64	100.00 ± 0.00	97.78 ± 0.84	89.11 ± 5.30	95.17 ± 1.10
QQWorld	仅端点	E1	37.78 ± 10.67	60.78 ± 3.15	16.56 ± 7.07	30.89 ± 15.12	36.50 ± 4.59
QQWorld	+SADL	E1	88.78 ± 0.84	100.00 ± 0.00	97.44 ± 0.38	99.44 ± 0.19	$\mathbf{96.42 \pm 0.22}$
QQWorld	仅端点	E5	90.11 ± 1.26	99.89 ± 0.19	97.67 ± 0.58	77.11 ± 22.04	91.19 ± 5.22
QQWorld	+SADL	E5	92.00 ± 1.15	100.00 ± 0.00	99.00 ± 0.58	99.67 ± 0.33	$\mathbf{97.67 \pm 0.08}$
QQWorld	仅端点	E10	92.56 ± 1.17	100.00 ± 0.00	97.33 ± 0.58	85.22 ± 22.13	93.78 ± 5.38

E.2.3 JOINT STRICT

Table 22: Joint four-task LeWM CLEAR-LeWM v0.8 Strict success (%). E1 and E5 compare equal training budgets; endpoint-only E10 is a longer-budget reference. LeWM uses Pure-CEM with $H = 5/RH = 5$. Endpoint-only uses a MATH encoder; +SADL uses a Flash encoder; both predictors use MATH. Values are mean ± sample SD over three training seeds.

Endpoint	Method	Epoch	PushT	Cube	Reacher	TwoRoom	Macro
SIGReg	Endpoint-only	E1	16.00 ± 4.93	9.56 ± 0.19	21.11 ± 3.56	10.33 ± 7.09	14.25 ± 3.44
SIGReg	+SADL	E1	61.78 ± 1.90	24.56 ± 2.52	82.67 ± 1.76	40.00 ± 1.45	**52.25±0.30**
SIGReg	Endpoint-only	E5	63.00 ± 3.61	26.67 ± 2.96	64.22 ± 19.33	11.78 ± 11.21	41.42 ± 4.50
SIGReg	+SADL	E5	62.22 ± 2.14	24.56 ± 2.52	72.44 ± 7.71	40.00 ± 1.45	**49.81±2.14**
SIGReg	Endpoint-only	E10	62.44 ± 7.32	26.33 ± 3.61	69.00 ± 24.29	17.22 ± 5.74	43.75 ± 6.94
VISReg	Endpoint-only	E1	35.56 ± 5.60	12.22 ± 2.59	30.00 ± 11.62	18.00 ± 6.66	23.94 ± 4.15
VISReg	+SADL	E1	54.11 ± 15.35	25.22 ± 6.36	81.67 ± 2.73	64.56 ± 15.91	**56.39±1.55**
VISReg	Endpoint-only	E5	73.78 ± 4.91	20.78 ± 0.77	82.22 ± 0.19	23.11 ± 7.63	49.97 ± 1.63
VISReg	+SADL	E5	71.78 ± 1.17	28.44 ± 1.35	83.22 ± 2.22	46.33 ± 0.33	**57.44±1.04**
VISReg	Endpoint-only	E10	76.11 ± 2.69	22.11 ± 0.84	68.89 ± 26.69	24.11 ± 1.50	47.81 ± 6.06
QQWorld	Endpoint-only	E1	21.11 ± 6.19	13.00 ± 3.18	26.33 ± 4.73	14.56 ± 8.44	18.75 ± 5.28
QQWorld	+SADL	E1	61.78 ± 11.48	30.33 ± 3.06	84.11 ± 2.46	47.89 ± 17.44	**56.03±1.74**
QQWorld	Endpoint-only	E5	66.22 ± 6.27	29.22 ± 3.29	79.56 ± 6.58	19.56 ± 9.22	48.64 ± 5.98
QQWorld	+SADL	E5	69.00 ± 2.00	29.44 ± 1.68	85.11 ± 0.84	37.78 ± 0.84	**55.33±0.71**
QQWorld	Endpoint-only	E10	72.22 ± 4.29	28.33 ± 2.40	74.11 ± 9.71	21.89 ± 3.95	49.14 ± 3.74

Table 23: Joint four-task INTACT CLEAR-LeWM v0.8 Strict success (%). E1 and E5 compare equal training budgets; endpoint-only E10 is a longer-budget reference. INTACT uses search-free Direct control. Endpoint-only uses a MATH encoder; +SADL uses a Flash encoder; both predictors use MATH. Values are mean ± sample SD over three training seeds.

Endpoint	Method	Epoch	PushT	Cube	Reacher	TwoRoom	Macro
SIGReg	Endpoint-only	E1	36.00 ± 7.23	43.33 ± 8.95	20.67 ± 7.54	24.33 ± 10.79	31.08 ± 2.04
SIGReg	+SADL	E1	77.56 ± 0.51	88.89 ± 1.17	98.00 ± 0.88	97.33 ± 0.67	**90.44±0.55**
SIGReg	Endpoint-only	E5	74.78 ± 4.02	89.44 ± 1.07	98.22 ± 0.51	37.33 ± 20.80	74.94 ± 3.81
SIGReg	+SADL	E5	84.33 ± 1.20	96.67 ± 5.20	98.56 ± 0.38	97.44 ± 0.96	**94.25±1.42**
SIGReg	Endpoint-only	E10	81.00 ± 2.19	88.67 ± 0.58	98.44 ± 0.19	59.33 ± 19.78	81.86 ± 4.32
VISReg	Endpoint-only	E1	42.56 ± 3.53	55.33 ± 12.55	54.78 ± 14.28	32.11 ± 14.05	46.19 ± 3.44
VISReg	+SADL	E1	74.89 ± 2.01	87.67 ± 1.20	97.89 ± 0.77	95.22 ± 1.02	**88.92±0.22**
VISReg	Endpoint-only	E5	78.11 ± 1.64	86.56 ± 1.17	99.44 ± 0.38	52.56 ± 7.07	79.17 ± 1.83
VISReg	+SADL	E5	83.56 ± 0.38	89.56 ± 0.19	98.44 ± 0.69	96.22 ± 1.17	**91.94±0.29**
VISReg	Endpoint-only	E10	81.78 ± 2.59	88.56 ± 0.51	98.56 ± 1.17	74.78 ± 6.06	85.92 ± 2.08
QQWorld	Endpoint-only	E1	40.56 ± 5.42	47.11 ± 6.11	33.89 ± 12.85	11.00 ± 4.10	33.14 ± 1.88
QQWorld	+SADL	E1	76.67 ± 2.03	88.22 ± 1.26	97.89 ± 0.69	96.11 ± 1.17	**89.72±0.96**
QQWorld	Endpoint-only	E5	79.11 ± 2.91	89.56 ± 0.69	99.00 ± 0.33	59.22 ± 28.55	81.72 ± 7.39
QQWorld	+SADL	E5	83.56 ± 2.69	96.89 ± 5.10	98.44 ± 0.51	96.22 ± 0.19	**93.78±1.25**
QQWorld	Endpoint-only	E10	81.33 ± 1.73	92.00 ± 0.58	98.56 ± 0.19	73.89 ± 27.35	86.44 ± 7.00

E.2.3 联合 Strict

表 22：联合四任务 LeWM CLEAR-LeWM v0.8 Strict 成功率（%）。E1、E5 为等训练预算比较，仅端点 E10 为长预算参考。LeWM 使用 Pure-CEM，配置 $H = 5/RH = 5$。仅端点用 MATH 编码器，+SADL 用 Flash 编码器；预测器均为 MATH。数值为三个训练种子的均值 ± 样本标准差。

端点	方法	轮次	PushT	Cube	Reacher	TwoRoom	宏平均
SIGReg	仅端点	E1	16.00 ± 4.93	9.56 ± 0.19	21.11 ± 3.56	10.33 ± 7.09	14.25 ± 3.44
SIGReg	+SADL	E1	61.78 ± 1.90	24.56 ± 2.52	82.67 ± 1.76	40.00 ± 1.45	**52.25±0.30**
SIGReg	仅端点	E5	63.00 ± 3.61	26.67 ± 2.96	64.22 ± 19.33	11.78 ± 11.21	41.42 ± 4.50
SIGReg	+SADL	E5	62.22 ± 2.14	24.56 ± 2.52	72.44 ± 7.71	40.00 ± 1.45	**49.81±2.14**
SIGReg	仅端点	E10	62.44 ± 7.32	26.33 ± 3.61	69.00 ± 24.29	17.22 ± 5.74	43.75 ± 6.94
VISReg	仅端点	E1	35.56 ± 5.60	12.22 ± 2.59	30.00 ± 11.62	18.00 ± 6.66	23.94 ± 4.15
VISReg	+SADL	E1	54.11 ± 15.35	25.22 ± 6.36	81.67 ± 2.73	64.56 ± 15.91	**56.39±1.55**
VISReg	仅端点	E5	73.78 ± 4.91	20.78 ± 0.77	82.22 ± 0.19	23.11 ± 7.63	49.97 ± 1.63
VISReg	+SADL	E5	71.78 ± 1.17	28.44 ± 1.35	83.22 ± 2.22	46.33 ± 0.33	**57.44±1.04**
VISReg	仅端点	E10	76.11 ± 2.69	22.11 ± 0.84	68.89 ± 26.69	24.11 ± 1.50	47.81 ± 6.06
QQWorld	仅端点	E1	21.11 ± 6.19	13.00 ± 3.18	26.33 ± 4.73	14.56 ± 8.44	18.75 ± 5.28
QQWorld	+SADL	E1	61.78 ± 11.48	30.33 ± 3.06	84.11 ± 2.46	47.89 ± 17.44	**56.03±1.74**
QQWorld	仅端点	E5	66.22 ± 6.27	29.22 ± 3.29	79.56 ± 6.58	19.56 ± 9.22	48.64 ± 5.98
QQWorld	+SADL	E5	69.00 ± 2.00	29.44 ± 1.68	85.11 ± 0.84	37.78 ± 0.84	**55.33±0.71**
QQWorld	仅端点	E10	72.22 ± 4.29	28.33 ± 2.40	74.11 ± 9.71	21.89 ± 3.95	49.14 ± 3.74

表 23：联合四任务 INTACT CLEAR-LeWM v0.8 Strict 成功率（%）。E1、E5 为等训练预算比较，仅端点 E10 为长预算参考。INTACT 用无搜索 Direct 控制。仅端点用 MATH 编码器，+SADL 用 Flash 编码器；预测器均为 MATH。数值为三个训练种子的均值 ± 样本标准差。

端点	方法	轮次	PushT	Cube	Reacher	TwoRoom	宏平均
SIGReg	仅端点	E1	36.00 ± 7.23	43.33 ± 8.95	20.67 ± 7.54	24.33 ± 10.79	31.08 ± 2.04
SIGReg	+SADL	E1	77.56 ± 0.51	88.89 ± 1.17	98.00 ± 0.88	97.33 ± 0.67	**90.44±0.55**
SIGReg	仅端点	E5	74.78 ± 4.02	89.44 ± 1.07	98.22 ± 0.51	37.33 ± 20.80	74.94 ± 3.81
SIGReg	+SADL	E5	84.33 ± 1.20	96.67 ± 5.20	98.56 ± 0.38	97.44 ± 0.96	**94.25±1.42**
SIGReg	仅端点	E10	81.00 ± 2.19	88.67 ± 0.58	98.44 ± 0.19	59.33 ± 19.78	81.86 ± 4.32
VISReg	仅端点	E1	42.56 ± 3.53	55.33 ± 12.55	54.78 ± 14.28	32.11 ± 14.05	46.19 ± 3.44
VISReg	+SADL	E1	74.89 ± 2.01	87.67 ± 1.20	97.89 ± 0.77	95.22 ± 1.02	**88.92±0.22**
VISReg	仅端点	E5	78.11 ± 1.64	86.56 ± 1.17	99.44 ± 0.38	52.56 ± 7.07	79.17 ± 1.83
VISReg	+SADL	E5	83.56 ± 0.38	89.56 ± 0.19	98.44 ± 0.69	96.22 ± 1.17	**91.94±0.29**
VISReg	仅端点	E10	81.78 ± 2.59	88.56 ± 0.51	98.56 ± 1.17	74.78 ± 6.06	85.92 ± 2.08
QQWorld	仅端点	E1	40.56 ± 5.42	47.11 ± 6.11	33.89 ± 12.85	11.00 ± 4.10	33.14 ± 1.88
QQWorld	+SADL	E1	76.67 ± 2.03	88.22 ± 1.26	97.89 ± 0.69	96.11 ± 1.17	**89.72±0.96**
QQWorld	仅端点	E5	79.11 ± 2.91	89.56 ± 0.69	99.00 ± 0.33	59.22 ± 28.55	81.72 ± 7.39
QQWorld	+SADL	E5	83.56 ± 2.69	96.89 ± 5.10	98.44 ± 0.51	96.22 ± 0.19	**93.78±1.25**
QQWorld	仅端点	E10	81.33 ± 1.73	92.00 ± 0.58	98.56 ± 0.19	73.89 ± 27.35	86.44 ± 7.00

F Structural Ablations and Mechanism Diagnostics

F.1 Supply branches and bridge potentials

All structural ablations use Flash, SIGReg weight .03, bridge weight $\lambda_S = .01$, and $\alpha = .5$. Excitation-only and Transmission-only replace k by C_{exc} or C_{tr}, respectively. Potential variants compare $[u]_+^2$, u^2, and $[u]_+$ for $u = \log(q+\epsilon) - \log(k+\epsilon)$. Bridge-only still evaluates detached demand but removes the endpoint regularizer's direct training gradient.

Table 24: E2 supply-branch and endpoint-gradient ablations under SIGReg. Entries are four-task CLEAR-LeWM v0.8 macro success, mean ± sample SD (%) over three training seeds, with three evaluation seeds per task. All use Flash.

Configuration	INTACT Mod.	INTACT Strict	LeWM Mod.	LeWM Strict
Excitation-only	97.14 ± 0.17	91.03 ± 0.61	62.78 ± 2.00	50.94 ± 3.26
Transmission-only	42.94 ± 45.45	38.33 ± 41.64	35.28 ± 27.30	26.97 ± 20.63
Bridge-only	47.00 ± 5.59	35.33 ± 1.86	11.72 ± 1.80	7.81 ± 0.49
Full SADL	$\mathbf{97.14 \pm 0.17}$	$\mathbf{92.06 \pm 1.04}$	$\mathbf{65.50 \pm 0.85}$	$\mathbf{53.03 \pm 1.42}$

The excitation-only branch matches Full SADL's INTACT E2 Moderate macro success (97.14%) and trails it by 2.72 percentage points in LeWM (62.78% versus 65.50%). Transmission-only is markedly weaker and more seed-variable in both models. Thus excitation is the dominant observed bottleneck in these four-task control runs; this does not make transmission capacity dispensable. Bridge-only falls to 47.00% in INTACT and 11.72% in LeWM on Moderate, showing that the path correction complements rather than replaces the direct endpoint constraint. In the independently constructed transmission-deficit setting of Table 28(B), Full SADL raises C_{tr} from .180 to .746 and held-out R^2 from $-.018$ to .535, whereas excitation-only leaves both essentially unchanged.

Table 25: Complete A2 bridge-potential ablation under SIGReg on four-task CLEAR-LeWM v0.8. Within each model and epoch, the canonical one-sided square $[u]_+^2$, bilateral square u^2, and one-sided linear $[u]_+$ use matched training and evaluation protocols. Entries are macro success mean ± sample SD (%) over three training seeds, with three evaluation seeds per task. LeWM uses Pure-CEM; INTACT uses Direct control.

Potential	Epoch	Moderate	Strict
LeWM			
Canonical $[u]_+^2$	E1	$\mathbf{64.39 \pm 1.44}$	$\mathbf{52.25 \pm 0.30}$
Bilateral u^2	E1	59.58 ± 4.38	52.11 ± 2.66
Linear $[u]_+$	E1	47.17 ± 5.55	39.92 ± 6.45
Canonical $[u]_+^2$	E2	$\mathbf{65.50 \pm 0.85}$	$\mathbf{53.03 \pm 1.42}$
Bilateral u^2	E2	61.00 ± 4.86	47.44 ± 6.22
Linear $[u]_+$	E2	53.53 ± 4.19	42.03 ± 6.65
INTACT			
Canonical $[u]_+^2$	E1	95.72 ± 0.05	$\mathbf{90.44 \pm 0.55}$
Bilateral u^2	E1	95.61 ± 0.47	89.97 ± 1.80
Linear $[u]_+$	E1	$\mathbf{96.06 \pm 0.34}$	89.44 ± 0.59
Canonical $[u]_+^2$	E2	$\mathbf{97.14 \pm 0.17}$	$\mathbf{92.06 \pm 1.04}$
Bilateral u^2	E2	93.97 ± 5.76	88.03 ± 7.90
Linear $[u]_+$	E2	$\mathbf{97.14 \pm 0.43}$	90.86 ± 0.67

Table 25 isolates potential shape. The canonical and bilateral squares have identical values and gradients for $u > 0$; only the bilateral potential continues to penalize the already-sufficient regime $u < 0$, while the canonical bridge is zero there by construction. At equal E1/E2 budgets, canonical exceeds bilateral in LeWM Moderate macro success by $4.81/4.50$ percentage points; the E1 gain is

F 结构消融与机制诊断

F.1 供给分支与桥接势函数

所有结构消融使用 Flash，SIGReg 权重 .03，桥接权重 $\lambda_S = .01$，以及 $\alpha = .5$。仅激发、仅传输分别以 C_{exc} 或 C_{tr} 替换 k。势函数变体对 $u = \log(q+\epsilon) - \log(k+\epsilon)$ 比较 $[u]_+^2$、u^2、$[u]_+$。仅桥接仍计算停止梯度的需求，但移除端点正则项的直接训练梯度。

表 24：SIGReg 下 E2 供给分支与端点梯度消融。数值为四任务 CLEAR-LeWM v0.8 宏平均成功率，三个训练种子的均值 ± 样本标准差（%），每任务三个评估种子。全部使用 Flash。

配置	INTACT Moderate	INTACT Strict	LeWM Moderate	LeWM Strict
仅激发	97.14 ± 0.17	91.03 ± 0.61	62.78 ± 2.00	50.94 ± 3.26
仅传输	42.94 ± 45.45	38.33 ± 41.64	35.28 ± 27.30	26.97 ± 20.63
仅桥接	47.00 ± 5.59	35.33 ± 1.86	11.72 ± 1.80	7.81 ± 0.49
完整 SADL	$\mathbf{97.14 \pm 0.17}$	$\mathbf{92.06 \pm 1.04}$	$\mathbf{65.50 \pm 0.85}$	$\mathbf{53.03 \pm 1.42}$

仅激发分支与完整 SADL 的 INTACT E2 Moderate 宏平均成功率相同（97.14%），在 LeWM 中则低 2.72 个百分点（62.78% 对 65.50%）。仅传输在两个模型中均明显更弱且种子波动更大。因此，激发是这些四任务控制运行中主要的已观测瓶颈；这不意味着传输容量可以省略。仅桥接的 Moderate 成功率降至 INTACT 47.00%、LeWM 11.72%，说明路径校正补充而非替代直接端点约束。在表 28(B) 独立构造的传输不足设置中，完整 SADL 将 C_{tr} 从 .180 提高至 .746，将保留集 R^2 从 $-.018$ 提高至 .535；仅激发几乎不改变两者。

表 25：四任务 CLEAR-LeWM v0.8、SIGReg 下完整 A2 桥接势函数消融。各模型和轮次内，标准单侧平方 $[u]_+^2$、双侧平方 u^2、单侧线性 $[u]_+$ 使用匹配训练与评估协议。数值为三个训练种子的宏平均成功率均值 ± 样本标准差（%），每任务三个评估种子。LeWM 用 Pure-CEM；INTACT 用 Direct。

势函数	轮次	Moderate	Strict
LeWM			
标准 $[u]_+^2$	E1	$\mathbf{64.39 \pm 1.44}$	$\mathbf{52.25 \pm 0.30}$
双侧 u^2	E1	59.58 ± 4.38	52.11 ± 2.66
线性 $[u]_+$	E1	47.17 ± 5.55	39.92 ± 6.45
标准 $[u]_+^2$	E2	$\mathbf{65.50 \pm 0.85}$	$\mathbf{53.03 \pm 1.42}$
双侧 u^2	E2	61.00 ± 4.86	47.44 ± 6.22
线性 $[u]_+$	E2	53.53 ± 4.19	42.03 ± 6.65
INTACT			
标准 $[u]_+^2$	E1	95.72 ± 0.05	$\mathbf{90.44 \pm 0.55}$
双侧 u^2	E1	95.61 ± 0.47	89.97 ± 1.80
线性 $[u]_+$	E1	$\mathbf{96.06 \pm 0.34}$	89.44 ± 0.59
标准 $[u]_+^2$	E2	$\mathbf{97.14 \pm 0.17}$	$\mathbf{92.06 \pm 1.04}$
双侧 u^2	E2	93.97 ± 5.76	88.03 ± 7.90
线性 $[u]_+$	E2	$\mathbf{97.14 \pm 0.43}$	90.86 ± 0.67

表 25 隔离势函数形状。当 $u > 0$ 时，标准与双侧平方值、梯度相同；仅双侧势函数继续惩罚已足够供给区域 $u < 0$，标准桥接在该区域按构造为零。相同 E1/E2 预算下，标准版本的 LeWM Moderate 宏平均成功率高于双侧 $4.81/4.50$ 个百分点；E1 增益为

positive for all three paired training seeds. INTACT is nearly tied at E1 (95.72% versus 95.61%), but canonical leads by 3.17 points at E2 (97.14% versus 93.97%). The large bilateral E2 seed SD in INTACT means this last mean difference is a robustness signal, not a uniform gain across seeds. The one-sided linear form remains competitive in INTACT but loses 17.22/11.97 points to canonical in LeWM Moderate success at E1/E2. Together, the matched A2 results support the practical value of a one-sided gate and the squared potential across the two control models.

In the fixed LeWM and INTACT probes, $C_{\text{exc}} < C_{\text{tr}}$, so Full SADL and excitation-only select the same instantaneous bottleneck: both bridge gradients repair C_{exc} at the same state. This equivalence concerns the observed state and bridge term, not necessarily the complete parameter trajectories of independently trained models. Persistent excitation growth explains the early expansion and convergence improvement. Transmission-only has high seed-level variance due to bimodal seed failures, not episode noise. Its E2 Moderate macro results by training seed are

$$\begin{aligned}\text{INTACT:}&\quad 95.42\%,\ 17.33\%,\ 16.08\%,\\ \text{LeWM:}&\quad 65.67\%,\ 12.83\%,\ 27.33\%.\end{aligned}$$

Excitation is the active bottleneck in this regime; an independent transmission-limited test follows in Section F.3.

F.2 Demand, supply, and gradients at fixed checkpoints

The audit contains 168 observations: two model families, seven configurations, three training seeds, and four tasks. Within each model–seed–task group, configurations use the same deterministic 64-sample probe; statistics are computed task-locally. Define deficit, active rate, and transmission-active rate by

$$\Delta = \left[\log \frac{q+\epsilon}{k+\epsilon}\right]_+, \quad p_{\text{act}} = \Pr(k < q), \quad p_{\text{tr,act}} = \Pr(C_{\text{tr}} < C_{\text{exc}},\ k < q).$$

The fixed-checkpoint diagnostics use E10/E5/E2; endpoint-only bridge gradients are computed post hoc as counterfactuals. Tables 26 and 27 report the observed states, whereas Table 24 reports the equal-budget E2 control ablation. All observed checkpoints have $C_{\text{exc}} < C_{\text{tr}}$; all 24 Full SADL snapshots have an active bridge.

Table 26: Demand, supply, and control success at fixed checkpoints. E is the audited epoch; capacity entries average 12 task–seed cells. $p_{\text{act}} = \Pr(q > k)$. SR is independently evaluated four-task CLEAR-LeWM v0.8 Moderate macro success (%; mean ± sample SD over three training seeds, with three evaluation seeds per task) at the listed epoch. The rows have different training budgets and are not an equal-budget SR comparison. † denotes a counterfactual bridge diagnostic on endpoint-only checkpoints; its SR is the actual endpoint-only result. Exc./Trans./Bilateral/Linear denote excitation-only, transmission-only, bilateral squared, and one-sided linear variants.

Setting	E	q_R	q_E	q	C_{exc}	C_{tr}	p_{act} (%)	SR (%)
LeWM								
Endpoint†	10	.8049	.6541	.7174	.3177	.6948	100.0	56.61 ± 6.49
Bridge-only	2	.0000	.0781	.0000	.0439	.1858	0.0	11.72 ± 1.80
Trans.	2	.5416	.3150	.3886	.2578	.5062	41.7	35.28 ± 27.30
Bilateral	2	.9496	.5307	.7098	.5882	.6944	100.0	61.00 ± 4.86
Linear	2	.9479	.5439	.7179	.6640	.7287	100.0	53.53 ± 4.19
Exc.	5	.9517	.6127	.7632	.6426	.7510	100.0	63.72 ± 0.92
Full SADL	5	.9521	.6133	.7637	.6430	.7509	100.0	60.78 ± 3.63
INTACT								
Endpoint†	10	.7381	.6366	.6789	.2622	.6918	100.0	90.44 ± 3.14
Bridge-only	2	.0000	.0791	.0000	.1196	.3486	0.0	47.00 ± 5.59
Trans.	2	.2947	.2171	.2225	.2687	.5673	25.0	42.94 ± 45.45
Bilateral	2	.9343	.5392	.7096	.5697	.7246	100.0	93.97 ± 5.76
Linear	2	.9497	.5591	.7286	.6710	.7539	100.0	97.14 ± 0.43
Exc.	5	.9524	.6411	.7812	.6384	.7964	100.0	97.50 ± 0.22
Full SADL	5	.9529	.6362	.7784	.6331	.7918	100.0	98.03 ± 0.13

全部三个配对训练种子均为正。INTACT 在 E1 几乎持平（95.72% 对 95.61%），但标准版本在 E2 领先 3.17 个百分点（97.14% 对 93.97%）。INTACT 双侧 E2 的种子标准差较大，因此最后这一均值差异是稳健性信号，而非所有种子统一改善。单侧线性形式在 INTACT 中仍具竞争力，但在 LeWM Moderate E1/E2 成功率中落后标准版本 17.22/11.97 个百分点。综合匹配 A2 结果，单侧门控与平方势函数在两个控制模型中具有实际价值。

固定 LeWM、INTACT 探针中有 $C_{\text{exc}} < C_{\text{tr}}$，因此完整 SADL 与仅激发选择相同的瞬时瓶颈：两者桥接梯度在同一状态修复 C_{exc}。该等价性针对观测状态与桥接项，不必适用于独立训练模型的完整参数轨迹。持续激发增长解释早期展开与收敛改善。仅传输的种子级方差大来自双峰种子失效，而非回合噪声。其 E2 Moderate 各训练种子宏平均结果为

$$\begin{aligned}\text{INTACT:}&\quad 95.42\%,\ 17.33\%,\ 16.08\%,\\ \text{LeWM:}&\quad 65.67\%,\ 12.83\%,\ 27.33\%.\end{aligned}$$

此设置中激发是活跃瓶颈；独立传输受限测试见 F.3 节。

F.2 固定检查点的需求、供给与梯度

审计包含 168 个观测：两类模型、七种配置、三个训练种子、四个任务。每个模型–种子–任务组内，各配置使用相同确定性 64 样本探针；统计量在任务局部计算。定义缺口、活跃率和传输活跃率为

$$\Delta = \left[\log \frac{q+\epsilon}{k+\epsilon}\right]_+, \quad p_{\text{act}} = \Pr(k < q), \quad p_{\text{tr,act}} = \Pr(C_{\text{tr}} < C_{\text{exc}},\ k < q).$$

固定检查点诊断使用 E10/E5/E2；仅端点桥接梯度作为反事实事后计算。表 26、27 报告观测状态，表 24 报告等预算 E2 控制消融。所有观测检查点有 $C_{\text{exc}} < C_{\text{tr}}$；完整 SADL 的全部 24 个快照均激活桥接。

表 26：固定检查点的需求、供给与控制成功率。E 是审计轮次；容量对 12 个任务–种子单元求平均。$p_{\text{act}} = \Pr(q > k)$。成功率为该轮次独立评估的四任务 CLEAR-LeWM v0.8 Moderate 宏平均成功率（%；三个训练种子的均值 ± 样本标准差，每任务三个评估种子）。各行训练预算不同，不能作为等预算成功率比较。† 表示仅端点检查点上的反事实桥接诊断；其成功率为实际仅端点结果。Exc./Trans./Bilateral/Linear 分别表示仅激发、仅传输、双侧平方及单侧线性变体。

设置	E	q_R	q_E	q	C_{exc}	C_{tr}	p_{act} (%)	成功率 (%)
LeWM								
端点†	10	.8049	.6541	.7174	.3177	.6948	100.0	56.61 ± 6.49
仅桥接	2	.0000	.0781	.0000	.0439	.1858	0.0	11.72 ± 1.80
传输	2	.5416	.3150	.3886	.2578	.5062	41.7	35.28 ± 27.30
双侧	2	.9496	.5307	.7098	.5882	.6944	100.0	61.00 ± 4.86
线性	2	.9479	.5439	.7179	.6640	.7287	100.0	53.53 ± 4.19
Exc.	5	.9517	.6127	.7632	.6426	.7510	100.0	63.72 ± 0.92
完整 SADL	5	.9521	.6133	.7637	.6430	.7509	100.0	60.78 ± 3.63
INTACT								
端点†	10	.7381	.6366	.6789	.2622	.6918	100.0	90.44 ± 3.14
仅桥接	2	.0000	.0791	.0000	.1196	.3486	0.0	47.00 ± 5.59
传输	2	.2947	.2171	.2225	.2687	.5673	25.0	42.94 ± 45.45
双侧	2	.9343	.5392	.7096	.5697	.7246	100.0	93.97 ± 5.76
线性	2	.9497	.5591	.7286	.6710	.7539	100.0	97.14 ± 0.43
Exc.	5	.9524	.6411	.7812	.6384	.7964	100.0	97.50 ± 0.22
完整 SADL	5	.9529	.6362	.7784	.6331	.7918	100.0	98.03 ± 0.13

Table 27: Bridge deficit and encoder-parameter gradients on the same fixed probe. Epoch identities and independently evaluated CLEAR-LeWM v0.8 Moderate SR (%; mean ± sample SD over three training seeds) match Table 26; different rows have different training budgets. G_T, G_B are task and bridge gradient norms; G_B/G_T averages per-cell ratios. Bridge-only has zero bridge gradient, so its cosine is undefined (n/a). † denotes a counterfactual bridge diagnostic.

Configuration	Δ	G_T	G_B	G_B/G_T (%)	$\cos(G_T, G_B)$	SR (%)
LeWM						
Endpoint†	.8180	.25432	.00756	3.843	-.0251	56.61 ± 6.49
Bridge-only	.0000	.00092	.00000	.000	n/a	11.72 ± 1.80
Trans.	.0233	15.74698	.00023	.017	.0987	35.28 ± 27.30
Bilateral	.1886	.16440	.00196	1.763	.0815	61.00 ± 4.86
Linear	.0790	.14240	.00408	3.697	.1885	53.53 ± 4.19
Exc.	.1740	.05307	.00080	1.852	.0588	63.72 ± 0.92
Full SADL	.1739	.05342	.00081	1.946	.0592	60.78 ± 3.63
INTACT						
Endpoint†	.9730	.69619	.01579	2.393	.1254	90.44 ± 3.14
Bridge-only	.0000	1.10121	.00000	.000	n/a	47.00 ± 5.59
Trans.	.0053	314.61248	.00003	.003	.0966	42.94 ± 45.45
Bilateral	.2223	.97499	.00455	.948	.2988	93.97 ± 5.76
Linear	.0829	.24236	.00472	2.123	.1361	97.14 ± 0.43
Exc.	.2032	.12964	.00110	.916	.0648	97.50 ± 0.22
Full SADL	.2083	.15412	.00119	.887	.0416	98.03 ± 0.13

The average bridge-to-task encoder-gradient norm ratio is $1.946 \pm 1.205\%$ for LeWM and $0.887 \pm 0.310\%$ for INTACT. These are pre-update parameter gradients on the same probe, $G_T = \|\nabla_\theta \mathcal{L}_{\text{task}}\|_2$ and $G_B = \|\nabla_\theta(\lambda_S \mathcal{S})\|_2$; the INTACT task term also contains action negative log-likelihood. For Full SADL, task–bridge cosine is 0.059 ± 0.173 in LeWM and 0.042 ± 0.131 in INTACT; bridge–endpoint cosine is 0.338 ± 0.197 and 0.320 ± 0.164, respectively, with SD across 12 task–seed cells. A small mid-path correction thus changes path geometry without overwhelming the task gradient. All Bridge-only snapshots have $q_R = 0$ and zero ideal demand; smoothed demand stays below available supply despite nonzero endpoint excitation, so the bridge gradient is zero.

F.3 Independent roles of the two supply capacities

Table 28(A) compares two initial capacity deficits. In separate paired ten-seed, 200-step tests, panels B/C construct $C_{\text{tr}} < q < C_{\text{exc}}$ and $C_{\text{exc}} < q < C_{\text{tr}}$, respectively. Each runs No SADL, Excitation-only, Transmission-only, and Full SADL; the table displays the ineffective single branch and Full, since the unshown paired arms coincide seed by seed in the corresponding controlled bottleneck. The three setups respectively optimize H, encoder weights, and a shared bias with fixed informative weights. B measures held-out readout; C holds the information channel fixed and reports H RMS. Full uses $k = \min(C_{\text{exc}}, C_{\text{tr}})$; each single-branch variant uses one capacity alone.

表 27：同一固定探针的桥接缺口与编码器参数梯度。轮次及独立评估的 CLEAR-LeWM v0.8 Moderate 成功率（%；三个训练种子的均值 ± 样本标准差）与表 26 一致；各行训练预算不同。G_T、G_B 为任务和桥接梯度范数；G_B/G_T 平均逐单元比值。仅桥接的桥接梯度为零，因此余弦未定义（n/a）。† 表示反事实桥接诊断。

配置	Δ	G_T	G_B	G_B/G_T (%)	$\cos(G_T, G_B)$	成功率 (%)
LeWM						
端点†	.8180	.25432	.00756	3.843	-.0251	56.61 ± 6.49
仅桥接	.0000	.00092	.00000	.000	n/a	11.72 ± 1.80
传输	.0233	15.74698	.00023	.017	.0987	35.28 ± 27.30
双侧	.1886	.16440	.00196	1.763	.0815	61.00 ± 4.86
线性	.0790	.14240	.00408	3.697	.1885	53.53 ± 4.19
Exc.	.1740	.05307	.00080	1.852	.0588	63.72 ± 0.92
完整 SADL	.1739	.05342	.00081	1.946	.0592	60.78 ± 3.63
INTACT						
端点†	.9730	.69619	.01579	2.393	.1254	90.44 ± 3.14
仅桥接	.0000	1.10121	.00000	.000	n/a	47.00 ± 5.59
传输	.0053	314.61248	.00003	.003	.0966	42.94 ± 45.45
双侧	.2223	.97499	.00455	.948	.2988	93.97 ± 5.76
线性	.0829	.24236	.00472	2.123	.1361	97.14 ± 0.43
Exc.	.2032	.12964	.00110	.916	.0648	97.50 ± 0.22
完整 SADL	.2083	.15412	.00119	.887	.0416	98.03 ± 0.13

桥接项与任务项的编码器梯度范数平均比值在 LeWM 中为 $1.946 \pm 1.205\%$，INTACT 中为 $0.887 \pm 0.310\%$。这些是在同一探针上的更新前参数梯度，$G_T = \|\nabla_\theta \mathcal{L}_{\text{task}}\|_2$ 与 $G_B = \|\nabla_\theta(\lambda_S \mathcal{S})\|_2$；INTACT 的任务项还含动作负对数似然。完整 SADL 的任务–桥接余弦在 LeWM 中为 0.059 ± 0.173，INTACT 中为 0.042 ± 0.131；桥接–端点余弦分别为 0.338 ± 0.197 和 0.320 ± 0.164，标准差来自 12 个任务–种子单元。因此，小幅中间路径校正能改变路径几何而不压过任务梯度。所有仅桥接快照均有 $q_R = 0$ 且理想需求为零；尽管端点激发非零，平滑需求仍低于可用供给，因此桥接梯度为零。

F.3 两种供给容量的独立作用

表 28(A) 比较两种初始容量不足。独立、配对的十种子 200 步测试中，B/C 分别构造 $C_{\text{tr}} < q < C_{\text{exc}}$ 与 $C_{\text{exc}} < q < C_{\text{tr}}$。每组运行无 SADL、仅激发、仅传输及完整 SADL；表中只显示无效的单分支与完整版本，因为未展示的配对分支在对应受控瓶颈中逐种子一致。三种设置分别优化 H、编码器权重，以及固定有效权重下的共享偏置。B 测量保留集读出；C 固定信息通道，报告 H RMS。完整版本使用 $k = \min(C_{\text{exc}}, C_{\text{tr}})$；各单分支版本只使用一种容量。

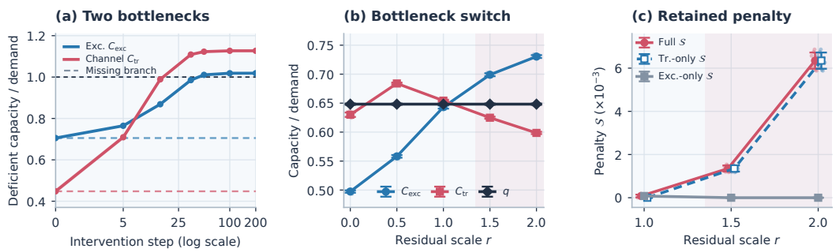


Figure 5: Independent capacity deficits and the transmission branch. (a) Matched-initialization controlled optimization of H: blue denotes excitation deficit, red transmission deficit; solid curves are Full SADL, dashed curves lack the needed branch. Curves show mean ± seed SD over 20 seeds; the horizontal dashed line marks capacity equal to demand. (b–c) Independent forward replay of an ImageNet-LT E100 checkpoint: endpoint and image mean are fixed while view-residual multiplier r varies. Pale dots show the same eight batches; filled dots and error bars show mean ± batch SD. The hollow blue squares in (c) are same-state Transmission-only counterfactual penalties, shifted horizontally only to separate overlapping error bars. At $r \geq 1.5$, $C_{\mathrm{tr}} < q < C_{\mathrm{exc}}$: Full and Transmission-only penalties coincide, whereas Excitation-only is zero. Panels (b–c) are forward-replay mechanism readouts, not retrained downstream success.

Table 28: Paired mechanism tests of two capacity deficits with fixed demand $q = .70$. (A) Direct optimization of H with 20 seeds. (B) Training encoder weights under a transmission deficit with held-out readout. (C) Training only a shared bias, with the informative weight channel fixed, under an excitation deficit. B/C use ten paired seeds each and display the ineffective single branch and Full SADL. Entries are initial → final mean ± seed SD.

(A) Controlled capacity optimization, 20 seeds

Initial deficit	Bridge target	C_{exc}, 0→200	C_{tr}, 0→200	$\mathcal{S}$, 0→200
Excitation	Transmission-only	.494 → .494 ± .002	.999 → .999 ± .000	.000 → .000
Excitation	Full SADL	.494 → **.713 ± .001**	.999 → .975 ± .001	.122 → .000
Transmission	Excitation-only	.976 → .976 ± .001	.314 → .314 ± .009	.000 → .000
Transmission	Full SADL	.976 → .978 ± .001	.314 → **.789 ± .003**	.646 → .000

(B) Transmission deficit, independent held-out readout, 10 seeds

Bridge target	C_{exc}, initial→final	C_{tr}, initial→final	Held-out R^2, initial→final
Excitation-only	.779 → .779 ± .003	.180 → .180 ± .005	−0.018 → −0.018±.002
Full SADL	.779 → .861 ± .003	.180 → **.746 ± .011**	−0.018 → .535 ± .018

(C) Excitation deficit, fixed weights/shared-bias training, 10 seeds

Bridge target	C_{exc}, initial→final	C_{tr}, initial→final	H RMS, initial→final
Transmission-only	.442 → .442 ± .002	.999 → .999 ± .000	2.234 → 2.234 ± .004
Full SADL	.442 → **.710 ± .002**	.999 → .999 ± .000	2.234 → 1.459 ± .003

Figure 5 juxtaposes the controlled deficits with a forward intervention on a real visual checkpoint. The former tests whether each capacity can independently become limiting; the latter tests whether the bridge still responds when transmission falls below demand. The same-state calculation shows why raising excitation alone cannot guarantee that the bridge has switched off: once transmission falls below demand, C_{tr} still sets the minimum capacity k.

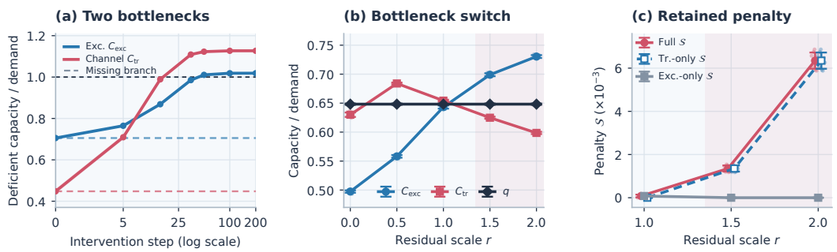


图 5：独立容量缺口与传输分支。(a) 匹配初始化下对 H 的受控优化：蓝色为激发不足，红色为传输不足；实线为完整 SADL，虚线缺少所需分支。曲线为 20 种子的均值 ± 种子标准差；水平虚线表示容量等于需求。(b–c) ImageNet-LT E100 检查点的独立前向重放：端点与图像均值固定，改变视图残差乘数 r。浅色点为相同八个批次；实心点和误差条为均值 ± 批次标准差。(c) 空心蓝方块为相同状态下仅传输分支的反事实惩罚，水平偏移仅用于分开重叠误差条。在 $r \geq 1.5$ 时，$C_{\mathrm{tr}} < q < C_{\mathrm{exc}}$：完整与仅传输惩罚相同，而仅激发惩罚为零。(b–c) 是前向重放的机制读数，不是重新训练后的下游成功率。

表 28：固定需求 $q = .70$ 下两种容量缺口的配对机制测试。(A) 20 种子直接优化 H。(B) 传输不足时训练编码器权重，以保留集读出评估。(C) 信息权重通道固定、激发不足时，仅训练共享偏置。B/C 各十个配对种子，展示无效单分支与完整 SADL。数值为初始 → 最终均值 ± 种子标准差。

(A) 受控容量优化，20 种子

初始缺口	桥接目标	C_{exc}, 0→200	C_{tr}, 0→200	$\mathcal{S}$, 0→200
激发	仅传输	.494 → .494 ± .002	.999 → .999 ± .000	.000 → .000
激发	完整 SADL	.494 → **.713 ± .001**	.999 → .975 ± .001	.122 → .000
传输	仅激发	.976 → .976 ± .001	.314 → .314 ± .009	.000 → .000
传输	完整 SADL	.976 → .978 ± .001	.314 → **.789 ± .003**	.646 → .000

(B) 传输不足，独立保留集读出，10 种子

桥接目标	C_{exc}，初始 → 最终	C_{tr}，初始 → 最终	保留集 R^2，初始 → 最终
仅激发	.779 → .779 ± .003	.180 → .180 ± .005	−0.018 → −0.018±.002
完整 SADL	.779 → .861 ± .003	.180 → **.746 ± .011**	−0.018 → .535 ± .018

(C) 激发不足，固定权重/共享偏置训练，10 种子

桥接目标	C_{exc}，初始 → 最终	C_{tr}，初始 → 最终	H RMS，初始 → 最终
仅传输	.442 → .442 ± .002	.999 → .999 ± .000	2.234 → 2.234 ± .004
完整 SADL	.442 → **.710 ± .002**	.999 → .999 ± .000	2.234 → 1.459 ± .003

图 5 将受控容量缺口与真实视觉检查点的前向干预并列。前者检验两种容量是否可独立成为限制因素；后者检验传输降到需求以下时桥接是否仍响应。同状态计算说明，单独提高激发不能保证桥接关闭：一旦传输低于需求，C_{tr} 仍决定最小容量 k。

F.4 Demand interpolation coefficient α

$\alpha = 0$ emphasizes external demand and is generally more aggressive; $\alpha = 1$ emphasizes realized endpoint demand and is more conservative. Table 29 gives realized demand and outcome for the same ImageNet-LT seed.

Table 29: Single-seed development ablation of demand interpolation α on ImageNet-LT (seed 42). q is the realized smoothed E20 demand, additionally capped by attainable supply in this image branch; online accuracy is at E100. $\alpha = 0$ favors the more aggressive external demand, whereas $\alpha = 1$ favors the conservative realized endpoint demand. In this setting, the highest observed result is at $\alpha = .5$.

Configuration	E20 demand q	E100 Top-1 (%)	vs. baseline (pp)
VISReg baseline	n/a	22.034	0.000
SADL $\alpha = 0$	.772	22.904	+0.870
SADL $\alpha = .25$	.772	22.348	+0.314
SADL $\alpha = .5$	.667	**23.026**	**+0.992**
SADL $\alpha = .75$	.554	22.144	+0.110
SADL $\alpha = 1$	.460	22.064	+0.030

F.5 Demand–Supply Geometry and Dual-Endpoint Constraints

We compare a simple H-side barrier, direct dual-endpoint regularization, and SADL. For centered H covariance Σ_H, the barrier penalizes low average feature standard deviation or excessive spectral concentration:

$$\mathcal{R}_{\text{bar}}(H) = [.10 - \overline{\sigma}(H)]_+^2 + [\lambda_{\max}(\Sigma_H)/\operatorname{tr}(\Sigma_H) - .35]_+^2. \tag{18}$$

The tested barrier configuration retains SIGReg(Z) at weight .03; its H weight is .03 in E1 and .003 thereafter. It is therefore an H-side regularizer added to an endpoint objective, not an H-only training objective. The dual recipe instead applies the same distributional penalty $\mathcal{R}$ at H and Z, together with the tested scale floors F_γ:

$$\mathcal{L}_{\text{dual}} = \mathcal{L}_{\text{task}} + \lambda_R[\mathcal{R}(H) + \mathcal{R}(Z)] + 0.02[F_{0.5}(H) + F_{1.0}(Z)]. \tag{19}$$

Here $\mathcal{R}$ is SIGReg, VISReg, or QQWorld; F_γ is the fixed scale-floor term with threshold γ. The three tested λ_R values are .02, 1.5, and 3.5, respectively. This is a *dual-plus-floor* control, not a pure two-endpoint penalty. SADL instead retains the selected endpoint loss $\lambda_D\mathcal{D}(Z)$ and applies the one-sided supply loss $\lambda_S\mathcal{S}(H, \text{sg}(Z); \text{sg}(q))$ only for a measured demand–supply deficit (Section 3).

Table 30 reports four-task Official control success for three training seeds per recipe. LeWM uses Pure-CEM and INTACT uses Direct; each task score is averaged over three evaluation seeds. The late checkpoint is E2 for dual LeWM and E5 for SADL LeWM and both INTACT recipes; LeWM late scores therefore are not equal-budget comparisons. The barrier, dual-plus-floor, and SADL recipes were trained in separate experimental matrices, so the table compares observed recipes rather than estimating a paired causal effect of replacing one loss term.

F.4 需求插值系数 α

$\alpha = 0$ 强调外部需求，通常更积极； $\alpha = 1$ 强调已实现端点需求，更保守。表 29 给出同一 ImageNet-LT 种子的已实现需求与结果。

表 29：ImageNet-LT 上需求插值 α 的单种子开发消融（种子 42）。q 为已实现的平滑 E20 需求，在该图像分支还以可达到供给为上限；在线准确率测于 E100。 $\alpha = 0$ 偏向更积极的外部需求，$\alpha = 1$ 偏向保守的已实现端点需求。该设置中观测最高结果位于 $\alpha = .5$。

配置	E20 需求 q	E100 Top-1（%）	相对基线（百分点）
VISReg 基线	n/a	22.034	0.000
SADL $\alpha = 0$	.772	22.904	+0.870
SADL $\alpha = .25$	.772	22.348	+0.314
SADL $\alpha = .5$	.667	**23.026**	**+0.992**
SADL $\alpha = .75$	.554	22.144	+0.110
SADL $\alpha = 1$	.460	22.064	+0.030

F.5 需求–供给几何与双端点约束

我们比较简单 H 侧屏障、直接双端点正则化与 SADL。对于中心化 H 协方差 Σ_H，屏障惩罚较低平均特征标准差或过度谱集中：

$$\mathcal{R}_{\text{bar}}(H) = [.10 - \overline{\sigma}(H)]_+^2 + [\lambda_{\max}(\Sigma_H)/\operatorname{tr}(\Sigma_H) - .35]_+^2. \tag{18}$$

测试屏障配置保留权重 .03 的 SIGReg(Z)；其 H 权重在 E1 为 .03，之后为 .003。因此它是在端点目标上加入 H 侧正则项，而非仅 H 的训练目标。双端点配方则在 H 、 Z 施加相同分布惩罚 $\mathcal{R}$ ，并加入所测试的尺度下限 F_γ：

$$\mathcal{L}_{\text{dual}} = \mathcal{L}_{\text{task}} + \lambda_R[\mathcal{R}(H) + \mathcal{R}(Z)] + 0.02[F_{0.5}(H) + F_{1.0}(Z)]. \tag{19}$$

这里 $\mathcal{R}$ 为 SIGReg、VISReg 或 QQWorld； F_γ 是阈值为 γ 的固定尺度下限项。三个测试的 λ_R 值分别为 .02、1.5、3.5。这是双端点加下限对照，而非纯双端点惩罚。SADL 则保留所选端点损失 $\lambda_D\mathcal{D}(Z)$，仅在测得需求–供给缺口时施加单侧供给损失 $\lambda_S\mathcal{S}(H,\text{sg}(Z);\text{sg}(q))$（第 3 节）。

表 30 报告每种配方三个训练种子的四任务官方控制成功率。LeWM 使用 Pure-CEM，INTACT 使用 Direct；每任务得分对三个评估种子平均。双端点 LeWM 的晚期检查点是 E2，SADL LeWM 及两种 INTACT 配方为 E5，因此 LeWM 晚期得分不是等预算比较。屏障、双端点加下限、SADL 配方在独立实验矩阵中训练，故该表比较观测配方，而非估计替换一个损失项的配对因果效应。

Table 30: Simple H-side barrier, direct dual-endpoint regularization with scale floors, and SADL in jointly trained four-task control. Official success rates are percentages; macro entries are mean ± sample SD over three training seeds. Superscripts in the late column give the checkpoint epoch.

Model	Endpoint	Recipe	E1 PushT	E1 Cube	E1 Reacher	E1 TwoRoom	E1 Macro	Late Macro
LeWM	SIGReg	Dual+floor	7.56	61.00	80.67	74.67	55.97 ± .17	$58.06 \pm .89^{E2}$
		SADL	28.78	70.22	79.89	73.00	**62.97 ± 1.25**	$70.64 \pm .77^{E5}$
	VISReg	Dual+floor	8.89	65.33	77.67	81.44	58.33 ± 1.09	$59.56 \pm .35^{E2}$
		SADL	53.56	66.44	78.44	88.11	**71.64 ± 2.96**	$78.58 \pm .98^{E5}$
	QQWorld	Dual+floor	7.22	64.44	78.89	71.67	55.56 ± 1.65	56.31 ± 2.12^{E2}
		SADL	45.67	67.44	78.00	82.56	**68.42 ± 4.19**	75.11 ± 1.34^{E5}
INTACT	SIGReg	H barrier	26.56	70.78	11.22	48.44	39.25 ± 12.74	48.00 ± 41.68^{E5}
INTACT	SIGReg	Dual+floor	42.11	99.11	94.78	98.00	83.50 ± 1.99	$95.81 \pm .43^{E5}$
		SADL	85.22	100.00	97.11	100.00	**95.58 ± .17**	$97.19 \pm .51^{E5}$
	VISReg	Dual+floor	61.11	99.44	96.67	97.56	88.69 ± 3.58	$96.67 \pm .36^{E5}$
		SADL	83.78	100.00	97.78	100.00	**95.39 ± .65**	$97.53 \pm .25^{E5}$
	QQWorld	Dual+floor	50.00	99.67	95.89	96.45	85.50 ± 1.38	$95.28 \pm .46^{E5}$
		SADL	86.00	100.00	96.56	100.00	**95.64 ± .32**	$97.56 \pm .17^{E5}$

The simple barrier is not uniformly ineffective: its three E5 training-seed macros are 28.67, 19.50, and 95.83%, but the aggregate remains $48.00 \pm 41.68\%$. At E2 its macro is $62.00 \pm 28.10\%$, compared with $96.56 \pm .17\%$ for three-seed SIGReg+SADL in the separate E2 mechanism audit (Table 3(b)). These are independent runs, not a paired loss-only ablation. Dual-endpoint regularization is also a viable control: at E5 it attains 95.28–96.67% INTACT Direct success. However, SADL is higher for all six model–endpoint pairs at E1, with the largest early gap on INTACT–SIGReg (83.50% versus 95.58%). The gain is most visible on PushT. At the matched INTACT E5 checkpoint, SADL remains higher by .86–2.28 percentage points; the margin has narrowed, but is not restricted to E1. These observations are consistent with adapting a path correction to measured demand and available supply rather than enforcing a fixed geometry at both ends, but the separately trained recipes do not isolate that mechanism on their own. Fixed-probe effective-rank, cosine, and prediction-loss evidence for SADL is in Table 2; the dual-plus-floor runs do not have an audited matched fixed-probe readout.

表 30：联合四任务控制中的简单 H 侧屏障、含尺度下限的直接双端点正则化及 SADL。官方成功率为百分比；宏平均为三个训练种子的均值 ± 样本标准差。晚期列的上标表示检查点轮次。

模型	端点	配方	E1 PushT	E1 Cube	E1 Reacher	E1 TwoRoom	E1 宏平均	晚期宏平均
LeWM	SIGReg	双端点 + 下限	7.56	61.00	80.67	74.67	55.97 ± .17	$58.06 \pm .89^{E2}$
		SADL	28.78	70.22	79.89	73.00	**62.97 ± 1.25**	$70.64 \pm .77^{E5}$
	VISReg	双端点 + 下限	8.89	65.33	77.67	81.44	58.33 ± 1.09	$59.56 \pm .35^{E2}$
		SADL	53.56	66.44	78.44	88.11	**71.64 ± 2.96**	$78.58 \pm .98^{E5}$
	QQWorld	双端点 + 下限	7.22	64.44	78.89	71.67	55.56 ± 1.65	56.31 ± 2.12^{E2}
		SADL	45.67	67.44	78.00	82.56	**68.42 ± 4.19**	75.11 ± 1.34^{E5}
INTACT	SIGReg	H 屏障	26.56	70.78	11.22	48.44	39.25 ± 12.74	48.00 ± 41.68^{E5}
INTACT	SIGReg	双端点 + 下限	42.11	99.11	94.78	98.00	83.50 ± 1.99	$95.81 \pm .43^{E5}$
		SADL	85.22	100.00	97.11	100.00	**95.58 ± .17**	$97.19 \pm .51^{E5}$
	VISReg	双端点 + 下限	61.11	99.44	96.67	97.56	88.69 ± 3.58	$96.67 \pm .36^{E5}$
		SADL	83.78	100.00	97.78	100.00	**95.39 ± .65**	$97.53 \pm .25^{E5}$
	QQWorld	双端点 + 下限	50.00	99.67	95.89	96.45	85.50 ± 1.38	$95.28 \pm .46^{E5}$
		SADL	86.00	100.00	96.56	100.00	**95.64 ± .32**	$97.56 \pm .17^{E5}$

简单屏障并非始终无效：三个 E5 训练种子的宏平均为 28.67、19.50、95.83%，但汇总仍为 $48.00 \pm 41.68\%$。其 E2 宏平均为 $62.00 \pm 28.10\%$，而独立 E2 机制审计中三种子 SIGReg+SADL 为 $96.56 \pm .17\%$（表 3(b)）。这些是独立运行，不是只改变损失的配对消融。双端点正则化也是可行对照：E5 的 INTACT Direct 成功率为 95.28–96.67%。然而，E1 的六组模型–端点配对中 SADL 均更高，最大早期差距在 INTACT–SIGReg（83.50% 对 95.58%）。增益在 PushT 最明显。在匹配的 INTACT E5 检查点，SADL 仍高 .86–2.28 个百分点；优势缩小，但不限于 E1。这符合根据测量需求和可用供给自适应校正路径、而非在两端强制固定几何的解释；不过分别训练的配方本身不能隔离该机制。SADL 的固定探针有效秩、余弦及预测损失证据见表 2；双端点加下限运行没有经过审计的匹配固定探针读数。

G Endpoint Distributions and Representation Geometry

G.1 Eight endpoint objectives and designable two-dimensional geometry

Let $Z \in \mathbb{R}^{T\times B\times d}$ and let v_j be random unit projection directions. Define $Q_\nu(Z)$ by sorting each projected batch against the quantiles of a reference distribution ν:

$$Q_\nu(Z) = \frac{1}{TBK}\sum_{t=1}^{T}\sum_{j=1}^{K}\sum_{i=1}^{B}\left((Z_t v_j)_{(i)} - F_\nu^{-1}\left(\frac{i-1/2}{B}\right)\right)^2.$$

For independent reference samples $y, y' \sim \nu$, the empirical energy distance is

$$E_\nu(Z) = \frac{1}{\sqrt{d}}\left[2\mathbb{E}\|z-y\|_2 - \mathbb{E}\|z-z'\|_2 - \mathbb{E}\|y-y'\|_2\right]_+.$$

At most 96 samples per temporal position are used to compute the energy distance.

Four Gaussian objectives. Write $\nu_G = \mathcal{N}(0, I)$, let $\widehat{\varphi}_Z$ be the empirical characteristic function, and let μ_j, σ_j denote coordinate-wise means and standard deviations. Set $\widetilde{Z} = (Z-\mu)/\operatorname{sg}(\sigma)$. The four objectives are

$$\mathcal{D}_{\mathrm{SIG}}(Z) \propto \left\langle |\widehat{\varphi}_Z(\omega) - e^{-\|\omega\|^2/2}|^2\right\rangle_\omega, \qquad \mathcal{D}_{\mathrm{QQ}}(Z) = Q_{\nu_G}(Z),$$
$$\mathcal{D}_{\mathrm{VIS}}(Z) = \left\langle (\sigma_j - 1)^2 + \mu_j^2\right\rangle_j + Q_{\nu_G}(\widetilde{Z}), \qquad \mathcal{D}_{\mathrm{EG}}(Z) = E_{\nu_G}(Z).$$

All share a Gaussian reference, but respectively constrain the characteristic function, separated scale and shape, raw sorted projections, and sample distances; their gradients and finite-sample behavior differ.

Four non-Gaussian or partially specified objectives. Let C be the sample covariance of the TB endpoint samples and $u = z/\|z\|_2$. VIC moments specifies no full probability density:

$$\mathcal{D}_{\mathrm{VIC}}(Z) = \left\langle \left[1 - \sqrt{\operatorname{Var}(Z_j) + 10^{-4}}\right]_+\right\rangle_j + \left\langle C_{ij}^2\right\rangle_{i\neq j} + \left\langle \mu_j^2\right\rangle_j.$$

Sphere, Student-t_5, and GMM-Ring use, respectively,

$$\mathcal{D}_{\mathrm{Sphere}}(Z) = \left\langle \left(\|z\|_2/\sqrt{d} - 1\right)^2\right\rangle_z + \left\langle e^{-2\|u_i - u_j\|_2^2}\right\rangle_{t,i\neq j},$$
$$\mathcal{D}_{t_5}(Z) = Q_{\nu_{t_5}}(Z), \qquad F^{-1}_{\nu_{t_5}}(p) = \sqrt{3/5}\, F^{-1}_{t_5}(p),$$
$$\mathcal{D}_{\mathrm{Ring}}(Z) = E_{\nu_{\mathrm{Ring}}}(Z), \qquad \nu_{\mathrm{Ring}} = W_{\#}\left[\frac{1}{16}\sum_{\ell=0}^{15}\mathcal{N}\left(2.5 m_\ell,\, 0.35^2 I\right)\right],$$
$$m_\ell = \cos\frac{2\pi\ell}{16}u_1 + \sin\frac{2\pi\ell}{16}u_2.$$

Here $W_{\#}$ analytically whitens the two-dimensional ring subspace and u_1, u_2 form an orthonormal basis. The two Sphere terms are added directly; Student-t_5 uses 128 projection directions. At $Z = 0$, Sphere sets $u = 0$ through its positive norm floor, and VISReg sets $\widetilde{Z} = 0$ through its positive standard-deviation floor; the cached $D_0 = \mathcal{D}(0)$ is evaluated with the same conventions.

G.2 Learned endpoint geometry and designable glyphs

The eight-objective control results are reported in Table 4 in the main text. For SIGReg and QQ-World, the first-step encoder-gradient calibration was performed separately for LeWM and INTACT, with VISReg anchored at $\lambda_D = 1.5$; the five additional objectives were likewise calibrated at the first step of seed 0. Each model-specific endpoint weight was then held fixed during training.

Figure 6 displays the corresponding E1 representations on held-out clips. Its first two rows show learned endpoint geometries under eight objectives; the third row is a separate demonstration of

G 端点分布与表征几何

G.1 八个端点目标与可设计二维几何

设 $Z \in \mathbb{R}^{T\times B\times d}$，$v_j$ 为随机单位投影方向。将每个投影批次排序后与参考分布 ν 的分位数比较，定义 $Q_\nu(Z)$：

$$Q_\nu(Z) = \frac{1}{TBK}\sum_{t=1}^{T}\sum_{j=1}^{K}\sum_{i=1}^{B}\left((Z_t v_j)_{(i)} - F_\nu^{-1}\left(\frac{i-1/2}{B}\right)\right)^2.$$

对于独立参考样本 y、$y' \sim \nu$，经验能量距离为

$$E_\nu(Z) = \frac{1}{\sqrt{d}}\left[2\mathbb{E}\|z-y\|_2 - \mathbb{E}\|z-z'\|_2 - \mathbb{E}\|y-y'\|_2\right]_+.$$

每个时间位置至多使用 96 个样本计算能量距离。

四个高斯目标。记 $\nu_G = \mathcal{N}(0, I)$，令 $\widehat{\varphi}_Z$ 为经验特征函数，$\widetilde{\ }$，μ_j、σ_j 为逐坐标均值和标准差。设 $Z = (Z-\mu)/\operatorname{sg}(\sigma)$。四个目标为

$$\mathcal{D}_{\mathrm{SIG}}(Z) \propto \left\langle |\widehat{\varphi}_Z(\omega) - e^{-\|\omega\|^2/2}|^2\right\rangle_\omega, \qquad \mathcal{D}_{\mathrm{QQ}}(Z) = Q_{\nu_G}(Z),$$
$$\mathcal{D}_{\mathrm{VIS}}(Z) = \left\langle (\sigma_j - 1)^2 + \mu_j^2\right\rangle_j + Q_{\nu_G}(\widetilde{Z}), \qquad \mathcal{D}_{\mathrm{EG}}(Z) = E_{\nu_G}(Z).$$

它们共享高斯参考，但分别约束特征函数、分离的尺度与形状、原始排序投影、样本距离；梯度和有限样本行为不同。

四个非高斯或部分指定的目标。设 C 为 TB 个端点样本的样本协方差，且 $u = z/\|z\|_2$。VIC 矩不规定完整概率密度：

$$\mathcal{D}_{\mathrm{VIC}}(Z) = \left\langle \left[1 - \sqrt{\operatorname{Var}(Z_j) + 10^{-4}}\right]_+\right\rangle_j + \left\langle C_{ij}^2\right\rangle_{i\neq j} + \left\langle \mu_j^2\right\rangle_j.$$

Sphere、Student-t_5、GMM-Ring 分别使用

$$\mathcal{D}_{\mathrm{Sphere}}(Z) = \left\langle \left(\|z\|_2/\sqrt{d} - 1\right)^2\right\rangle_z + \left\langle e^{-2\|u_i - u_j\|_2^2}\right\rangle_{t,i\neq j},$$
$$\mathcal{D}_{t_5}(Z) = Q_{\nu_{t_5}}(Z), \qquad F^{-1}_{\nu_{t_5}}(p) = \sqrt{3/5}\, F^{-1}_{t_5}(p),$$
$$\mathcal{D}_{\mathrm{Ring}}(Z) = E_{\nu_{\mathrm{Ring}}}(Z), \qquad \nu_{\mathrm{Ring}} = W_{\#}\left[\frac{1}{16}\sum_{\ell=0}^{15}\mathcal{N}\left(2.5 m_\ell,\, 0.35^2 I\right)\right],$$
$$m_\ell = \cos\frac{2\pi\ell}{16}u_1 + \sin\frac{2\pi\ell}{16}u_2.$$

这里 $W_{\#}$ 解析白化二维环子空间，u_1、u_2 构成正交归一基。两个 Sphere 项直接相加；Student-t_5 使用 128 个投影方向。当 $\sim Z = 0$ 时，Sphere 通过正范数下限设定 $u = 0$，VISReg 通过正标准差下限设定 $Z = 0$；缓存的 $D_0 = \mathcal{D}(0)$ 使用相同约定计算。

G.2 学习所得端点几何与可设计字形

八目标控制结果见正文表 4。SIGReg 与 QQWorld 的首步编码器梯度校准对 LeWM、INTACT 分别进行，VISReg 锚定为 $\lambda_D = 1.5$；其他五个目标也在种子 0 的第一步校准。随后各模型专用端点权重在训练中固定。

图 6 展示保留片段上的对应 E1 表征。前两行为八个目标下学习的端点几何；第三行单独展示

designable two-dimensional glyph endpoints. The learned representations have no visually obvious collapse in these visualizations, although neither a low-dimensional projection nor control SR alone establishes a zero incidence of strict collapse.

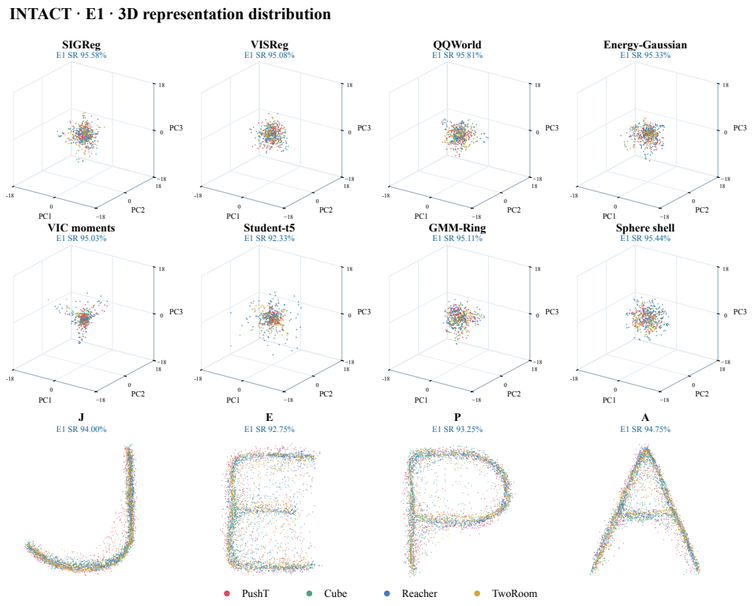


Figure 6: Real E1 representations with four-task color coding; all particles are vector objects. The first two rows show independent 3-D PCA projections of INTACT+SADL z on a fixed probe; blue SR values are three-seed Official Direct means. The third row shows independently trained raw 2-D glyph representations; blue SR values are single-seed Direct macro success, evaluated with 100 episodes per task, and are therefore not directly comparable to those above. The image illustrates learned local geometry, not the original high-dimensional density.

G.3 Eight-endpoint latent-geometry diagnostic

The diagnostic uses E1 INTACT checkpoints from three training seeds (0, 42, 3072); Official success is the corresponding three-seed aggregate. For each task, 128 segments with eight positions yield 4,096 pooled observations. A reference subset of 1,024 fits whitening, and the remaining 3,072 observations are analyzed. With nonzero covariance spectrum, we adapt the entropy-based effective rank of Roy & Vetterli (2007) to covariance eigenvalues:

$$r_{\text{eff}}(U) = \exp\left(-\sum_i p_i \log p_i\right), \qquad p_i = \frac{\lambda_i(\Sigma_U)}{\operatorname{tr}(\Sigma_U)},$$

where $0 \log 0 = 0$; raw effective rank is computed before whitening. We also report leading-eigenvalue share and average absolute excess kurtosis across random standardized projections:

$$\eta_1 = \frac{\lambda_{\max}(\Sigma_Z)}{\operatorname{tr}(\Sigma_Z)}, \qquad \bar{\kappa} = \langle |\operatorname{kurt}(Zv)| \rangle_v.$$

Each checkpoint fits its own whitening transform, while observation indices, 32 random projection directions, and the split are shared. Four-task observations are pooled within each seed before

可设计的二维字形端点。这些可视化中的学习表征没有明显可见坍塌，但低维投影或控制成功率都不能单独证明严格坍塌发生率为零。

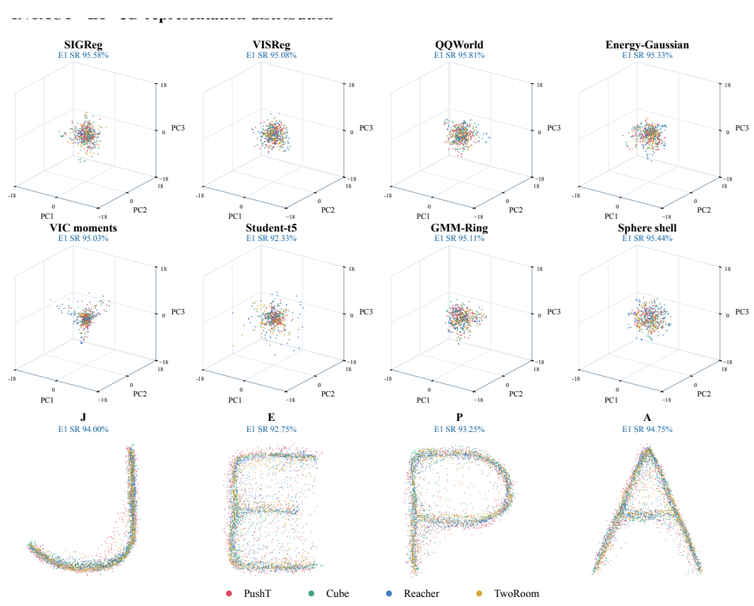


图 6：按四任务着色的真实 E1 表征；全部粒子均为矢量对象。前两行展示固定探针上 INTACT+SADL z 的独立三维 PCA 投影；蓝色成功率为三种子官方 Direct 均值。第三行为独立训练的原始二维字形表征；蓝色成功率是单种子 Direct 宏平均，每任务评估 100 回合，故不能与上方直接比较。图中展示学习得到的局部几何，而非原始高维密度。

G.3 八端点潜空间几何诊断

诊断使用三个训练种子（0、42、3072）的 E1 INTACT 检查点；官方成功率为对应三种子汇总。每任务 128 段、每段八个位置，共汇集 4,096 个观测。1,024 个参考观测拟合白化，其余 3,072 个用于分析。在非零协方差谱上，我们将 Roy 与 Vetterli（2007）基于熵的有效秩改写到协方差特征值：

$$r_{\text{eff}}(U) = \exp\left(-\sum_i p_i \log p_i\right), \qquad p_i = \frac{\lambda_i(\Sigma_U)}{\operatorname{tr}(\Sigma_U)},$$

其中 $0 \log 0 = 0$；原始有效秩在白化前计算。我们还报告首特征值占比，以及随机标准化投影的平均绝对超额峰度：

$$\eta_1 = \frac{\lambda_{\max}(\Sigma_Z)}{\operatorname{tr}(\Sigma_Z)}, \qquad \bar{\kappa} = \langle |\operatorname{kurt}(Zv)| \rangle_v.$$

各检查点拟合自身白化变换，共享观测索引、32 个随机投影方向与划分。各种子内汇集四任务观测，然后

reporting mean and sample standard deviation across seeds. Training uses within-task covariance, whereas pooled covariance obeys

$$\mathrm{Cov}(Z) = \mathbb{E}_T[\mathrm{Cov}(Z \mid T)] + \mathrm{Cov}_T(\mathbb{E}[Z \mid T]).$$

Function readout uses 32 smooth random Fourier functions $f_j(x) = \sin(a_j \omega_j^\top x + b_j)$ in each checkpoint's whitened coordinates. Directions, frequencies, and phases are shared, while x changes with the representation; these are isomorphic coordinate-defined function classes, not a shared physical-state target. For 128 fixed queries, $N = 128$ or 1,024 training samples, and eight independent resamples, kNN with $n_{\mathrm{nbr}} = 15$ and fixed-bandwidth RBF predictions are averaged before integrated squared bias is computed. Table 31 gathers the eight-objective E1 geometry, projected Fisher proxy, kNN bias, and control outcomes.

G.4 Asymptotic endpoint optimality and finite-training attainability

LeJEPA analyzes linear and nonlinear probe risk under fixed covariance scale, smooth full-dimensional densities, and unknown downstream tasks. Its Fisher-information argument selects an *isotropic Gaussian* endpoint, not an arbitrary anisotropic Gaussian. SADL does not dispute this endpoint choice; it asks whether the encoder–projector path can produce and transmit enough sample variation to attain it within a training budget T. Conceptually, the evaluation levels can be separated as

$$R_T(p; \mathcal{F}) \;\approx\; R_{\text{asymptotic probe}}(p) + \Delta_{\text{path/optimization},T}(p) + \Delta_{\text{task}}(p; \mathcal{F}), \qquad p \in \mathcal{P}_T.$$

Here $\mathcal{P}_T$ is the set of endpoints attainable in T steps and $\mathcal{F}$ is the task family. The expression distinguishes asymptotic probing, finite-training path effects, and task fit, with interactions assigned to the final term. Consequently, asymptotic, task-unknown Gaussian optimality does not automatically imply system-level optimality for a restricted path, finite budget, and specific control tasks.

We first calibrate synthetic distributions at matched total covariance, observation indices, smooth function family, and kNN-15 readout, then examine learned E1 endpoints. For synthetic Student-t_5, the full-dimensional Fisher ratio has the closed-form benchmark

$$\frac{J(t_\nu)}{J(\mathcal{N}(0, I_d))} = \frac{\nu(\nu + d)}{(\nu - 2)(\nu + d + 2)} = 1.650 \quad (d = 192,\ \nu = 5).$$

Synthetic full Fisher ratios are 1.000 for Gaussian, 1.650 for Student-t_5, and 2.283 for a two-scale full-dimensional Gaussian mixture, consistent with LeJEPA's theorem. Sphere lies on a singular lower-dimensional manifold outside its full-density assumptions.

The eight-objective E1 three-seed experiment reveals a distinct finite-time perspective (Table 31): the mean projected Fisher proxy is $1.220 \pm .083$ for GMM-Ring versus $1.287 \pm .053$ for SIGReg; finite-sample b_{1024} is $.1590 \pm .0025$ for Sphere versus $.1762 \pm .0049$ for SIGReg, and is lower in all three paired seeds. Together with Table 4, this shows that SADL's anti-collapse behavior and high control success do not depend on a Gaussian endpoint. LeJEPA's asymptotic Fisher conclusion remains applicable under its stated conditions. All geometry entries are E1 measurements; E5 appears only for control success.

报告跨种子均值与样本标准差。训练使用任务内协方差，而汇集协方差满足

$$\mathrm{Cov}(Z) = \mathbb{E}_T[\mathrm{Cov}(Z \mid T)] + \mathrm{Cov}_T(\mathbb{E}[Z \mid T]).$$

函数读出在各检查点的白化坐标中使用 32 个平滑随机 Fourier 函数 $f_j(x) = \sin(a_j \omega_j^\top x + b_j)$。方向、频率、相位共享，而 x 随表征变化；这是同构的坐标定义函数类，并非共享物理状态目标。对 128 个固定查询、$N = 128$ 或 1,024 个训练样本及八次独立重采样，先平均使用 $n_{\mathrm{nbr}} = 15$ 的 kNN 与固定带宽 RBF 预测，再计算积分平方偏差。表 31 汇总八目标 E1 几何、投影 Fisher 代理、kNN 偏差及控制结果。

G.4 渐近端点最优性与有限训练可达性

LeJEPA 在固定协方差尺度、平滑全维密度及未知下游任务条件下分析线性与非线性探针风险。Fisher 信息论证选择各向同性高斯端点，而非任意各向异性高斯。SADL 不反对这一端点选择，而是追问编码器–投影器路径能否在预算 T 内产生并传输足够样本变化以实现它。概念上，可将评估层级分开为

$$R_T(p; \mathcal{F}) \;\approx\; R_{\text{asymptotic probe}}(p) + \Delta_{\text{path/optimization},T}(p) + \Delta_{\text{task}}(p; \mathcal{F}), \qquad p \in \mathcal{P}_T.$$

这里 $\mathcal{P}_T$ 为 T 步内可达到的端点集合，$\mathcal{F}$ 为任务族。该表达式区分渐近探针、有限训练路径效应和任务适配，将交互作用归入末项。因此，未知任务下渐近高斯最优性，不自动意味着受限路径、有限预算和特定控制任务的系统层面最优性。

我们先在匹配总协方差、观测索引、平滑函数族和 kNN-15 读出下校准合成分布，再检查学习的 E1 端点。合成 Student-t_5 的全维 Fisher 比值具有闭式基准

$$\frac{J(t_\nu)}{J(\mathcal{N}(0, I_d))} = \frac{\nu(\nu + d)}{(\nu - 2)(\nu + d + 2)} = 1.650 \quad (d = 192,\ \nu = 5).$$

合成完整 Fisher 比值：高斯为 1.000，Student-t_5 为 1.650，双尺度全维高斯混合为 2.283，符合 LeJEPA 定理。Sphere 位于奇异低维流形，不满足其全密度假设。

八目标 E1 三种子实验揭示不同的有限时间视角（表 31）：GMM-Ring 平均投影 Fisher 代理为 $1.220 \pm .083$，SIGReg 为 $1.287 \pm .053$；Sphere 有限样本 b_{1024} 为 $.1590 \pm .0025$，SIGReg 为 $.1762 \pm .0049$，且全部三个配对种子均更低。结合表 4，这说明 SADL 的防坍塌与高控制成功率不依赖高斯端点。LeJEPA 的渐近 Fisher 结论在其既定条件下仍然适用。全部几何数值在 E1 测量；E5 仅报告控制成功率。

Table 31: Three-training-seed E1 geometry, finite-sample readout, and Official Direct control across eight endpoints. All models are jointly trained four-task INTACT+SADL; each seed pools 4,096 positions from the same fixed four-task observations. Geometry and bias are mean ± sample seed SD; E1/E5 SR also aggregates three training seeds. The two panels use identical endpoint order. $10^3\mathrm{SW}^2$ denotes squared sliced-Wasserstein distance to a Gaussian reference, multiplied by 10^3. Geometry and readout are measured at E1; the E5 column contains only control SR from E5 checkpoints.

A. Path and endpoint geometry: mean ± seed SD

Endpoint	$r_{\mathrm{eff}}(h)$	$r_{\mathrm{eff}}(z)$	η_1 (%)	$\bar{\kappa}$	$\widehat{J}_{\mathrm{proj}}$
SIGReg	118.85 ± 1.19	58.68 ± 1.77	3.45 ± 0.22	$.573 \pm .103$	$1.287 \pm .053$
VISReg	122.11 ± 1.09	60.44 ± 1.26	3.34 ± 0.06	$.473 \pm .074$	$1.250 \pm .077$
QQWorld	120.25 ± 0.90	58.82 ± 0.76	3.47 ± 0.11	$.604 \pm .063$	$1.213 \pm .004$
Energy-Gaussian	112.86 ± 1.69	48.10 ± 0.56	4.26 ± 0.08	$.677 \pm .098$	$1.259 \pm .077$
VIC moments	106.17 ± 1.53	66.42 ± 0.35	3.01 ± 0.22	$1.035 \pm .158$	$1.330 \pm .066$
Sphere	102.61 ± 1.29	42.30 ± 0.60	4.96 ± 0.15	$.920 \pm .062$	$1.251 \pm .081$
Student-t_5	89.91 ± 3.03	35.11 ± 1.27	6.38 ± 0.47	$2.840 \pm .319$	$1.660 \pm .135$
GMM-Ring	112.46 ± 0.34	48.29 ± 0.57	4.20 ± 0.10	$.657 \pm .066$	$1.220 \pm .083$

B. Distribution distance, function readout, and control: mean ± seed SD

Endpoint	$10^3\mathrm{SW}^2$	b_{128}	b_{1024}	E1 SR (%)	E5 SR (%)
SIGReg	5.19 ± 0.97	$.2558 \pm .0056$	$.1762 \pm .0049$	$95.58 \pm .00$	$97.19 \pm .51$
VISReg	4.02 ± 0.46	$.2505 \pm .0049$	$.1716 \pm .0044$	$95.08 \pm .17$	$97.53 \pm .25$
QQWorld	5.37 ± 0.98	$.2552 \pm .0045$	$.1745 \pm .0035$	$95.81 \pm .13$	$97.56 \pm .17$
Energy-Gaussian	5.88 ± 0.58	$.2483 \pm .0020$	$.1683 \pm .0012$	$95.33 \pm .63$	$97.39 \pm .42$
VIC moments	9.43 ± 1.58	$.2836 \pm .0182$	$.2287 \pm .0167$	$95.03 \pm .99$	$96.97 \pm .19$
Sphere	8.35 ± 0.77	$.2399 \pm .0085$	$.1590 \pm .0025$	$95.44 \pm .82$	$97.44 \pm .10$
Student-t_5	29.76 ± 3.33	$.2804 \pm .0052$	$.2052 \pm .0043$	92.33 ± 1.04	$96.56 \pm .55$
GMM-Ring	5.57 ± 0.32	$.2403 \pm .0061$	$.1661 \pm .0037$	$95.11 \pm .35$	$97.81 \pm .38$

Table 32: LeWM+SADL E1 path and endpoint diagnostics underlying the upper half of Table 4. Each training seed pools 4,096 positions from the same fixed four-task observations as the INTACT probe in Table 31. Entries are mean ± sample SD across three training seeds. $\widehat{J}_{\mathrm{proj}}$ is a projected Fisher proxy, not the full-dimensional asymptotic functional; b_{1024} is finite-sample kNN integrated squared bias after per-checkpoint whitening.

Endpoint	$r_{\mathrm{eff}}(h)$	$r_{\mathrm{eff}}(z)$	$\widehat{J}_{\mathrm{proj}}$	b_{1024}
SIGReg	116.71 ± 2.84	48.84 ± 2.23	$1.289 \pm .049$	$.1772 \pm .0075$
VISReg	123.62 ± 2.15	55.24 ± 2.65	$1.184 \pm .020$	$.1717 \pm .0027$
QQWorld	114.11 ± 9.39	48.18 ± 3.52	$1.179 \pm .031$	$.1699 \pm .0020$
Energy-Gaussian	103.69 ± 10.34	40.40 ± 2.31	$1.161 \pm .037$	$.1712 \pm .0019$
VIC moments	105.05 ± 2.75	64.58 ± 2.00	$1.308 \pm .117$	$.2228 \pm .0157$
Sphere	95.48 ± 2.14	$33.57 \pm .93$	$1.189 \pm .025$	$.1660 \pm .0022$
Student-t_5	84.26 ± 1.06	30.48 ± 1.29	$1.591 \pm .093$	$.2034 \pm .0184$
GMM-Ring	106.73 ± 2.29	39.83 ± 1.86	$1.232 \pm .063$	$.1756 \pm .0042$

表 31：八个端点的三训练种子 E1 几何、有限样本读出与官方 Direct 控制。模型均为联合四任务 INTACT+SADL；每个种子汇集相同固定四任务观测的 4,096 个位置。几何与偏差为均值 ± 种子样本标准差；E1/E5 成功率也汇总三个训练种子。两个面板的端点顺序相同。$10^3\mathrm{SW}^2$ 表示与高斯参考的平方切片 Wasserstein 距离，乘以 10^3。几何与读出在 E1 测量；E5 列仅包含 E5 检查点的控制成功率。

A. 路径与端点几何：均值 ± 种子标准差

端点	$r_{\mathrm{eff}}(h)$	$r_{\mathrm{eff}}(z)$	η_1 (%)	$\bar{\kappa}$	$\widehat{J}_{\mathrm{proj}}$
SIGReg	118.85 ± 1.19	58.68 ± 1.77	3.45 ± 0.22	$.573 \pm .103$	$1.287 \pm .053$
VISReg	122.11 ± 1.09	60.44 ± 1.26	3.34 ± 0.06	$.473 \pm .074$	$1.250 \pm .077$
QQWorld	120.25 ± 0.90	58.82 ± 0.76	3.47 ± 0.11	$.604 \pm .063$	$1.213 \pm .004$
能量–高斯	112.86 ± 1.69	48.10 ± 0.56	4.26 ± 0.08	$.677 \pm .098$	$1.259 \pm .077$
VIC 矩	106.17 ± 1.53	66.42 ± 0.35	3.01 ± 0.22	$1.035 \pm .158$	$1.330 \pm .066$
球面	102.61 ± 1.29	42.30 ± 0.60	4.96 ± 0.15	$.920 \pm .062$	$1.251 \pm .081$
Student-t_5	89.91 ± 3.03	35.11 ± 1.27	6.38 ± 0.47	$2.840 \pm .319$	$1.660 \pm .135$
GMM-Ring	112.46 ± 0.34	48.29 ± 0.57	4.20 ± 0.10	$.657 \pm .066$	$1.220 \pm .083$

B. 分布距离、函数读出与控制：均值 ± 种子标准差

端点	$10^3\mathrm{SW}^2$	b_{128}	b_{1024}	E1 成功率（%）	E5 成功率（%）
SIGReg	5.19 ± 0.97	$.2558 \pm .0056$	$.1762 \pm .0049$	$95.58 \pm .00$	$97.19 \pm .51$
VISReg	4.02 ± 0.46	$.2505 \pm .0049$	$.1716 \pm .0044$	$95.08 \pm .17$	$97.53 \pm .25$
QQWorld	5.37 ± 0.98	$.2552 \pm .0045$	$.1745 \pm .0035$	$95.81 \pm .13$	$97.56 \pm .17$
能量–高斯	5.88 ± 0.58	$.2483 \pm .0020$	$.1683 \pm .0012$	$95.33 \pm .63$	$97.39 \pm .42$
VIC 矩	9.43 ± 1.58	$.2836 \pm .0182$	$.2287 \pm .0167$	$95.03 \pm .99$	$96.97 \pm .19$
球面	8.35 ± 0.77	$.2399 \pm .0085$	$.1590 \pm .0025$	$95.44 \pm .82$	$97.44 \pm .10$
Student-t_5	29.76 ± 3.33	$.2804 \pm .0052$	$.2052 \pm .0043$	92.33 ± 1.04	$96.56 \pm .55$
GMM-Ring	5.57 ± 0.32	$.2403 \pm .0061$	$.1661 \pm .0037$	$95.11 \pm .35$	$97.81 \pm .38$

表 32：表 4 上半部分对应的 LeWM+SADL E1 路径与端点诊断。每个训练种子从与表 31 INTACT$\widehat{\ }$ 探针相同的固定四任务观测汇集 4,096 个位置。数值为三个训练种子的均值 ± 样本标准差。J_{proj} 是投影 Fisher 代理，而非全维渐近泛函；b_{1024} 是逐检查点白化后的有限样本 kNN 积分平方偏差。

端点	$r_{\mathrm{eff}}(h)$	$r_{\mathrm{eff}}(z)$	$\widehat{J}_{\mathrm{proj}}$	b_{1024}
SIGReg	116.71 ± 2.84	48.84 ± 2.23	$1.289 \pm .049$	$.1772 \pm .0075$
VISReg	123.62 ± 2.15	55.24 ± 2.65	$1.184 \pm .020$	$.1717 \pm .0027$
QQWorld	114.11 ± 9.39	48.18 ± 3.52	$1.179 \pm .031$	$.1699 \pm .0020$
能量–高斯	103.69 ± 10.34	40.40 ± 2.31	$1.161 \pm .037$	$.1712 \pm .0019$
VIC 矩	105.05 ± 2.75	64.58 ± 2.00	$1.308 \pm .117$	$.2228 \pm .0157$
球面	95.48 ± 2.14	$33.57 \pm .93$	$1.189 \pm .025$	$.1660 \pm .0022$
Student-t_5	84.26 ± 1.06	30.48 ± 1.29	$1.591 \pm .093$	$.2034 \pm .0184$
GMM-Ring	106.73 ± 2.29	39.83 ± 1.86	$1.232 \pm .063$	$.1756 \pm .0042$

Table 33: LeWM E1 endpoint-only versus +SADL under the same Official Pure-CEM protocol. Both configurations use a Flash encoder, MATH predictor, shared four-task training, and the same endpoint coefficient for each objective. Pure-CEM uses 300 samples, 30 iterations, $H = 5$, and $RH = 5$ without actor warm-start. Task columns show endpoint-only success rates (%); macro columns are mean ± sample SD across three training seeds, each averaged over three evaluation seeds of 100 episodes per task. Δ is the difference of the two macro means in percentage points.

Endpoint	PushT_0	Cube_0	Reacher_0	TwoRoom_0	Macro_0	$\text{Macro}_{\text{SADL}}$	Δ pp
SIGReg	3.44	49.89	20.00	36.44	27.44 ± 3.44	71.56 ± 1.27	+44.12
VISReg	53.00	55.11	27.33	55.33	47.69 ± 5.14	71.58 ± 4.67	+23.89
QQWorld	19.22	52.56	18.78	40.33	32.72 ± 11.23	74.36 ± 2.90	+41.64
Energy-Gaussian	3.78	45.11	14.78	36.89	25.14 ± 0.67	70.25 ± 2.35	+45.11
VIC moments	5.67	47.44	15.56	29.56	24.56 ± 4.21	67.53 ± 6.09	+42.97
Sphere	34.33	54.33	23.22	47.11	39.75 ± 3.54	75.33 ± 1.36	+35.58
Student-t_5	3.11	45.78	13.44	36.00	24.58 ± 1.78	62.81 ± 3.99	+38.23
GMM-Ring	3.00	43.44	12.78	36.00	23.81 ± 3.71	70.75 ± 1.23	+46.94

表 33：相同官方 Pure-CEM 协议下，LeWM E1 仅端点与 +SADL 的比较。两配置均使用 Flash 编码器、MATH 预测器、共享四任务训练，各目标的端点系数相同。Pure-CEM 使用 300 个样本、30 轮迭代、 $H = 5$ 与 $RH = 5$ ，不使用 actor 热启动。任务列为仅端点成功率（%）；宏平均列为三个训练种子的均值 ± 样本标准差，各种子先对三个评估种子求平均，每任务每评估种子 100 回合。 Δ 是两宏均值之差，单位百分点。

端点	PushT_0	Cube_0	Reacher_0	TwoRoom_0	宏平均$_0$	宏平均$_{\text{SADL}}$	Δ 个百分点
SIGReg	3.44	49.89	20.00	36.44	27.44 ± 3.44	71.56 ± 1.27	+44.12
VISReg	53.00	55.11	27.33	55.33	47.69 ± 5.14	71.58 ± 4.67	+23.89
QQWorld	19.22	52.56	18.78	40.33	32.72 ± 11.23	74.36 ± 2.90	+41.64
能量–高斯	3.78	45.11	14.78	36.89	25.14 ± 0.67	70.25 ± 2.35	+45.11
VIC 矩	5.67	47.44	15.56	29.56	24.56 ± 4.21	67.53 ± 6.09	+42.97
球面	34.33	54.33	23.22	47.11	39.75 ± 3.54	75.33 ± 1.36	+35.58
Student-t_5	3.11	45.78	13.44	36.00	24.58 ± 1.78	62.81 ± 3.99	+38.23
GMM-Ring	3.00	43.44	12.78	36.00	23.81 ± 3.71	70.75 ± 1.23	+46.94

H Image Classification Evaluation

We test whether the same path-capacity correction helps image representations across long-tailed natural images (ImageNet-LT), galaxies (Galaxy10), small natural images (CIFAR-10), and the Imagenette subset of ImageNet. Each comparison pairs the endpoint-only baseline with the same model plus SADL, holding the data, architecture, optimization, and evaluation protocol fixed. SADL estimates excitation and realized demand from the mean representation of each image's augmented views (the *image-mean* variant) in all four datasets. Table 34 reports fixed-epoch classification accuracy: an online head for ImageNet-LT, Galaxy10, and Imagenette, and a held-out ridge readout for CIFAR-10. Each mean and standard deviation is over independent training seeds; the last column counts seeds improved by SADL.

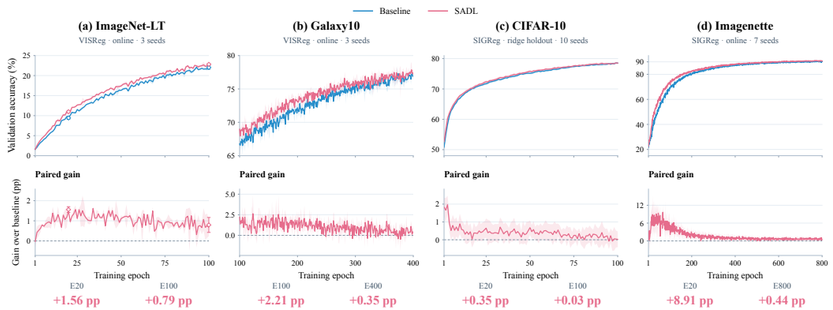


Figure 7: Training trajectories and paired accuracy gains across four image-classification datasets. Top: classification accuracy; bottom: SADL minus the paired baseline, in percentage points. Lines and bands show means and standard deviations over paired training seeds (3, 3, 10, and 7, respectively). ImageNet-LT, Galaxy10, and Imagenette use online heads; CIFAR-10 uses a held-out ridge readout.

Table 34: Image-classification accuracy (%). Baseline and SADL use matched training seeds and the same endpoint objective. Δ is the mean paired gain in percentage points; "wins" counts positive seed-wise differences.

Dataset (n)	Endpoint	Epoch	Baseline	+SADL	Δ	Wins
ImageNet-LT (3)	VISReg	100	22.02 ± 0.41	$\mathbf{22.81 \pm 0.20}$	$+0.794$	3/3
Galaxy10 (3)	VISReg	400	76.89 ± 0.45	$\mathbf{77.23 \pm 0.53}$	$+0.348$	3/3
CIFAR-10 (10)	SIGReg	1	50.81 ± 0.73	$\mathbf{52.64 \pm 0.80}$	$+1.830$	10/10
		10	66.87 ± 0.40	$\mathbf{67.76 \pm 0.67}$	$+0.894$	9/10
		50	75.16 ± 0.36	$\mathbf{75.84 \pm 0.36}$	$+0.680$	9/10
		100	78.50 ± 0.30	78.52 ± 0.32	$+0.026$	4/10
Imagenette (7)	SIGReg	5	24.23 ± 1.49	$\mathbf{27.68 \pm 1.43}$	$+3.450$	7/7
		20	35.60 ± 4.48	$\mathbf{44.51 \pm 3.13}$	$+8.906$	7/7
		100	69.34 ± 1.99	$\mathbf{73.15 \pm 1.89}$	$+3.807$	7/7
		800	90.25 ± 0.17	$\mathbf{90.69 \pm 0.31}$	$+0.444$	6/7

ImageNet-LT and Galaxy10. On ImageNet-LT, ViT-S/8 with VISReg is trained for 100 epochs using a global batch of 256, four 128×128 global crops, and 4,096 projection directions. At E100, SADL improves the matched three-seed online accuracy by 0.794 points. Galaxy10 uses the same backbone and endpoint family; its three-seed gain is 2.207 points at E100 and remains positive at E400 (Table 34). The fixed E400 comparison uses each arm's own E100 model and optimizer state.

H 图像分类评估

我们检验相同路径容量校正是否改善长尾自然图像（ImageNet-LT）、星系（Galaxy10）、小型自然图像（CIFAR-10）及 ImageNet 的 Imagenette 子集表征。每项比较将仅端点基线与加入 SADL 的同一模型配对，固定数据、架构、优化和评估协议。在四个数据集中，SADL 都从各图像增强视图的平均表征估计激发和已实现需求（图像均值变体）。表 34 报告固定轮次分类准确率：ImageNet-LT、Galaxy10、Imagenette 使用在线头部，CIFAR-10 使用保留集岭回归读出。均值、标准差来自独立训练种子；最后一列统计 SADL 改善的种子数。

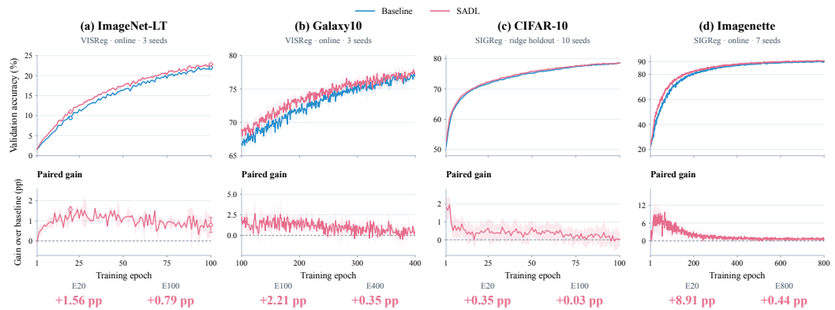


图 7：四个图像分类数据集的训练轨迹与配对准确率增益。上：分类准确率；下：SADL 减配对基线，单位百分点。曲线、带状区域为配对训练种子的均值与标准差（依次为 3、3、10、7 个）。ImageNet-LT、Galaxy10、Imagenette 用在线头部；CIFAR-10 用保留集岭回归读出。

表 34：图像分类准确率（%）。基线与 SADL 使用匹配训练种子、相同端点目标。 Δ 为平均配对增益，单位百分点；"wins"统计逐种子差异为正的数量。

数据集（n）	端点	轮次	基线	+SADL	Δ	Wins
ImageNet-LT (3)	VISReg	100	22.02 ± 0.41	$\mathbf{22.81 \pm 0.20}$	$+0.794$	3/3
Galaxy10 (3)	VISReg	400	76.89 ± 0.45	$\mathbf{77.23 \pm 0.53}$	$+0.348$	3/3
CIFAR-10 (10)	SIGReg	1	50.81 ± 0.73	$\mathbf{52.64 \pm 0.80}$	$+1.830$	10/10
		10	66.87 ± 0.40	$\mathbf{67.76 \pm 0.67}$	$+0.894$	9/10
		50	75.16 ± 0.36	$\mathbf{75.84 \pm 0.36}$	$+0.680$	9/10
		100	78.50 ± 0.30	78.52 ± 0.32	$+0.026$	4/10
Imagenette (7)	SIGReg	5	24.23 ± 1.49	$\mathbf{27.68 \pm 1.43}$	$+3.450$	7/7
		20	35.60 ± 4.48	$\mathbf{44.51 \pm 3.13}$	$+8.906$	7/7
		100	69.34 ± 1.99	$\mathbf{73.15 \pm 1.89}$	$+3.807$	7/7
		800	90.25 ± 0.17	$\mathbf{90.69 \pm 0.31}$	$+0.444$	6/7

ImageNet-LT 与 Galaxy10。ImageNet-LT 上的 ViT-S/8 VISReg 训练 100 轮，全局批量 256，四个 128×128 全局裁剪及 4,096 个投影方向。E100 时 SADL 将匹配三种子在线准确率提高 0.794 个百分点。Galaxy10 使用相同骨干与端点族，E100 三种子增益为 2.207 个百分点，E400 仍为正（表 34）。固定 E400 比较使用各组自身的 E100 模型与优化器状态。

CIFAR-10 and Imagenette. CIFAR-10 pairs a ResNet-18 SIGReg baseline with image-mean SADL and the same held-out ridge readout over ten seeds. The E1 gain is 1.830 points, and nine of ten seeds still improve at E50; by E100 the mean accuracies are nearly equal. The Imagenette baseline follows the LeJEPA ViT-S/8 SIGReg recipe (Balestriero & LeCun, 2025), with four global views and a global batch of 256. Seven paired seeds show the largest gain at E20 (+8.906 points), and the gain remains positive at E800 (+0.444 points). These results show the clearest and most consistent advantage in the early training regime, with improvements also present at the fixed final checkpoint on ImageNet-LT, Galaxy10, and Imagenette.

Image-specific demand. With n image means and encoder width d_h, the covariance rank is at most $r = \min(n-1, d_h)$. We cap demand at

$$q_{\max} = \frac{r \log(1+\gamma d_h/r)}{d_h \log(1+\gamma)}, \qquad q_{\text{img}} = \min\{q_{\max}, \widetilde{q}_\alpha\}.$$

As in the general demand interpolation, the smoothed image demand is

$$\widetilde{q}_\alpha = \begin{cases} q_R, & \alpha = 0, \\ (q_R+\epsilon)^{1-\alpha}(q_E+\epsilon)^\alpha, & 0 < \alpha < 1, \\ q_E, & \alpha = 1, \end{cases} \qquad \mathcal{S}^{\text{img}} = \left[\log(q_{\text{img}}+\epsilon) - \log(k+\epsilon)\right]_+^2,$$

where $\epsilon = 10^{-6}$ and the demand is detached. For ImageNet-LT, $n = 256$, $d_h = 384$, and $\gamma = 9$, giving $q_{\max} \simeq 0.7723$. In the matched five-point interpolation sweep, $\alpha = 0.5$ gives the highest E100 online accuracy.

Which capacity limits the image path? Unlike the measured LeWM and INTACT fixed probes, which are excitation-limited ($C_{\text{exc}} < C_{\text{tr}}$), the separate ImageNet-LT E100 view-residual replay can become transmission-limited ($C_{\text{tr}} < q < C_{\text{exc}}$; Figure 5). This bottleneck switch motivates retaining both capacities in k; the classification comparisons above use the image-mean protocol.

CIFAR-10 与 Imagenette。CIFAR-10 在十个种子上将 ResNet-18 SIGReg 基线与图像均值 SADL 配对，使用相同保留集岭回归读出。E1 增益为 1.830 个百分点；E50 十个种子中仍有九个改善；E100 平均准确率几乎相同。Imagenette 基线遵循 LeJEPA ViT-S/8 SIGReg 配方（Balestriero 与 LeCun，2025），使用四个全局视图与全局批量 256。七个配对种子的最大增益在 E20（+8.906 个百分点），E800 仍为正（+0.444 个百分点）。结果显示训练早期优势最清晰、最一致，同时 ImageNet-LT、Galaxy10、Imagenette 固定最终检查点也有改善。

图像专用需求。对于 n 个图像均值与编码器宽度 d_h，协方差秩至多为 $r = \min(n-1, d_h)$。将需求上限设为

$$q_{\max} = \frac{r \log(1+\gamma d_h/r)}{d_h \log(1+\gamma)}, \qquad q_{\text{img}} = \min\{q_{\max}, \widetilde{q}_\alpha\}.$$

与一般需求插值一样，平滑图像需求为

$$\widetilde{q}_\alpha = \begin{cases} q_R, & \alpha = 0, \\ (q_R+\epsilon)^{1-\alpha}(q_E+\epsilon)^\alpha, & 0 < \alpha < 1, \\ q_E, & \alpha = 1, \end{cases} \qquad \mathcal{S}^{\text{img}} = \left[\log(q_{\text{img}}+\epsilon) - \log(k+\epsilon)\right]_+^2,$$

其中 $\epsilon = 10^{-6}$，需求停止梯度。对于 ImageNet-LT，$n = 256$、$d_h = 384$、$\gamma = 9$，得到 $q_{\max} \simeq 0.7723$。匹配的五点插值扫描中，$\alpha = 0.5$ 给出最高 E100 在线准确率。

哪种容量限制图像路径？与测量的 LeWM、INTACT 固定探针受激发限制（$C_{\text{exc}} < C_{\text{tr}}$）不同，独立 ImageNet-LT E100 视图残差重放可转为受传输限制（$C_{\text{tr}} < q < C_{\text{exc}}$；图 5）。这一瓶颈切换支持在 k 中保留两种容量；上述分类比较使用图像均值协议。

I Computational Cost and Inference Workload

I.1 Short-Run SDPA Hardware Matrix

We benchmark the incremental training cost of SADL across MATH and Flash attention in the encoder and predictor. Here M and F denote MATH and Flash SDPA, respectively; B denotes endpoint-only LeWM and S denotes the same model with full SADL. Both arms use the same SIGReg endpoint weight 0.03, SADL bridge weight 0.01 where applicable, and $\alpha = 0.5$. The fixed protocol uses BF16, gradient checkpointing, batch size 256 per task, 10 data-loader workers, and 40 optimizer steps. PushT runs on one GPU; four-task training uses four GPUs, one task per rank. The reported step time is $(t_{30} - t_{10})/20$, including data loading, optimization, and synchronization; peak device-wide memory is sampled by `nvidia-smi` every 500 ms. Each non-dagger cell averages three independent repetitions.

Table 35: Training runtime and peak memory on RTX 4090, H200, and H20. Each B/S pair gives endpoint-only/SADL means; $\Delta_t = 100(t_S/t_B - 1)$, where positive means measured SADL time is higher. Times are ms/step and memory is GiB. M/M for H200 and H20 uses earlier same-protocol measurements; H20 M/M† pools four hosts and is not host-matched to the other H20 rows. Three-repeat SDs and telemetry are archived.

		Four tasks, four GPUs			*PushT, one GPU*		
Hardware	Enc./pred.	Step B/S	Δ_t (%)	Peak B/S	Step B/S	Δ_t (%)	Peak B/S
RTX 4090	M/M	2528/2442	-3.4	23.38/23.28	2352/2326	-1.1	22.37/22.38
	M/F	2428/2422	-0.3	23.51/23.51	2283/2300	$+0.7$	22.24/22.24
	F/M	1996/1856	-7.0	17.40/17.42	1694/1669	-1.5	16.10/16.10
	F/F	1861/1865	$+0.2$	17.21/17.18	1631/1671	$+2.5$	15.85/15.87
H200	M/M	1195/1131	-5.3	25.11/25.26	1148/1112	-3.1	22.48/22.87
	M/F	1065/1018	-4.4	25.18/24.94	1070/1023	-4.4	22.79/22.55
	F/M	654/569	-13.0	19.97/19.46	622/637	$+2.4$	17.12/17.14
	F/F	579/516	-11.0	19.18/19.13	577/616	$+6.6$	16.87/16.90
H20	M/M†	1355/1354	-0.1	24.24/24.18	1398/1353	-3.3	22.26/22.26
	M/F	1323/1335	$+0.9$	24.77/24.49	1398/1323	-5.4	22.40/22.38
	F/M	1077/921	-14.4	19.28/19.03	1133/918	-19.0	16.92/16.93
	F/F	1078/871	-19.2	19.09/18.76	1063/1053	-0.9	16.69/16.70

Across the 24 within-backend B/S comparisons in Table 35, the median measured SADL step-time change is -3.2%; six comparisons are positive, with a maximum of $+6.6\%$. These short runs therefore show no systematic added per-step latency from the bridge, rather than evidence that the bridge itself accelerates an optimizer step. Flash in the encoder accounts for most of the hardware benefit: in four-task endpoint-only runs, M/M to F/M reduces step time from 2528 to 1996 ms and peak memory from 23.38 to 17.40 GiB on RTX 4090, and from 1195 to 654 ms and 25.11 to 19.97 GiB on H200. Enabling Flash in both modules further reaches 1861 ms/17.21 GiB and 579 ms/19.18 GiB, respectively. The corresponding H20 trend is consistent but uses the historical M/M reference from different hosts. The RTX 4090 M/M third repeat completed all 40 steps and wrote its measurements, despite a post-training distributed shutdown error documented in the archive.

I.2 Search Workload of the Control Interfaces

INTACT's Direct interface performs no candidate-action search. The Official records for the alternative endpoint experiments contain zero candidate-action evaluations and zero planning-cost calls. LeWM Pure-CEM uses 300 candidates per iteration and 30 iterations, corresponding to 9,000 candidate evaluations per planning invocation.

SADL adds a training-only objective and retains these existing control interfaces. It introduces no additional inference network or candidate-search procedure.

I 计算成本与推理工作量

I.1 短期 SDPA 硬件矩阵

我们在编码器和预测器的 MATH、Flash 注意力组合下评测 SADL 增量开销。M/F 分别表示 MATH/Flash SDPA；B 表示仅端点 LeWM，S 表示加入完整 SADL 的同一模型。两组 SIGReg 端点权重均为 0.03；适用时桥接权重为 0.01，且 $\alpha = 0.5$。固定协议使用 BF16、梯度检查点、每任务批量 256、10 个数据加载工作进程及 40 个优化步。PushT 用单 GPU；四任务训练用四 GPU，每个 rank 一个任务。单步时间为 $(t_{30} - t_{10})/20$，包含加载、优化与同步；nvidia-smi 每 500 毫秒采样设备总峰值显存。每个不带匕首标记的单元平均三次独立重复。

表 35：RTX 4090、H200、H20 的训练时间与峰值显存。每组 B/S 给出仅端点/SADL 均值；$\Delta_t = 100(t_S/t_B - 1)$，正值表示 SADL 实测时间更高。时间单位毫秒/步，显存 GiB。H200、H20 的 M/M 使用早期同协议测量；H20 M/M† 汇集四主机，与其他 H20 行不匹配主机。三次重复标准差及遥测已存档。

		四任务，四 *GPU*			*PushT*，一 *GPU*		
硬件	编码器/预测器	单步 B/S	Δ_t (%)	峰值 B/S	单步 B/S	Δ_t (%)	峰值 B/S
RTX 4090	M/M	2528/2442	-3.4	23.38/23.28	2352/2326	-1.1	22.37/22.38
	M/F	2428/2422	-0.3	23.51/23.51	2283/2300	$+0.7$	22.24/22.24
	F/M	1996/1856	-7.0	17.40/17.42	1694/1669	-1.5	16.10/16.10
	F/F	1861/1865	$+0.2$	17.21/17.18	1631/1671	$+2.5$	15.85/15.87
H200	M/M	1195/1131	-5.3	25.11/25.26	1148/1112	-3.1	22.48/22.87
	M/F	1065/1018	-4.4	25.18/24.94	1070/1023	-4.4	22.79/22.55
	F/M	654/569	-13.0	19.97/19.46	622/637	$+2.4$	17.12/17.14
	F/F	579/516	-11.0	19.18/19.13	577/616	$+6.6$	16.87/16.90
H20	M/M†	1355/1354	-0.1	24.24/24.18	1398/1353	-3.3	22.26/22.26
	M/F	1323/1335	$+0.9$	24.77/24.49	1398/1323	-5.4	22.40/22.38
	F/M	1077/921	-14.4	19.28/19.03	1133/918	-19.0	16.92/16.93
	F/F	1078/871	-19.2	19.09/18.76	1063/1053	-0.9	16.69/16.70

表 35 的 24 组后端内 B/S 比较中，SADL 实测单步时间变化中位数为 -3.2%；六组为正，最大为 $+6.6\%$。因此，这些短运行显示桥接没有系统性额外单步延迟，而不能据此声称桥接本身加快优化步。编码器 Flash 提供大部分硬件收益：四任务仅端点运行从 M/M 改为 F/M，RTX 4090 时间由 2528 降至 1996 毫秒，峰值显存由 23.38 降至 17.40 GiB；H200 时间由 1195 降至 654 毫秒，显存由 25.11 降至 19.97 GiB。两模块均启用 Flash 后，分别进一步达到 1861 毫秒/17.21 GiB 与 579 毫秒/19.18 GiB。H20 趋势一致，但使用不同主机的历史 M/M 参考。RTX 4090 M/M 第三次重复完成全部 40 步并写出测量，尽管随后发生分布式关闭错误，已记录于存档。

I.2 控制接口的搜索工作量

INTACT Direct 接口不搜索候选动作。其他端点实验的官方记录均含零次候选动作评估、零次规划代价调用。LeWM Pure-CEM 每轮 300 个候选、30 轮迭代，每次规划共评估 9,000 个候选。

SADL 只增加训练目标，保留既有控制接口，不增加推理网络或候选搜索过程。